Panasonic

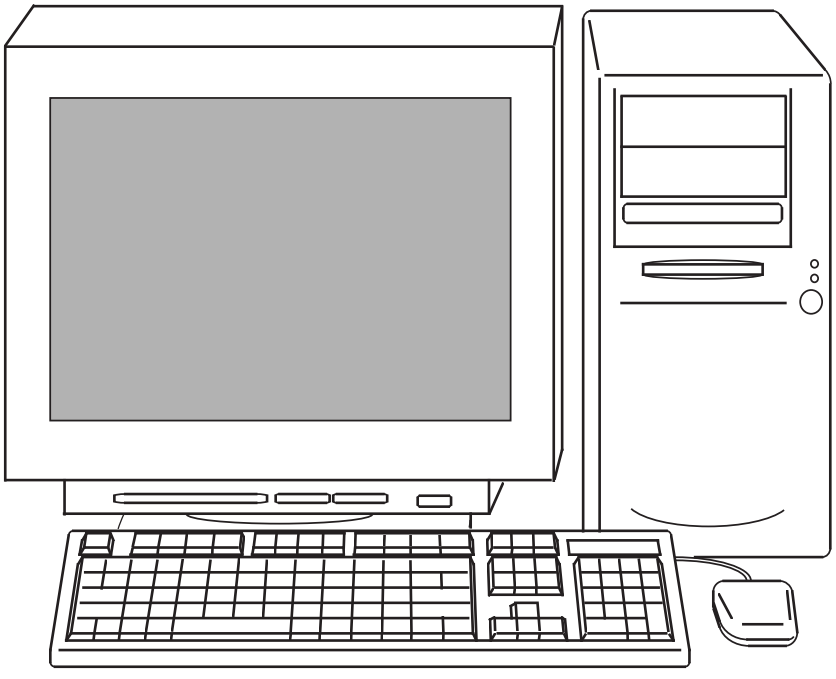
Operating Instructions

Production Monitoring & Dispatch

MES for Electronics Assembly

PanaCIM-EE

Model No. **NM-EJS3B**



Thank you for purchasing the “Panasonic product.”
- Read the operating instructions carefully for proper and safe usage.
- Before placing the machine in service, be sure to read “Safety Precautions.”
- Store the operating instructions carefully.

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Original Instructions

For using this system:
This system is designed with a purpose of visualization of work productivity, more efficiency and quality warranty in electronic component mounting process. Use this system solely for the purpose.

Do not use this system for other purposes.

Operating machines:
Any person who operates the machine should receive a special training.

Any operator who has not received the training is not allowed to operate.

For training, contact Panasonic Smart Factory Solutions Co., Ltd., Sales company or Sales agent.

⚠ Warning

Do not operate the machine without receiving special training



Risk of injury due to improper operation.

Prohibited

●About training, conduct in your company or contact us.

For machine training in Europe, you may contact Panasonic Industry Europe:

Panasonic Industry Europe GmbH FA Division
Robert-Koch-Strasse 100, 85521 Ottobrunn Germany
Phone number: 0049-(0)89-453541192

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About the Instruction

■ Operating instruction contents

The following operating instructions are included in the DVD-ROM attached to this machine. The instructions are divided according to work or function. Use one that is appropriate for the purchased module (application) or your purpose.

∗ BM series machine is included in the explanation of this manual, but its functions are not supported.

Type	Description
<General>	
Introduction	Overview of machine and basic setting/operation/analysis/utility.
Basic Operation (PanaCIM Web Application)	PanaCIM Web application's basic operation.
<Modules>	
Production Analysis	Collect and analyze machine's running information and display its statistics on real time basis to support productivity improvements (recognition/pickup error rate improvement, machine trouble stop time reduction) for mounting line.
Traceability	Combine PCB information read by a barcode reader on machine or conveyor and production record or material/part information of machine to manage history of PCBs for quality control and market support.
Material Verification	Verify material/part information and recipes while setting up material/parts in production setup/model change/splicing/part exhaust to prevent wrong material/part set up.
Product Changeover and Control	Auto download recipe and adjust lane width based on PCB ID for quicker model change.
Production Monitoring	Retrieve production information and operation information on real time basis and display line status on the monitor and PDA to provide operators with instructions for events such as error or part exhaust. This is to reduce mount line stop time or to prepare in advance for better productivity.
Enterprise Link	A function to output the data collected from machines during production (material verification data, trace data, operation performance data) in the standard format so external applications can reuse the data. Part information can be loaded also in the standard format.

Type	Description
Material Control	A function to provide the location and the number of parts or feeders stocked in the production or preparation area. MSD function controls MSD parts' expired date.
Component Supply Navigator	The Component Supply Navigator optimizes the supply sequence to provide a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line occurred due to component shortage and contribute to manpower saving.
Maintenance	For the purpose of reducing time spent on maintenance planning / control operations, asset (machines, heads and feeders) profiles can be controlled on the floor, or the maintenance history of each asset can be recorded.

License CD name of modules is as follows.

Module Name	License CD Name
Production Analysis	Production Analysis
Traceability	Traceability
Material Verification	Material Verification
Product Changeover and Control	Product Changeover and Control
Production Monitoring	Production Monitoring & Dispatch
Enterprise Link	Enterprise Link
Material Control Base	Material Control Base
Material Control Allocation	Material Control Allocation
Material Control MSD	Material Control MSD
Component Supply Navigator	Component Supply Navigator
Maintenance	Maintenance

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1. OVERVIEW

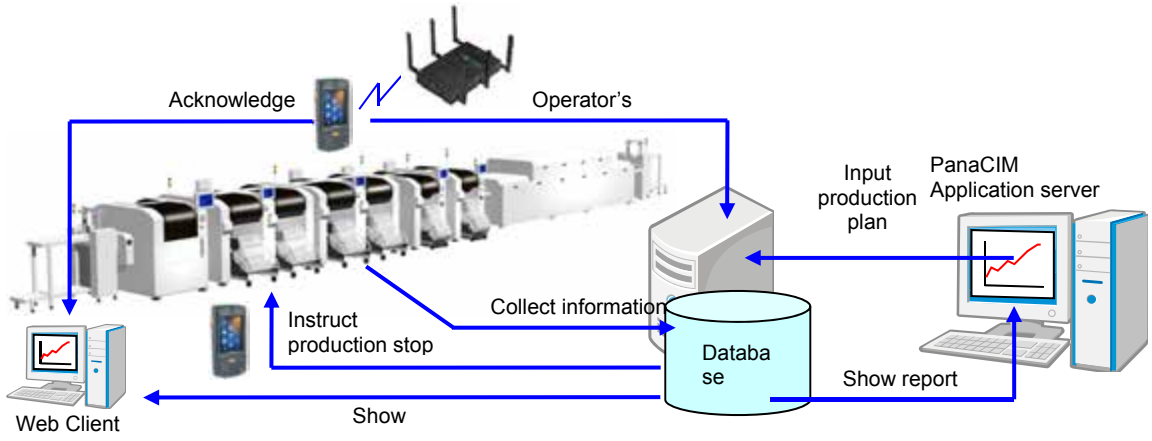
This chapter explains the overview of PanaCIM-EE / Production Monitoring & Dispatch.

1.1 Overview

PanaCIM-EE (PanaCIM Enterprise Edition: hereinafter called PanaCIM) PMD allows helps users to retrieve real-time production and operation information, and to display the line status on monitors and mobile scanners to instruct operators to solve errors or part exhaust which will reduce the time of mount line stop or part preparation and improve productivity.

Main Function *1

Monitor lines of the entire factory
Collect real time events and display necessary warnings
Create work order
Display changeover information
Display operator's response to events



* 1 To use PMD modules, you will need Material Verification Module (license).
To show machines' operational information, Production Analysis (license) is required.

PMD Module functions

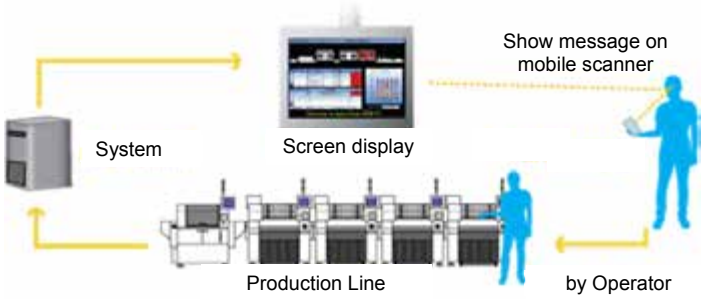
Main Function	Detailed Function		Description	Necessary license ※	
	Category	Sub-category		PMD	Material Control Basic
Line monitor	Show operation (ANDON)		Calculate remaining count based on production record	○	—
Display events	Show events and warnings		Collect real time events and display necessary warnings		—
	Show operator's response		Input work acknowledgement to the events and notify other operators		—
Work order coordination	Create work order		Input and instruct planned production count to workers	○	○
	Show work order status		Show the actual status of work order	○	○
Changeover	Show changeover information		Show changeover information when it needed		

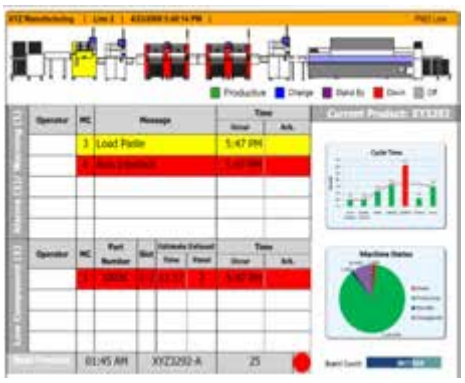
＊ It also requires Material Verification Module(license).

PMD Report (Overview)

PMD Report		Necessary license
PMD Event History Report	Show event history	PMD license
PMD Event Summary Report	Show summary of an event	PMD license
Work in Progress Report	Show Work in Progress	Material Control basic license
Next Work Order Report	Instruct operator the next work	Material Control basic license
Line downtime report	Show machine's down time	PMD license

PMD screen sample



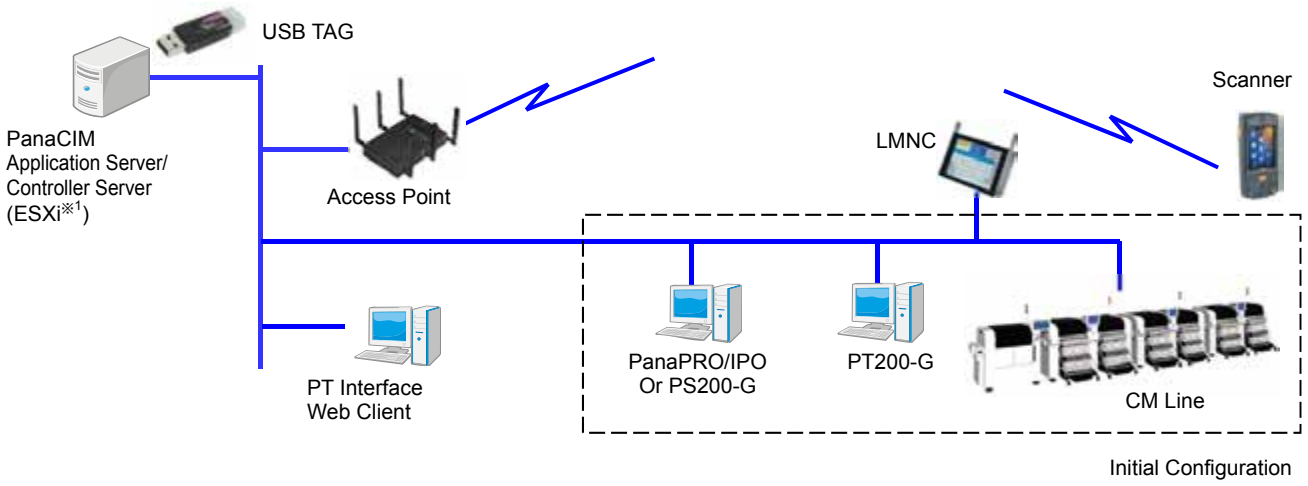


1.2 PanaCIM-EE/PMD System Configuration

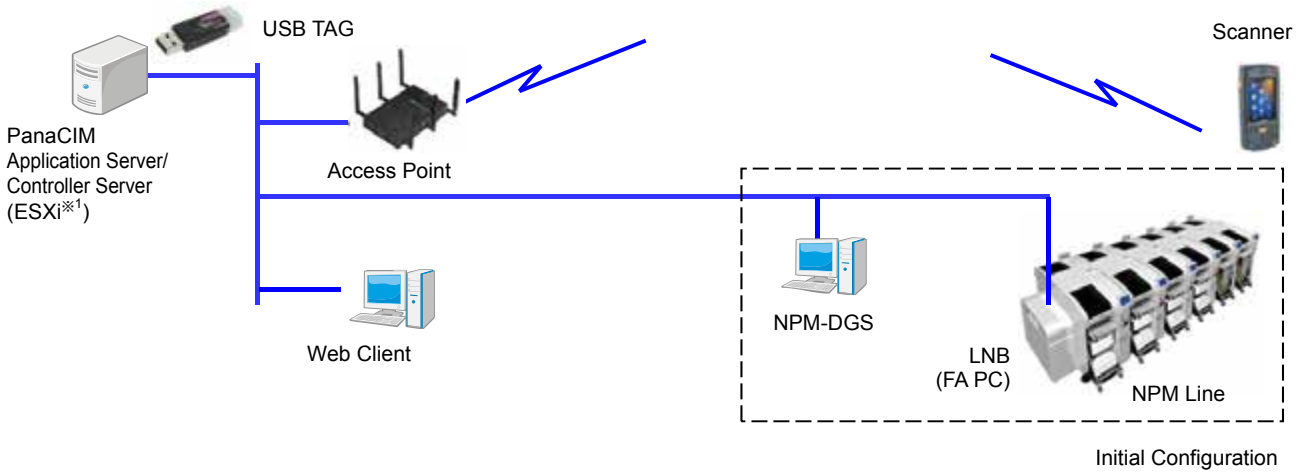
Small/Large scale system configurations including PanaCIM are shown below.

System Configuration with PanaCIM (Small scale)

CM Line



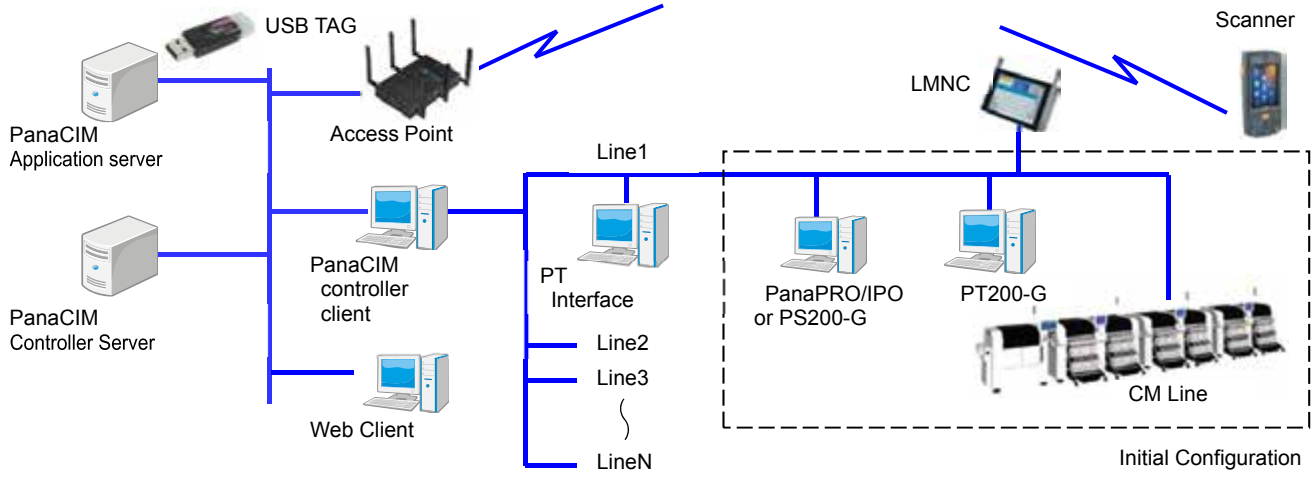
NPM Line*2



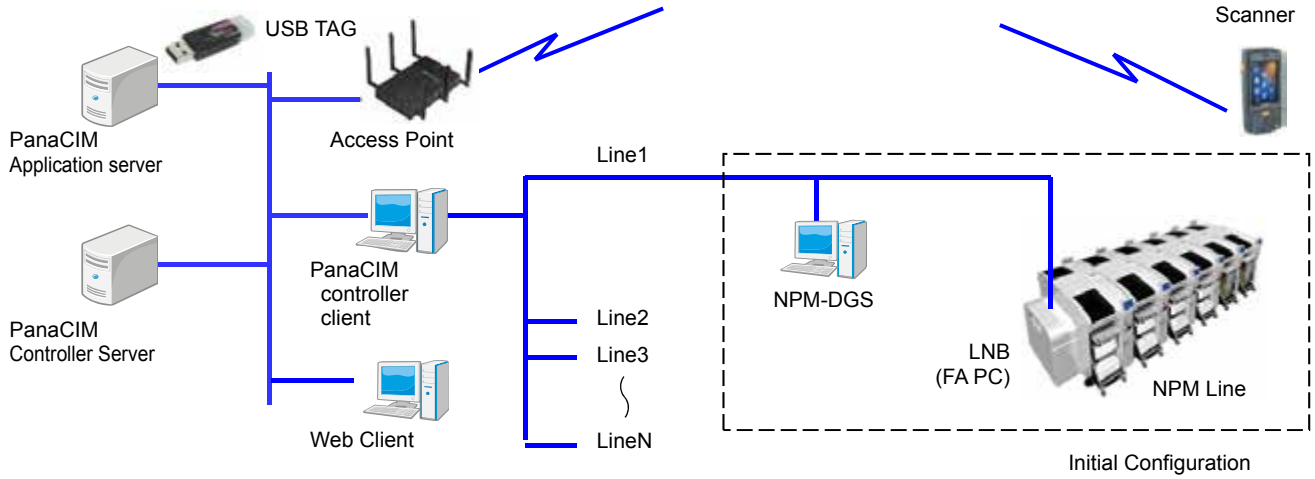
* 1 ESXi: Virtual tool which allows multiple applications to be run on a server.
* 2 For NPM Line: NPM series machine includes AM100/SPD/SPG/SPV.

System Configuration with PanaCIM (Large scale)

CM Line



NPM Line [*1]



* 1 For NPM Lines: AM100/SPD/SPG/SPV machines included as NPM-series machine.

1.3 Functional Overview

PMD Functions

PanaCIM-EE/PMD Functions (1/3)

General			
	Necessary license		PMD, Material Verification, Production Analysis
	Applicable machines		CM602, CM402, CM401,CM232, CM212, CM101, DT401, NPM-S,NPM-D,NPM-D2,NPM-D3,NPM-W,NPM-W2,NPM-W2S,NPM-TT, NPM-TT2,NPM-DX,AM-100,SP70,SPD,SPG,SPV, third party's machine (SPI,OVEN,AOI)
Monitor function			
	Production information/Real time event monitor		
		Real time event display rule	Machine events and material verification events are displayed separately.
			Events are displayed in the chronological order.
			Error events are displayed before warning ones.
			Each of error or warning events are individually displayed in chronological order.
		Displayed event(s)	The number of event(s) displayed can be changed from Properties setting (lines).
			If more events the max number occurred, the total event count will be displayed.
		Event kind	Events to be displayed or their warning level color depends on their type.
	Customized screen		
		Controller	Custom message, clock, text, horizontal line, image, vertical line, vertical arrows, horizontal arrow and others.
		Layout	Set controller on the canvas as you wish.
		View	Change controller view.
		Display	
	Monitor device		Monitor display on the machines which work with VGA.

	Visual control panel controller name	Displayed
KPI	Cycle time	Cycle time
	Cycle Time by M/C	Route name
		Cycle time by machine
	Operational ratio	Actual operational ratio
		Last one hour operational ratio
		Total operational ratio (Operational Ratio: Operational ratio of a day)
	Operational Ratio by M/C	Route name
		Operational ratio by machine
	Panel Trace	Route name
		PCB barcode
	Pickup ratio	Machine name
		Feeder address
		Pickup ratio
	Produced count	Produced count

	Visual control panel controller name	Displayed
Chart	Operational ratio	Actual operational ratio (in graph)
		Last hour operational ratio (in graph)
		Total operational ratio (in graph)
	Actual cycle time	Actual cycle time (in graph)
	Produced count	Produced count (in meter)

PanaCIM-EE/PMD Functions (2/3)

Event notify function			
	Send to visual control panel		Send machine events to visual control panel and display them.
	Send to mobile scanner		Send machine events to mobile scanner and display them.
Custom message			
	Display on visual control panel		Display a custom message of any line at a specified interval.
Acknowledge function (mobile scanner)			
	Setting	Select line	Select a line events of which you want to receive.
			You can select multiple lines.
		Select event	Select events to receive (machine events only, part supply events only, or both)
	Operation	Acknowledge events	Acknowledge and response to the events displayed on a mobile scanner.
		Event acknowledge clear	Clear event acknowledgement you have acknowledged once from mobile scanner.
		Event force end	Change the displayed event status to complete forcibly from a mobile scanner.
			You can select a reason from pre-defined ones for force end.
		Start calculating line stop time	Start calculating line top time from a mobile scanner.
		Stop calculating line stop time	End calculating line top time from a mobile scanner.
	Record	Event acknowledgement response record	Record the operator ID who acknowledged the event and its time stamp.
		Event acknowledgement clear record	Record the operator ID who cleared the event and its time stamp.
		Event force end record	Record the operator ID who ended the event forcibly and its time stamp.

PanaCIM-EE/PMD Functions (3/3)

Create work order (Schedule input is a function of Material Control)			
	Display schedule		Display production schedule based on the work orders.
	Schedule display unit		Production schedule can be displayed on daily or weekly basis.
	Manage work order		Add/edit/delete/import work orders.
			Data to be input when you add a work order: work order name, route name, product name, mount side, mix name, line tact time, pattern(s)/panel, planned pattern(s), production start time, production end time, changeover time and others.
Show changeover information			
	Setting	Select line	Select a line changeover information of which it to be displayed on the visual control panel.
		Set display timing	Set timing to display changeover information.
			Select one of the following timings.
			Remaining production count (PCBs): Notify based on the remaining count (of PCBs) to finish the on-going production.
			Remaining production count (%): Notify based on the remaining production count (5) to finish the on-going production.
			Remaining production time: Notify based on the remaining time to finish the on-going production.
	Display		Based on the work order entered at the schedule, show the next product name on the visual control panel.
Report function (viewable on Web application)			
	PMD event history report		Includes the following data. ID, line name, machine name, location, event details, message, product name, operator, occurred time, acknowledged time, time to acknowledge, handling started time, time required to start handling, handling end time, time required to finish handling, total time, force clear, force clear reason.
	PMD Event Summary Report		Includes the following data. ID, line mame, event descriptions, count, total time to acknowledge, average time to acknowledge, total time to finish and average time to finish
	Work in Progress report		Includes the following data. ID, line name, product name, cycle time, count, planned start time, actual start time, expected end time, comment.
	Next work order report		Includes the following data. ID, line name, product name, cycle time, count, planned start time, expected start time, expected end time, comment.
	Line downtime report		Includes the following data. ID, line name, start time, end time, period, (DD:MM:HH:SS), reason, operator.
Others			
	Operator display image input		Input operator’s image to be displayed on a mobile scanner.

■ Report for production monitoring and dispatch

Reports for production monitoring and dispatch are displayed using the PanaCIM Web application.

Note: To view the reports, refer to the PanaCIM Introduction manual.

The table below is a list of the selectable reports for production monitoring and dispatch.

Production Reports	
PMD Event History Report	Displays all the events that occurred on the selected machine/route over a selected period.
PMD Event Summary Report	Displays the events that occurred on the selected machine/route grouped by Event Type.
Work In Progress Report	Provides the list of Work Orders currently running on the factory.
Upcoming Work Orders Report	Displays the list of Work Orders scheduled for Production.
Route Down Time Detail Report	Displays the Route Down Times for the selected period, along with the reason for the down time and the operator who initiated it.

■ PMD functional overview

Application	Overview
Visual Control Panel (VCP)	Select the **Visual Control Panel** Application from the Production Monitoring & Dispatch link in the PanaCIM EE Launchpad to start this function. This Application displays the real-time production status of a route in a visual display unit, and provides real time updates for the following: • Machine Status • Production Events and Exceptions • Changeover Details • Operator Acknowledgements • Key Performance Indicators
Manage Work Orders	Select the **Manage Work Orders** Application from the Chart & Report link in the PanaCIM EE Launchpad to start this function. Use this Application view work orders in real-time. When PMD receives real-time notifications from the database (i.e. work order start times, end times, and shifts in schedule), data is refreshed so users are always aware of the current status of the work order(s).
Template Design	Select the **Design** Application from the Production Monitoring & Dispatch link in the PanaCIM EE Launchpad to start this function. Use this application to design and customize screens according to your own factory requirements. Your customized templates can then be used to view real-time and statistical data in a visually appealing and highly informative format. It includes the positioning of various controls, binding with Routes and Equipment, Colors, Fonts, and various Properties. Create and save as many templates as you wish for future use.
Template Explorer	Select the **Template Explorer** Application from the Production Monitoring & Dispatch link in the PanaCIM EE Launchpad to start this function. Use this Application to map previously designed templates to different Routes or Equipment quickly. Also, the import/export of templates allows the templates to be applied to other users' Web applications as well. Existing templates can be overwritten and new templates can be created with this Application.

Mobile Scanner	When a machine encounters production exceptions or events they are dispatched to the scanner for you view and acknowledge them. If PMD is configured, the PMD Events screen will display after operator login.  Activities preformed on the Mobile Scanner are reflected in the Visual Control Panel. You will use the Mobile Scanner to: • Set Machine Events • Configure Routes for Event Types • Configure Route Downtime • Fulfill Part Supply Events

■ PMD functional overview (Mobile Scanner)

When a machine encounters production exceptions or events they are dispatched to the scanner for you view and acknowledge them.
If PMD is configured, the PMD Events screen displays after operator login. Activities preformed on the Mobile Scanner are reflected in the Visual Control Panel.

When the PMD license has been enabled, the PMD icon will be displayed.
PMD events can be checked via the following screens.



Refer to the table below for a PMD Events scanner screen settings and descriptions.



When a machine encounters an event, it is displayed as shown at left.
An error event that will cause machine stop will be displayed on red background, and a warning event will be displayed on yellow background.

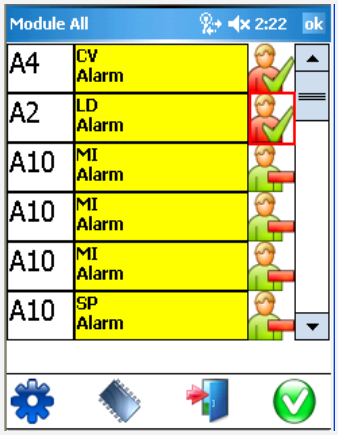
Operator uses the toggle button to select the event whose problem to be eliminated and press  to dispatch the action status of the event concerned to the scanner. See below for details of the Toggle button.

R10 Abbreviation of the line name

M1
AXIS ERROR An event that has occurred

Toggle (See below for details)







Toggle. Use to change the status of an event. The following icons can be toggled to using this button:



Acknowledge an event. Notify operators and the PMD system that you *will* be working on this event.



Un-acknowledge an event. Notify operators and the PMD system that you *will not* be working on this event.



Close an event. Used to clear the event forcibly from the Mobile Scanner and VCP window.



Confirm settings made when toggling.



Displays the settings screen.





Displays the part scan screen.

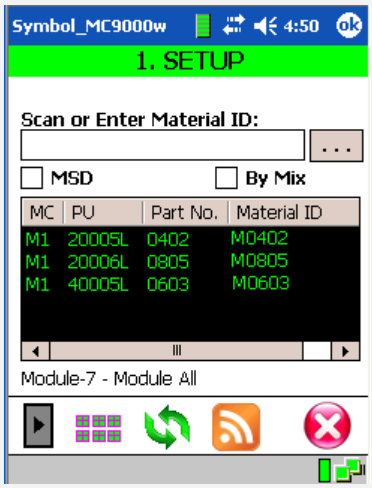


Log out. Note that the scanner may automatically log you out based on a pre-configured time.

■ Fulfill Part Supply Events

Refer to the action table below to fulfill Part Supply events.

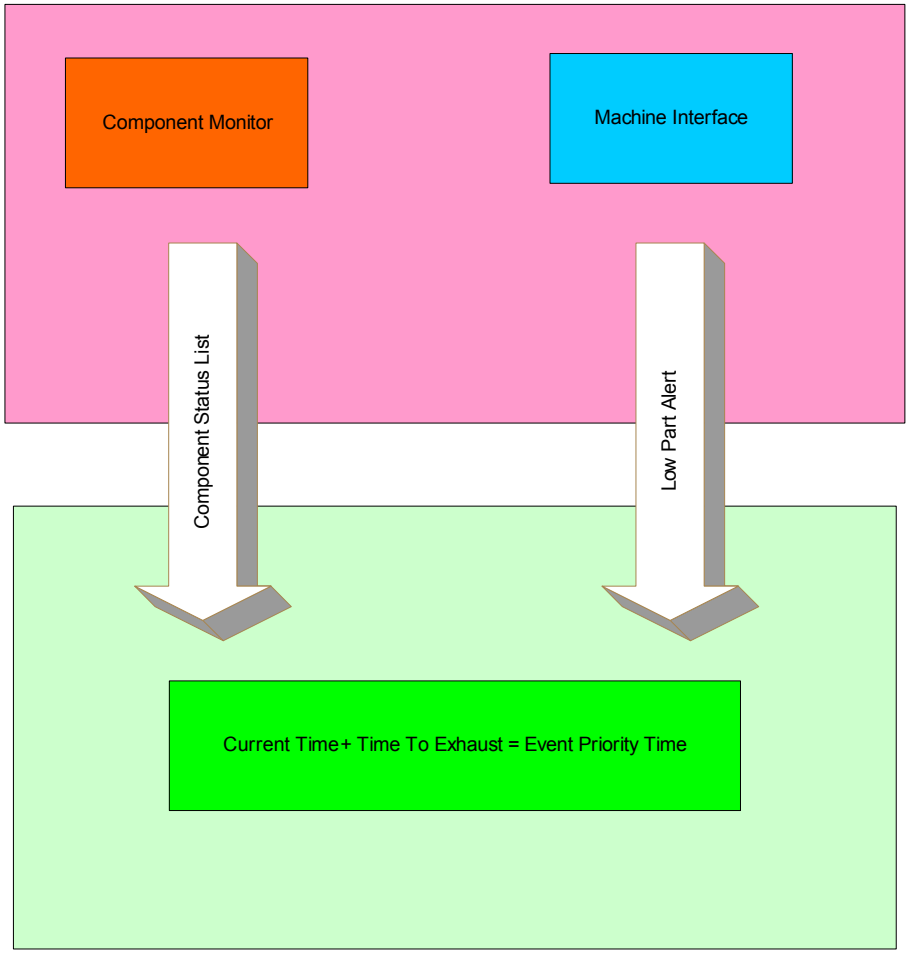
Note: Accessing the Part Scan menu provides a single scanner interface for both PMD Event Handling and Material Verification/Material Control operations.

Step	Procedure
1	Select  in the PMD Events screen. 
2	The main **Part Scan** screen displays.   Click this icon to return to the **PMD Events** screen. The  icon in the lower right corner of the screen indicates that the scanner is connected to the PanaCIM host. If not connected, the icon is displayed in blue.

1.4 Event Prioritization

The prioritization of Andon events is based on the time to exhaust. Currently the priority is based on first in first displayed. This change will add time to exhaust component to the prioritization. Thus, a splice warning for a part will show up ahead of another part if its reel will exhaust sooner, even though the system might have received it at a later time.

The system interfaces with component monitor to evaluate time to exhaust priority. Thus **MC** license is a requirement for this feature to work. The internal architecture for this feature is shown as below. The following diagram shows how event priority is calculated. The Event priority time is added to splice warning message that VCP and Scanner use to order their display.



The Andon Monitor registers a request with the Component Monitor for Low Component Alert list for all equipment that is licensed for PMD. This data is cached in the Andon Monitor and updated when new data is received. When splice warning event is received from the Machine Interface this event is infused with the expected time to exhaust received from the component monitor list then an Andon Event generated and propagated.

There is however a race condition that exists in the model when events are triggered by machine events. Since the Component Monitor only propagates its data periodically the relevant data may not be, and probably will not be, available for evaluation at the point in time that the Andon Monitor receives the same event. The Component Monitor and Andon Monitor will receive this event at the same time but data has yet to be synced up between the two processes.

The response to the request is delayed because Component Monitor could be busy processing many other events. This could delay the event response time to the scanner and PMD VCP. System load could cause delays from time of event reception to Andon event propagation.

2. SETTINGS

This chapter explains a setup of PanaCIM-EE / Production Monitoring & Dispatch.

2.1 Overview

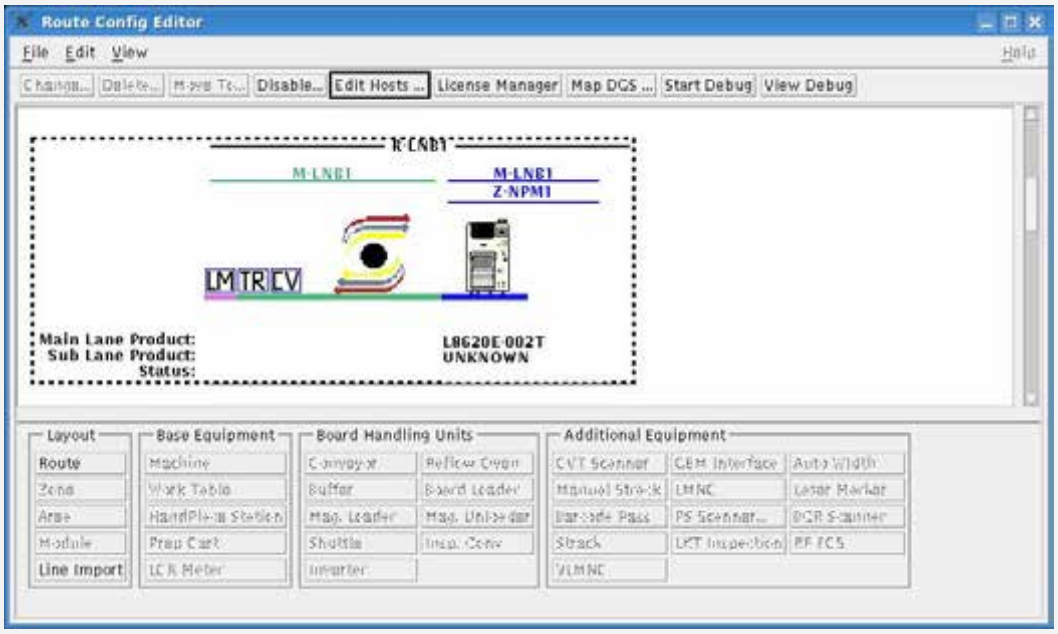
Before performing PMD operations, the PMD Module must be configured for use. This section explains the settings related to the PMD screen and notifications.

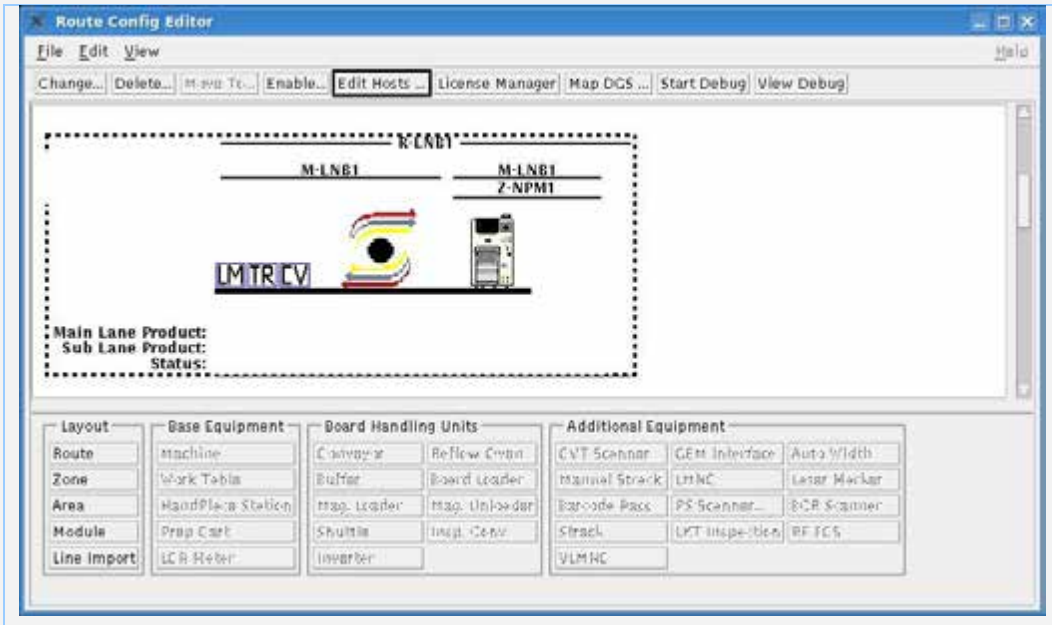
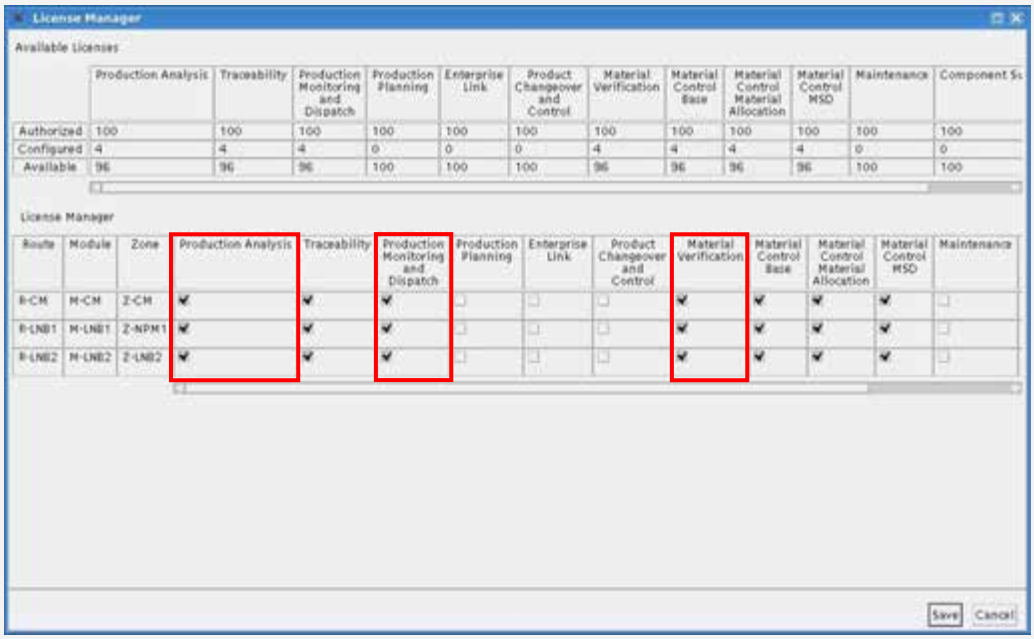
2.2 Settings – CIMC Controller

2.2.1 Configuration Licenses

Ensure the below guidelines are met when configuring PMD licenses. If you need further guidance to configure these licenses, please reference the PanaCIM Base document for complete instructions to configure licenses using Route Editor.

- Scroll down to the Route that is to be controlled by **PMD** and check the following:
  - Production Monitoring and Dispatch
  - Material Verification
  - Production Analysis

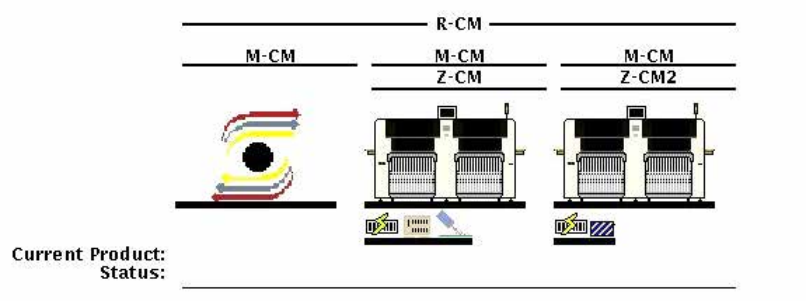
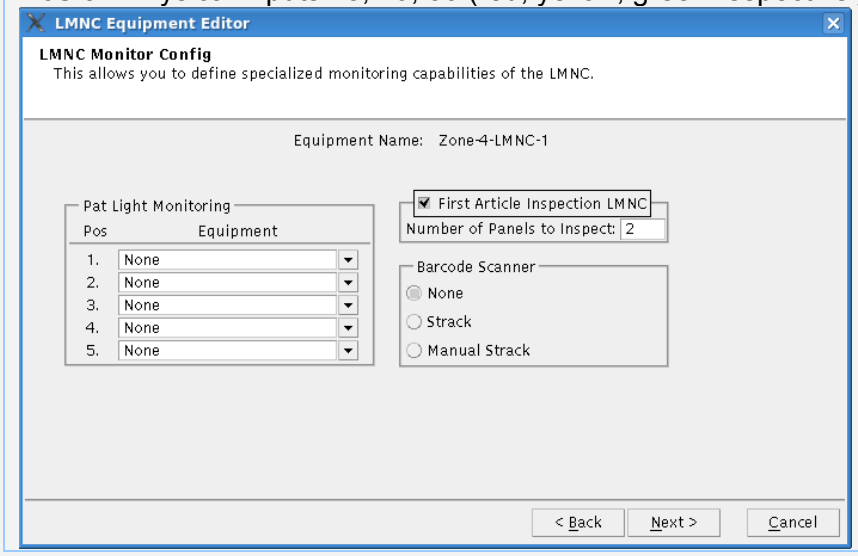
Step	Procedure
1	Open **Route Editor** from PanaCIM controller application tool bar. 
2	Select the route to change the license configuration and click **Disable**. 
3	From the menu on the toolbar, select **License Manager** to launch the License Manager.

Step	Procedure
	
4	Under **Available licenses**, select the license for modules and zones. To use the PanaCIM/Production Monitoring & Dispatch, place a check mark in the following checkboxes: • **Production Monitoring & Dispatch, Material Verification,** and **Production Analysis:** Enable the selected license module by zone.  **Note:** The lines available in the Route Editor appear in green. To change the configuration of the license, disable the lines by removing the check mark in the box.
5	After license configuration has been completed, click **Save** and exit the License Manager.

2.2.2 Pat Light Configuration

This function is not eligible for product support
PMD supports Pat Light Monitoring for third-party machine such as Conveyors and Reflow.

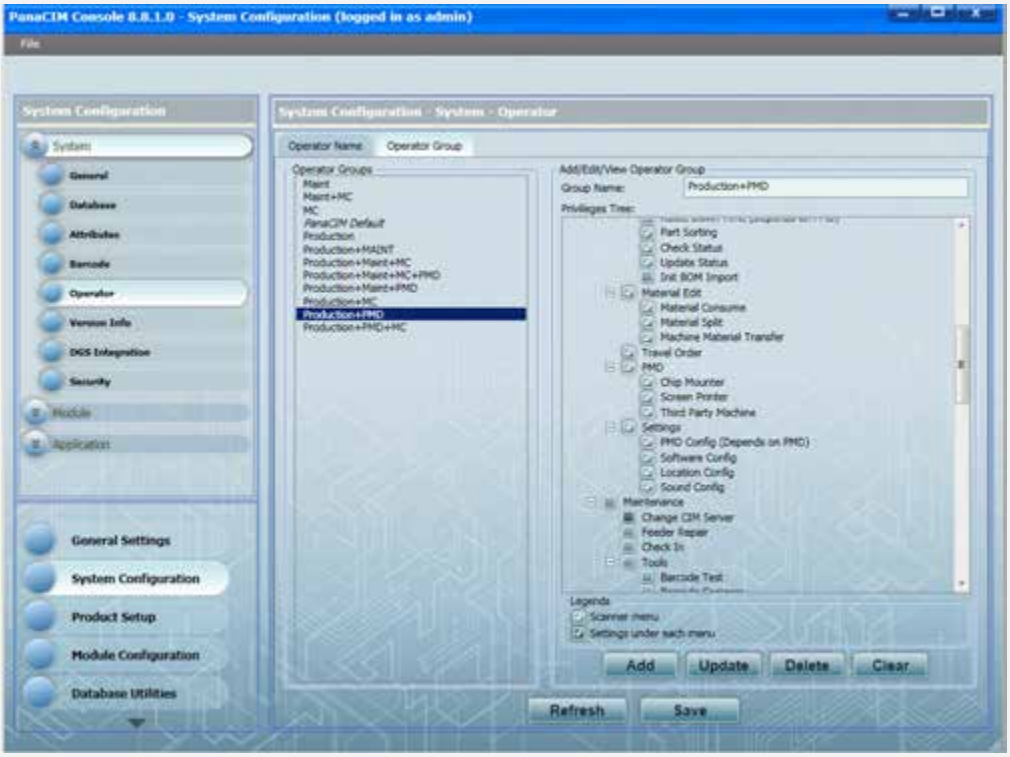
Refer to the action table below to configure the LMNC for Pat Light Monitoring.

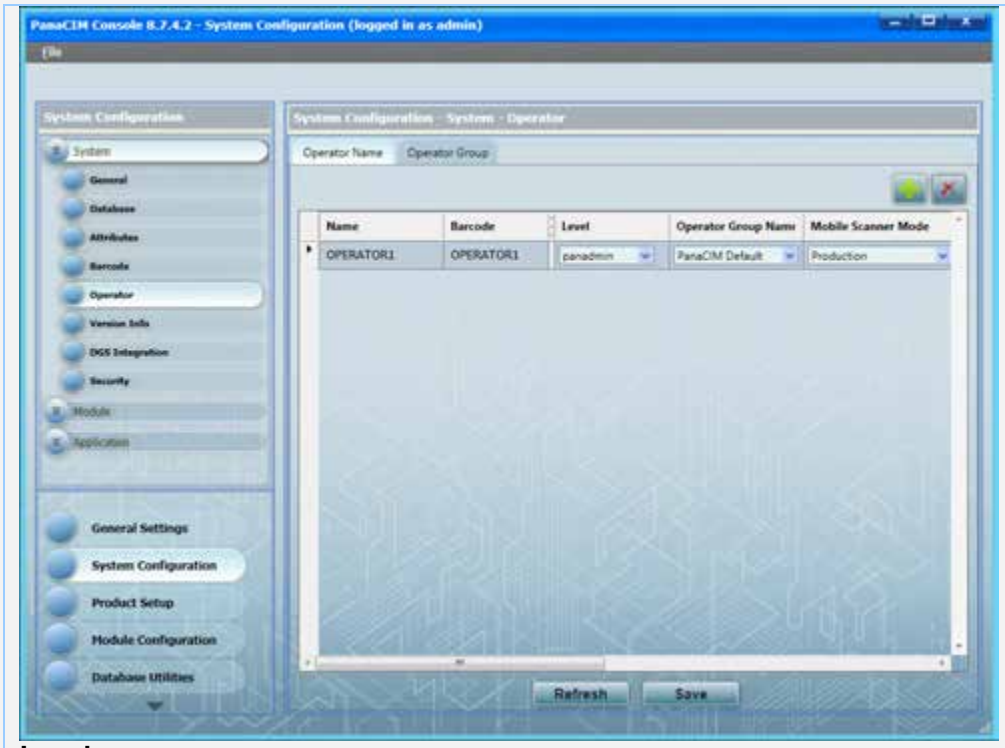
Step	Procedure
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the **LMNC icon** in the appropriate Route. 
3	Click **Next** through each of the LMNC Equipment Editor pages until you reach the **LMNC Monitor Config** page.
4	Select the **Equipment Positions** where Pat Light monitoring will be configured using the drop-down selections. **Note:** Drop-down selections are limited to how it is hard-wired into the LMNC. Required selections for each position include: **Pos 1** - Physical Inputs 16, 17, 18 (red, yellow, green respectively) **Pos 2** - Physical Inputs 19, 20, 21 (red, yellow, green respectively) **Pos 3** - Physical Inputs 22, 23, 24 (red, yellow, green respectively) **Pos 4** - Physical Inputs 25, 26, 27 (red, yellow, green respectively) **Pos 5** - Physical Inputs 28, 29, 30 (red, yellow, green respectively) 
5	Click **Next** and **Save**.

2.3 Configuration on Application Server Side

2.3.1 Configuration of Mobile Scanner Operator

To dislay the PMD screen on the mobile scanner, follow the steps below to configure the settings. This setting must be performed operator by operator.

Step	Procedure
1	Select **PanaCIM Console**->**System Configuration**->**System**->**Operator**.
2	Click the **Operator Group** tab.
3	Select **an operator group** you want to change in the **Operator Group** and check the checkboxes of **PDM** in **Privileges tree** according to the condition shown below.  <Chip Mounter>Event for the placement machine is shown in the PMD monitor. <Screen Printer>Event for the printer is shown in the PMD monitor. <Third party machine> Event for the third party machine is shown in the PMD monitor. Checking all checkboxes will show all events.
4	Upon completion of setting, press **Save.**
5	Repeat the above steps per operator group**.**
6	Click the **Operator name** tab.
7	Type **Name** and **Barcode** (operator ID), and select **Level**, **Operator Group Name** and **Mobile Scanner Mode** from the drop-down list.

Step	Procedure
	 **Level:** Select an operation level of a mobile scanner. Select from among panadmin (Administrator), panasupr(Super) and panaoper(Operator) **Operator Group Name**: Select an operator group that PMD is checked in the **Operator Group** tab. **Mobile Scanner Mode:** Select a mobile scanner mode. Select mainly used mode from among Production, Maintenance and Material.
8	Repeat the above steps for each operator and click **Save**.

<Reference>

Once an operator registerd in above step logs in the mobile scanner, the PMD screen will automatically appear.

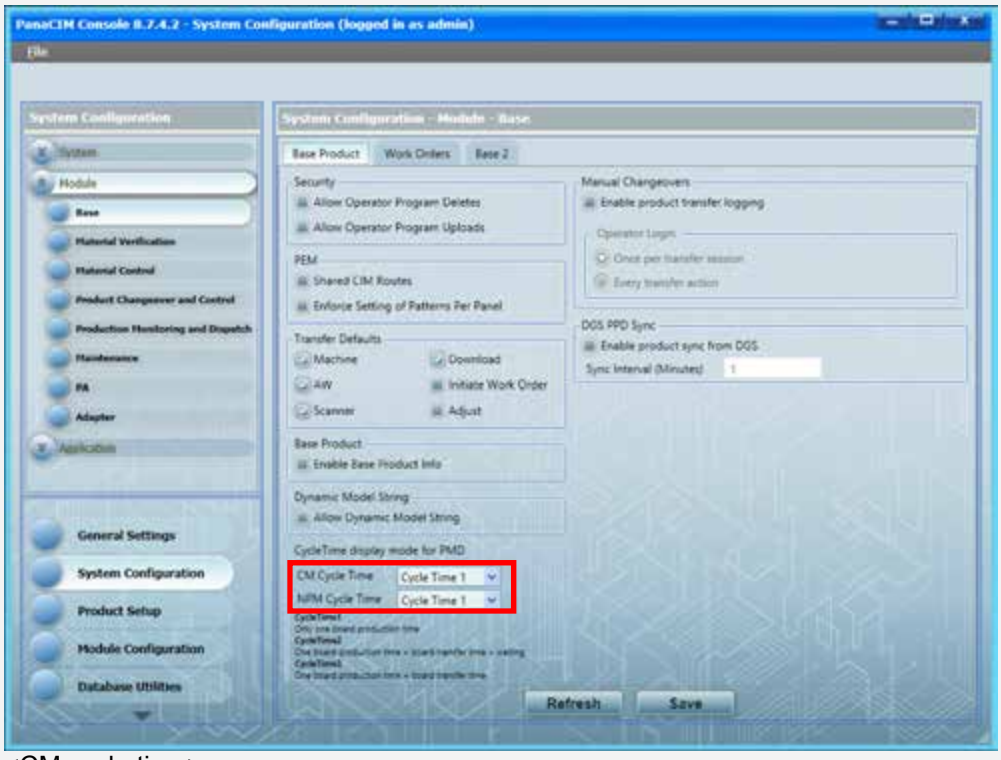
Also when the PMD function is enabled, the PMD icon appears on the Production menu.



2.3.2 Setting of Cycle Time Display

The cycle time displayed on the cycle controller and machine cycle controller of PMD can be selected from among “Only one board production time”, “One board production time + board transfer time + waiting” and “Once board poduction time + board trabsfer time”.

Note) This setting exists in PanaCIM console only. It does not exist on the CIMC controller side.

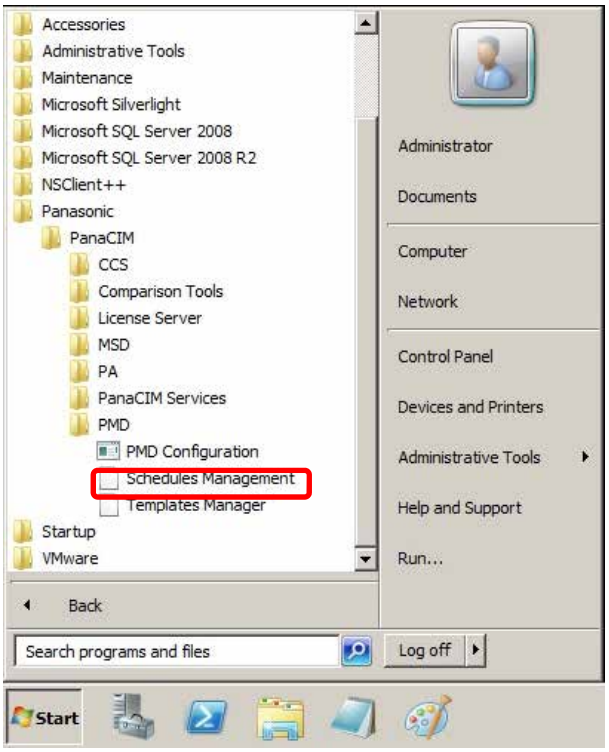
Step	Operation
1	Select **PanaCIM Console->System Configuration->Module->Base->Base Product**.
2	Select the values of Cycle Tme display mode for PMD from the values below.  <CM cycle time> Cycle time 1: Displays one board production time only Cycle time 2: Displays one obard production time + board transfer time + waiting <NPM cycle time > Cycle time 1: Displays one board production time only Cycle time 2: Displays one board production time + board transfer time + waiting Cycle time 3: Displays one board production time + board transfer time Note) CM and NPM are both set to **Cycle Time 1** as default.
3	Click **Save** to save changes.

2.4 Settings – Application Server (PMD Configuration)

To start the production monitoring, application server configuration and client PC (only when client PC is used) configuration are required.
Both configuration procedures are almost same. The client system setting window has fewer tabs than the server setting window.

Note: The following sections explain the settings to configure the application server. Settings for the client system are the same except that the number of tabs is less than that for the application server.

Select Start→Programs→Panasonic→PMD→PMD Configuration in order to launch the server and the client configuration tool.



2.4.1 Configuration Changeover Notification

The changeover notification signals the approaching end of the current production. Notifications provide your changeover team adequate buffer time to prepare for the upcoming production.

Typical changeover activities include:
- Loading the feeders with correct parts
- Downloading the program to the machines
- Scanning parts with, and verifying scanners
- Verifying and checking the Line Manager Node Controller (CM/DT and SP)
- Trial runs and check the machines.

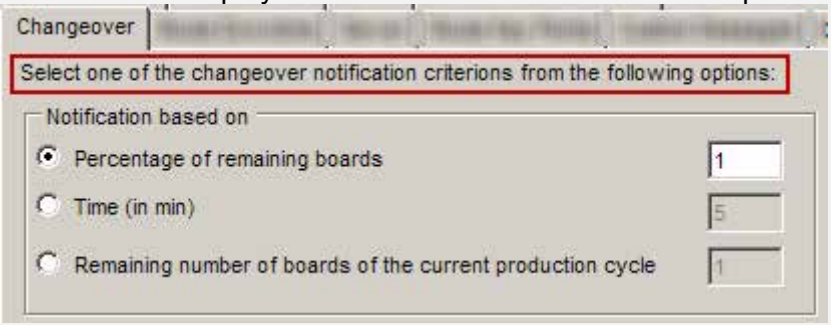
The condition for changeover notification can be configured based on:
- Percentage of remaining boards
- Time (in min)
- Remaining number of boards of the current production cycle

Notifications will display in the Visual Control Panel (VCP) when any one of the selected conditions (with the configured value) is fulfilled.

When you specify Changeover Notification criteria on the server machine, it is applicable for all Routes and will be overridden when configured for the Client.

Access the Changeover tab to:
- Configure the Changeover Notification Criteria.
- Configure the Changeover End Points. (*This function is not eligible for product support.)
- Delete the Changeover End Points. Changeover endpoint is the PC that is configured to view changeover information. (*This function is not eligible for product support.)

Refer to the action table below to configure Changeover Notifications.

Step	Procedure
1	Click the **Changeover** tab.
2	Select a **Notification** option (you may only select one). The changeover notification will display in the VCP based on the selected option.  • **Percentage of remaining boards**. Changeover notification for the upcoming product will display when the percentage of remaining boards reaches the level entered for this option. The minimum possible value is 1 and the maximum value is 99. A default value of 1 is displayed for this option. • **Time (in min)**. Changeover notification will display in minutes prior to the actual changeover where *n* is the time entered for this option. The minimum value is 5 minutes and the maximum value is 120 minutes. A default value of 5 minutes is displayed for this option. • **Remaining number of boards of the current production cycle**. Changeover notification is displayed when the remaining number of boards to be produced for the production cycle reaches the value entered

Step	Procedure
	for this option. The minimum value is 1 and the maximum value is 99. A default value of 1 is displayed for this option.
3	Click **OK** to save changes.

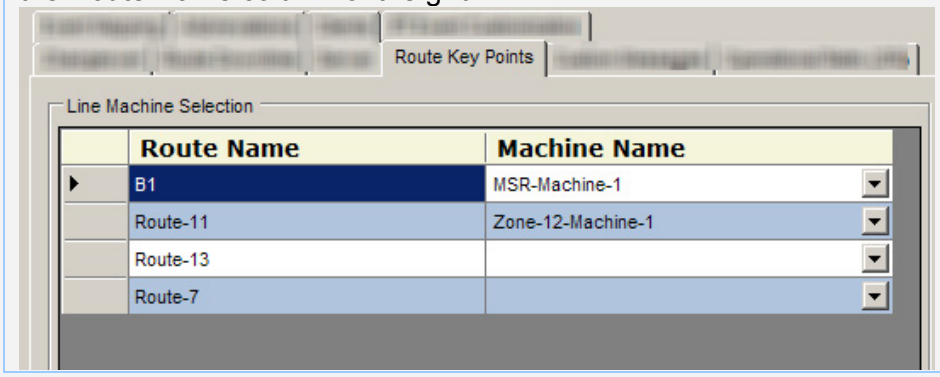
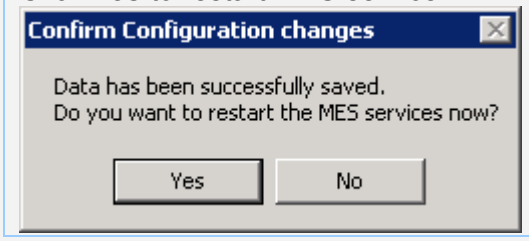
2.4.2 Configuration Route Key Points

Defining a Route Key Point is required as it is the point from which all Route based data is calculated. This includes Operation Ratio, Production Completed/Started events, Completed Panel Counts, etc.

*For iLNB, you can also configure the settings of the third party machines (SPI/AOI/Reflow furnace).
Configuring these settings allows you to show the operational ratio, cycle time, planned number of boards and board quality ratio of the third party machine as line data.

Note: If a key machine is not defined, PMD system picks up the last placement machine as the key machine.

Refer to the action table below to configure Route Key Points.

Step	Procedure
1	Click the **Route Key Points** tab and select a **Route**. **Note:** Route Names cannot be entered, modified or deleted if visible below the Route Name column of the grid. 
2	In the **Machine Name** column, click the drop-down to select a machine name. This machine will be the key machine for the selected route.
3	Click **OK**.
4	Click **Yes** to restart MES service. 
5	Click **OK** to save.

2.4.3 Define the Operational Ratio

Select the Operational Ratio (OR) tab define the range for the target and minimum expected Operational Ratios (ORs) for a Route.

Note: The range above the configured Target OR is displayed in green. The range between the configured Minimum Expected OR and the Target OR is displayed in yellow. Anything below the configured Minimum Expected OR is displayed in red.



Post production time will be recorded when the Product changes. If the same product is run under different work orders, Post Production time will *not* be recorded.

Changeover time will be captured automatically; this is time that occurs between two consecutive production runs.

The Changeover process can be described as:

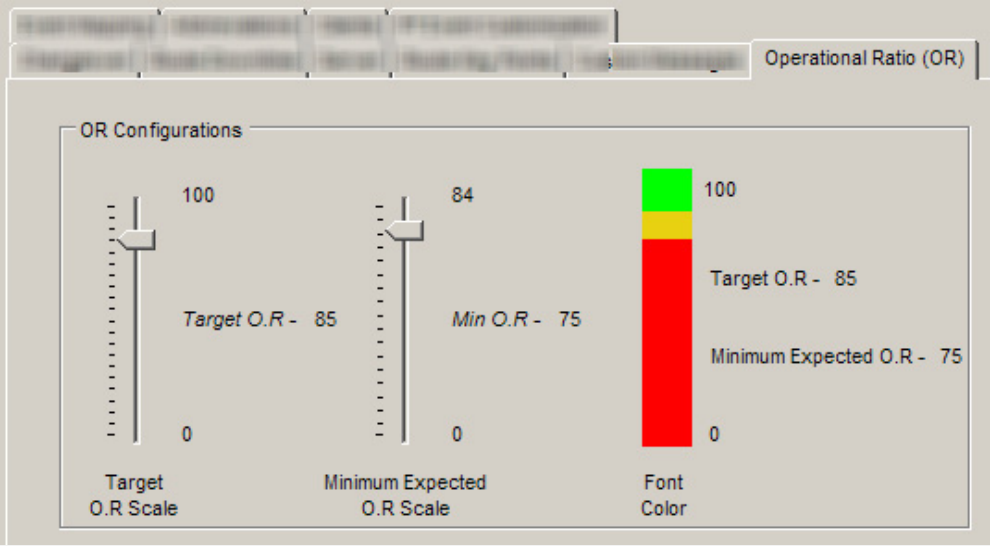
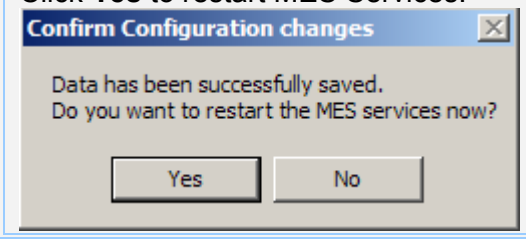
1. Product X production completes and an end is issued on the current production run. This point in the process is considered as the beginning of post-production time and includes various activities such as cart preparation, material procurement, cleaning feeders, maintenance, etc.
2. A new program is downloaded to the Machine from the PT200/DGS.
3. This point is considered as the end of post production time.
4. The width is adjusted.
5. A Changeover is issued to the next Machine in the Module.
6. Production begins for Product Y.

Reference the table below for common Operational Ratio calculations. Formulas used are based on the available data elements.

1	**Hourly OR by Line for n Products** Σ (Panels Produced for product n) within an hour / Σ (Load Time for the product n per hr/Default cycle time of product n where n = 1 to n products)])
2	**Cumulative OR by Line for n Products** This calculation includes the OR over each hour for all products during the specified period.
3	**Hourly OR by Line for Current Product** (Actual Panels Produced for the given product)/[(Load Time for the current product/Default cycle time)]
4	**Cumulative OR by Line for Current Product** Σ (Actual Panels Produced for the given product) / Σ (Load Time for the current product/Default cycle time)])
5	**Hourly OR by Machine for n Products** (Real Run Time / Total Run Time) for all Products during the hour Where Total Run Time = (Real Run Time (Productive Time) + Downtime + Standby Time) from the State Model

6	**Cumulative OR by Machine for n Products** (Real Run Time / Total Run Time) for all products during all hours specified Where Total Run Time = (Real Run Time (Productive Time) + Downtime + Standby Time) from the State Model
7	**Hourly OR by Machine for Current Product** (Real Run Time / Total Run Time) for only current product during the hour Where Total Run Time = (Real Run Time (Productive Time) + Downtime + Standby Time) from the State Model
8	**Cumulative OR by Machine for Current Product** (Real Run Time / Total Run Time) for only current product during all hours the product has been running Where Total Run Time = (Real Run Time (Productive Time) + Downtime + Standby Time) from the State Model

Refer to the action table below to define the Operational Ratio range.

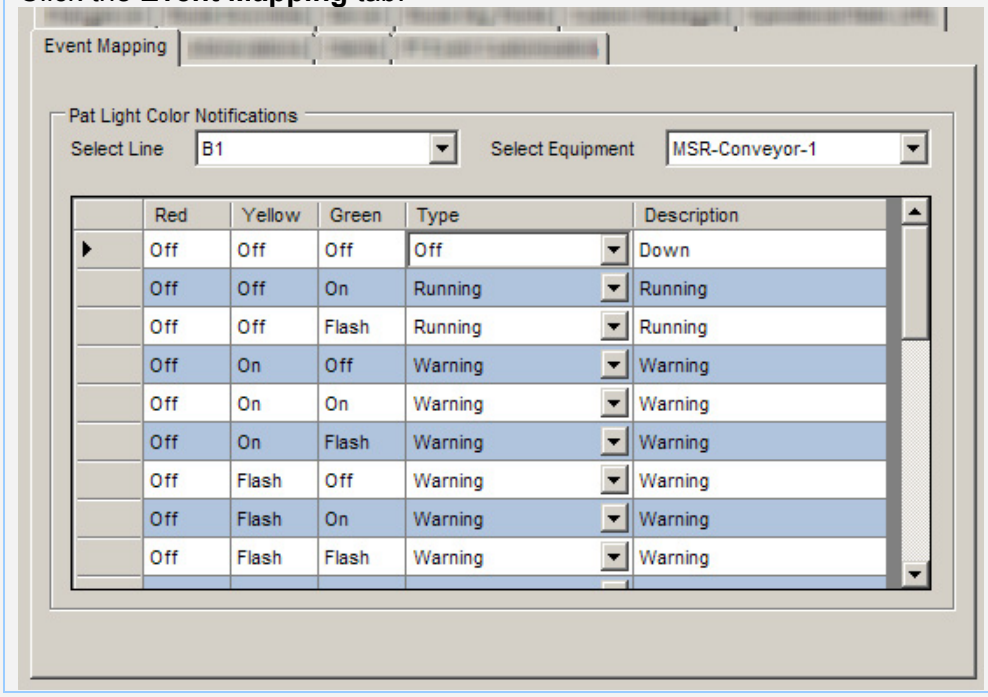
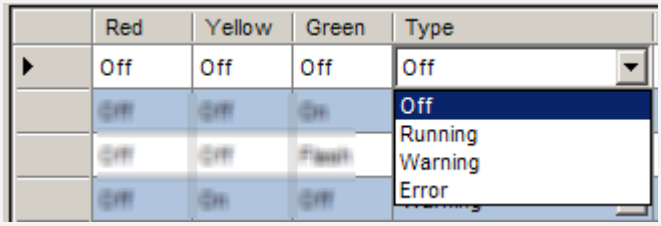
Step	Procedure
1	Click the **Operational Ratio** tab.  Specify the font color to indicate the operation ratio.
2	The **Target Operational Ratio** (T.O.R) is the ratio that you would like to run the production with. The configurable range is from 0 to 100. The default value for target OR is 85. Drag the slider on the scale to set the T.O.R. As you drag the slider, T.O.R value changes.
3	The **Minimum Operational Ratio** (M.O.R) is the ratio that you would at least like to run the production. Minimum expected control becomes active only after the planned target is set. The default value for minimum expected OR is 75. Drag the slider on the scale to set the M.O.R. As you drag the slider, M.O.R value changes. *M.O.R. should be less than T.O.R.*
4	Click **OK**.
5	Click **Yes** to restart MES Services. 

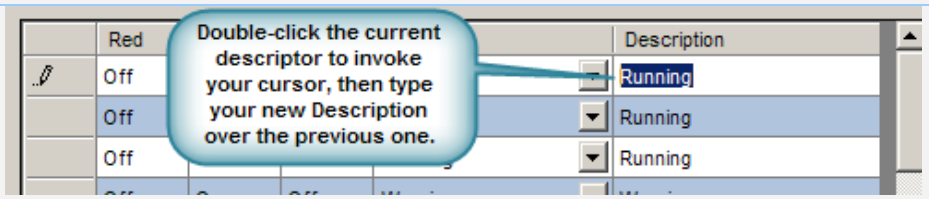
6	Click **OK** to save.

2.4.4 Configuration Event Mapping Options

Refer to the action table below to configure event mapping options. Select the Event Mapping tab to:

- Allow the PMD system to interpret and display real time machine events based on the Pat light color signals from third party (non-Panasonic) machines.
- Define up to 27 unique third-party machine events for interpretation and display in the VCP.

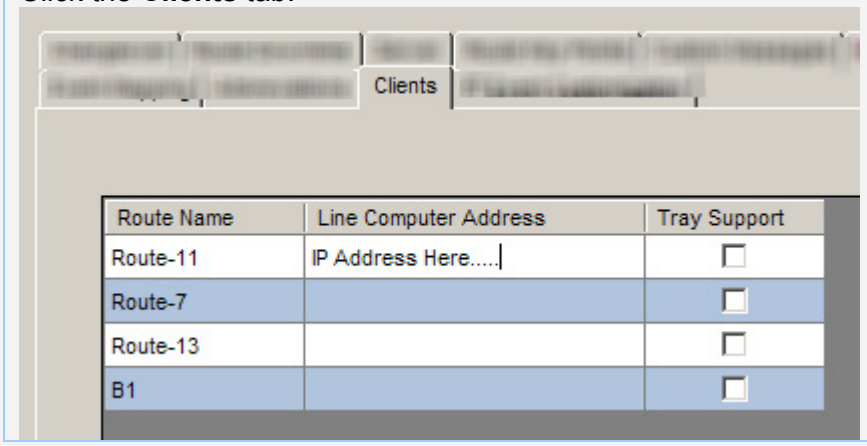
Step	Procedure
1	Click the **Event Mapping** tab. 
2	In the **Select Line** field click the drop-down to select a Route.
3	In the **Select Equipment** field click the drop-down to select the Equipment corresponding to the Route selected.
4	A grid below the **Select Line** field and Select Equipment field displays events and their descriptions corresponding to various Pat light color combinations.
5	To modify the event type for a combination of red, yellow and green Pat lights: a. Select a **Type** from the drop-down and modify the type of event signaled by the Pat lights as desired (Off, Running, Warning, and Error).  b. In the **Description** column, modify the message corresponding to the combination of Red, Yellow and Green Pat lights as desired. During production, whenever the Pat light color combination from the third party machine matches with the above-configured condition, the corresponding user defined message is displayed on the VCP. Each third party machine employs a three-light signal tower to notify the PMD system.

Step	Procedure
	 c. Click **OK**.
6	Click **Yes** to restart MES Services. 
7	Click **OK** to save.

2.4.5 Configuration Clients

Use the Clients tab to set the IP addresses of clients.

Refer to the action table below to add the Client IP Address.

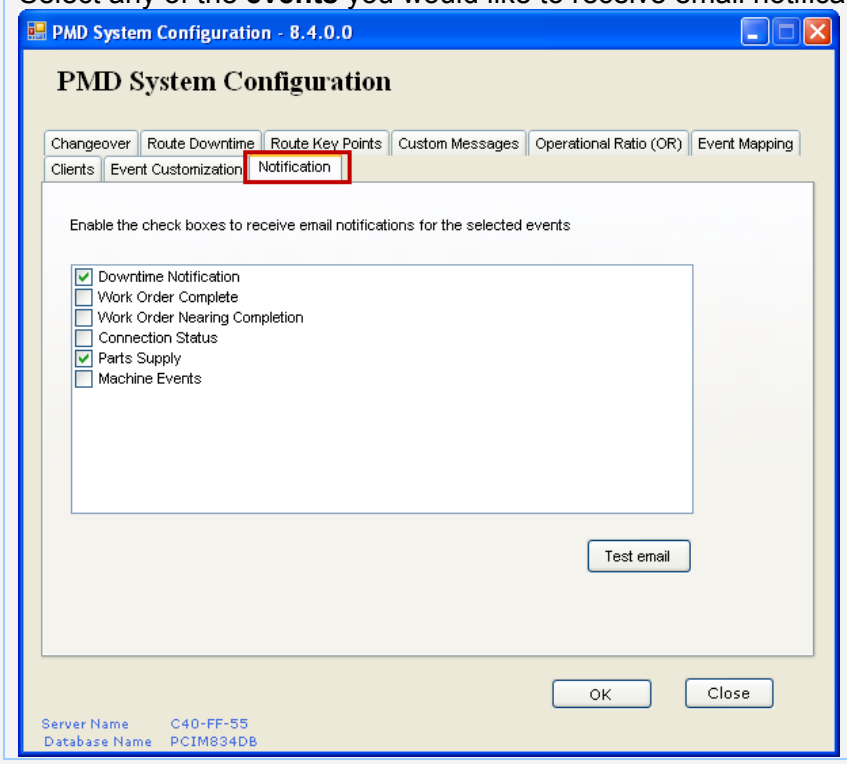
Step	Procedure
1	Click the **Clients** tab. 
2	Enter the **IP Address** in the Line Computer Address field of the Client that will do the processing for the Route (Route Name column). Select the Tray Support checkbox if you will be scanning trays for the route.
3	Click **OK**.

2.4.6 Configure Notification

Use the Configure Notifications tab to configure any PMD related notifications. PMD will trigger an email based on the following events, which can be enabled or disabled.

- Downtime Notification.
- Work Order Complete
- Work Order Nearing Completion
- Connection Status
- Parts Supply
- Short Stop

Note: Work Order Nearing Completion checkbox is based on the settings you configured in the Changeover Notifications tab (Notifications Based On section).

Step	Procedure
1	Click the **Notification** tab.
2	Select any of the **events** you would like to receive email notifications for. 
3	Click the **Test email** button to ensure you will receive the notification(s).
4	Click **OK** to save changes.

3. FUNCTION – Mobile Scanner

This chapter explains the operation on mobile scanner.

Use the following PanaCIM application to configure the mobile scanner of the production monitor.

- Mobile scanner

3.1 Overview

When a machine encounters production exceptions or events they are dispatched to the scanner for you view and acknowledge them.

Refer to the table below for a PMD Events scanner screen settings and descriptions.

<Example of Machine Event>



<Example of Part Supply event>



3.2 Settings

Types of events (machine status other than production or part supply event) and a route for which events are displayed on the Mobile Scanner screen can be set.

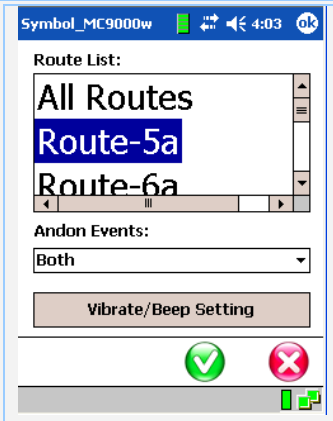
If a Route/Event type has not been configured, the scanner will not display PMD events in the Events window. Access the Mobile Scanner PMD Config screen to configure/ select the Route and Event type(s) operators should receive.

The following Event Types can be selected:

- Part Supply only: Only Part Supply event will be displayed.
- Short Stop only: Only Short Stop event will be displayed.
- Both: Both Part Supply and Short Stop events will be displayed.

Refer to the action table below to configure Routes for event types.

Step	Procedure
1	Launch **Production** Main Menu screen. Click the Settings icon. 
2	Following screen appears. Click the PMD Config icon. 
3	Following screen appears. Select a Route from the **Route List** for which Andon Events need to be configured.



Andon events can be configured for all the routes by selecting **All Routes** from the **Route List**.

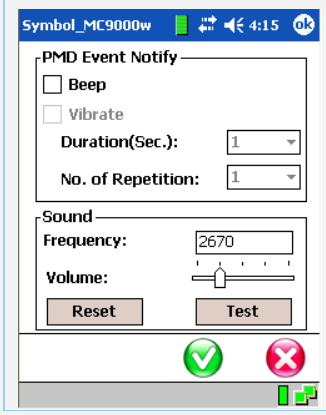
4 Select the desired event type from the Andon Events drop-down list.



The following Event Types can be selected:

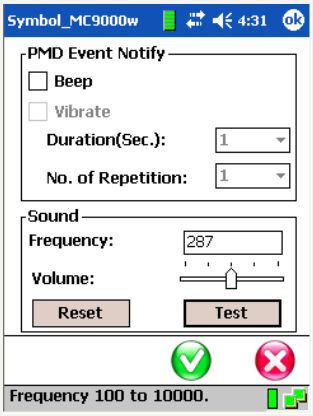
- Part Supply only: Part Supply event
- Short Stop only: Short Stop event
- Both: Both Part Supply and Short Stop events

5 Click the **Vibrate/Beep Setting** button to notify operator of PMD event by generating the beep sound on the scanner or via vibration of a scanner.



6 When the **Beep** checkbox is checked, you can set the frequency of the beep sound by entering a value in the Frequency textbox. You can adjust the volume of the beep by moving the slider.

Click the **Test** button to check that beep sound is produced. It is to be noted that if sound frequency value entered is outside the range of frequencies supported by the scanner, a message **"Frequency <lower limit> to <upper**

	limit>" is displayed as shown below:  If the scanner supports vibration, you can check the **vibrate** checkbox and set the duration in seconds for which scanner vibrates and number of times vibrations are repeated. In the screen shown in step 5, **vibrate** checkbox is disabled because this particular scanner does not support vibration. **Note:** Both **Vibrate** and **Beep** modes can be enabled.
7	Click  to save the settings.

3.3 Operation

Notify operators and the PMD system of your involvement in a particular event. Scanners configured to receive events for this Route will update with the new event status.

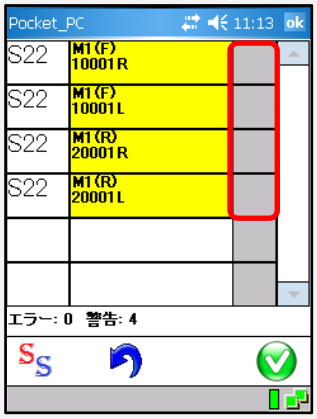
If you choose to acknowledge an event, the Visual Control Panel will display your operator image. If you unacknowledged an event, the VCP will remove your image from the display.

After a machine event has been resolved (i.e. the machine or route is running) it is automatically cleared from both the Scanner and Visual Control Panel. Sometimes an event will not clear automatically from the Scanner or the VCP after resolution, so the scanner can be used to manually remove it from the event screen.

The following tables show the procedures for acknowledging, un-acknowledging, and closing events.

3.3.1 Acknowledge events

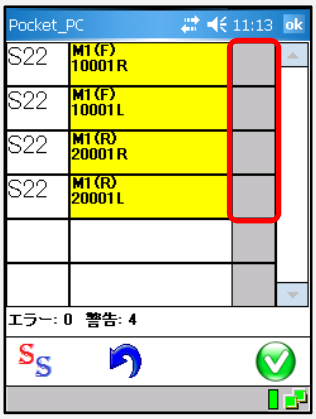
Refer to the table below to acknowledge events.

Step	Procedure
1	Click the **toggle** button next to the event. 
2	When  is displayed, click  to acknowledge the event that you selected as an operator who is in charge of the event. 

3	The route number for the acknowledged event turns **GREEN** indicating that the event has been acknowledged by an operator. 

3.3.2 Un-Acknowledge Events

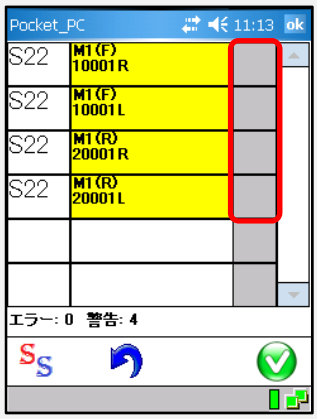
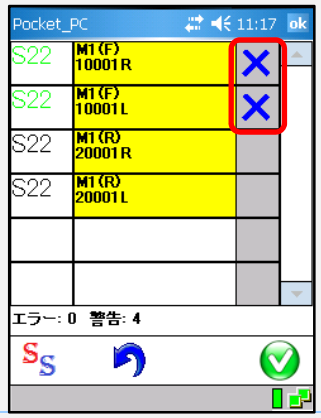
Refer to the table below to un-acknowledge events that are displayed on the mobile scanner.

Step	Procedure
1	Click the **toggle** button next to the event. 
2	Click the **toggle** button to  and then click  . Then, the operator cannot acknowledge the selected event any more.  When an event cannot be acknowledged by an operator (i.e. emergency, switching tasks), other operators can be registered to acknowledge it.

3	When an event has been unacknowledged, the Route number will turn to **BLACK** from **GREEN**. 

3.3.3 Close PMD Events

Refer to the table below to close PMD events using the Mobile Scanner

Step	Procedure
1	Click the **toggle** button next to the event. 
2	To manually close a PMD event, select the event to close and change the icon next to the event to  and click  . 
3	The event is removed from the Display and Acknowledgement subsystems.

4. FUNCTION – Visual Control Panel

This chapter explains each operation of VCP related to the PMD.

This chapter explains following PanaCIM application.

- Visual Control Panel

4.1 Overview

The PMD Template displays real time information of production events to an operator visually using VCP: <Information to be notified>

- Displays the current running program.
- Displays the System time.
- Indicates that the VCP is connected with the PMD server.
- Indicates machines and equipment are responding to the PMD events.
- Displays Machine Locks and reason for the lock
- Displays the Cycle Time for each machine and equipment. The CycleTime (sec) area toggles between machine cycle times and production operational ratios. It indicates 3 types of operational ratios in percentages:
  - Total operational ratio for the day starting from midnight until the present moment.
  - Hourly operational ratio for the current hour.
  - Last hour operation ratio.
- Displays the equipment abbreviation in the MC column (i.e. UL for Un-Loader and CV for Conveyor)
- Displays parts supply events such as Low Components, Alarms, and Warnings. A Parts supply event refers to the condition when parts are either exhausted or close exhaust on the table/tray corresponding to a machine. Part exhaust can lead to stoppage of production.

  - Part supply/stop events follow the following color codes:

    Yellow indicates a warning message for the machine/table.

    
    Red indicates an error message for the machine/table.

    
    Smiley (or operator image) indicates an acknowledged event
- Indicates the time elapsed since the parts supply event occurred.
- Indicates the time elapsed since the short stop/quality check event occurred. The Short supply event refers to the condition when the machine stops working for a small interval. Any interval above this will be considered as downtime. This interval is configurable. Typically, short stop events are machine based and not based on parts supply.
- Indicates the next product.
- Indicates the quantity of the next product.
- Indicates the estimated quantity of the next product.
- Indicates the ratio of current board count to the scheduled board count.
- The PMD window displays production events in the following method:

  - High priority items appear at the top of the list.
  - An operator image appears upon acknowledgement of a task. If the operator unacknowledged a task, the image is removed from the display monitor, but the timer continues. **Note:** The operator has to acknowledge an event using the scanner, and fix it to clear the event from the queue in the VCP.
  - Task timer starts immediately upon receipt of task.
  - System stops the timer and removes the task upon completion from the task list.
  - The machines change color based on the state (red/yellow).
  - The system reorders the event list each time a task is received or completed.

4.2 Settings

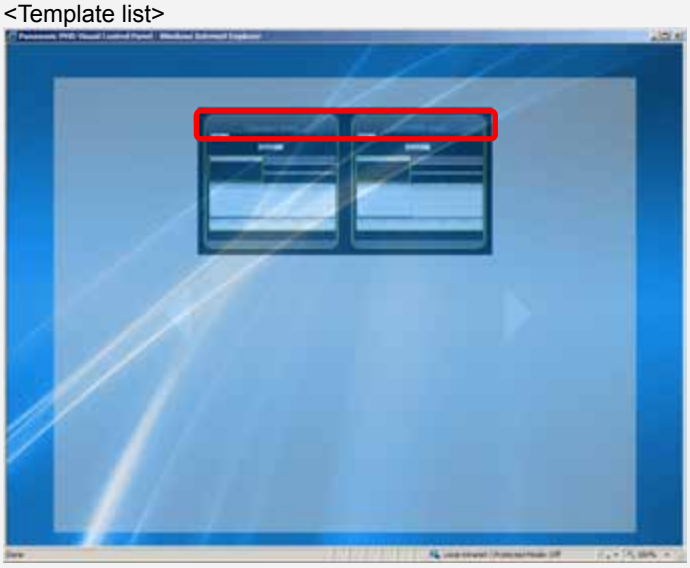
Use template designer to set controls (information) to be displayed on Visual Control Panel (VCP).

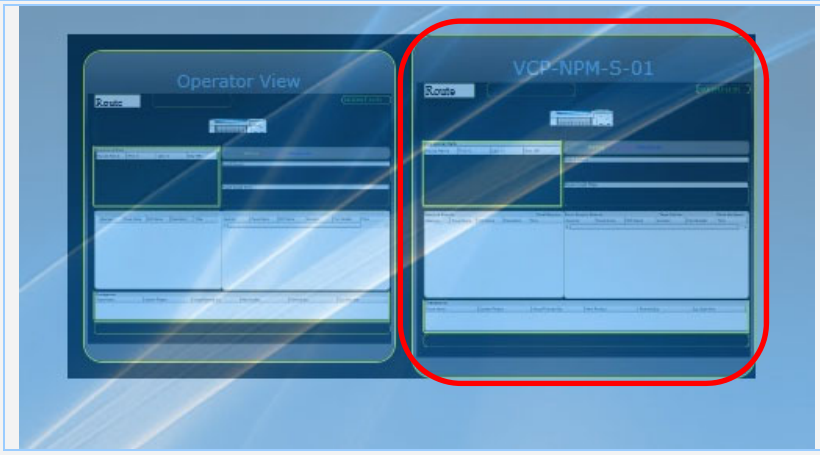
* See [5. Function-Template Designer] for more details.

4.3 Operation

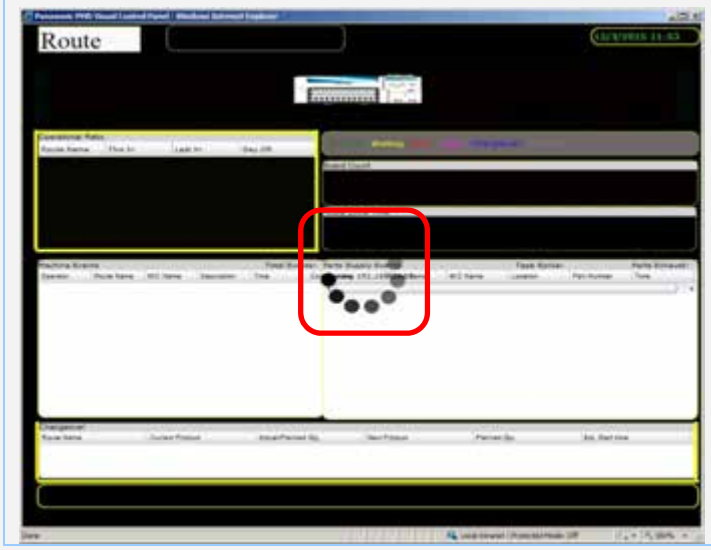
4.3.1 Start

The following table shows how to start Visual Control Panel (VCP) with a set template.

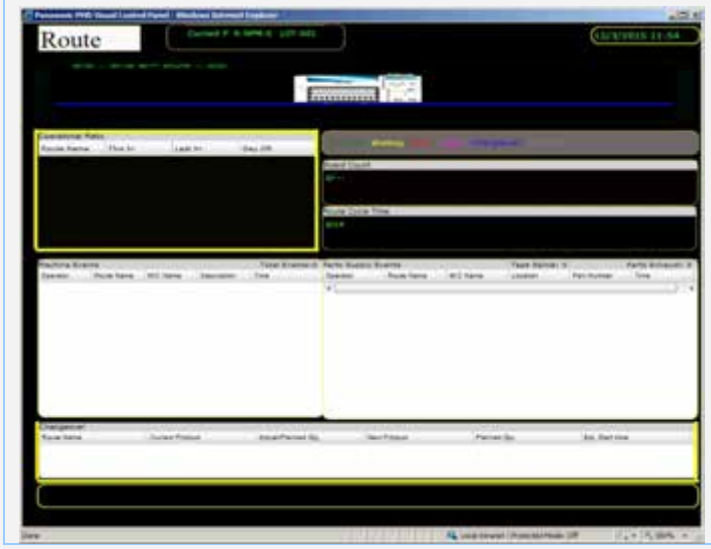
Step	Operation
1	Start WebApplication and select <Visual control panel> from <PMD>. 
2	Check the template name at PMD template list, and click the template you want to see. <Template list>  <Select template>



3 Visual Control Panel starts with the selected template.



4 The information appears with the selected template.



4.3.2 Display

PMD operation runs using various applications and utilities.

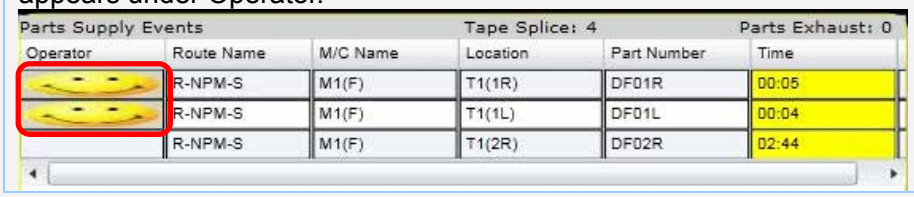
Set events appears on Visual Control Panel (CVP).
The latest event information appears for events of the selected route.

After the event is acknowledged on a mobile scanner, an icon for the operator appears on Visual Control Panel (CVP). If acknowledge is canceled from the mobile scanner, the icon disappears.

After machine events are solved (when machine or route is running), it disappears automatically from Visual Control Panel (CVP). However you can also manually clear (end) from mobile scanner when it is not cleared automatically under some circumstance.

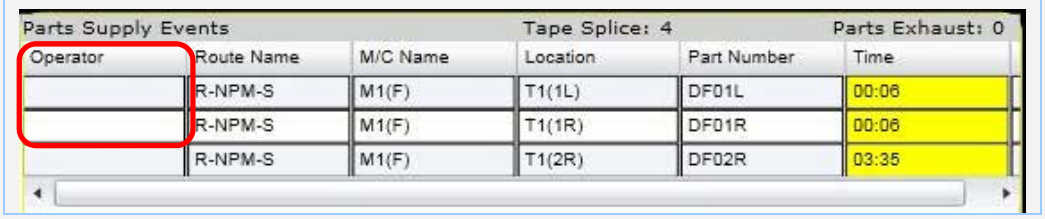
■ Register Event Acknowledgement

The following table shows how to acknowledge an event displayed on a mobile scanner.

Step	Operation
1	After [Warning Parts Empty] detected on a machine, an event message appears at [Chip Supply] control area. 
2	After acknowledged it on PMD screen of a mobile scanner, the operator icon appears under Operator. 

■ Cancel (Clear) Event Acknowledgement

The table below shows how to cancel/clear event acknowledgement.

Step	Operation
1	Acknowledge [Warning Parts Empty] on a mobile scanner' PMD(PMD) screen, and the operator icon appears. 
2	After you cancel (clear) acknowledgement from the PMD screen, the icon disappears. 

Parts Supply Events — Tape Splice: 4 — Parts Exhaust: 0

Operator	Route Name	M/C Name	Location	Part Number	Time
	R-NPM-S	M1(F)	T1(1L)	DF01L	00:06
	R-NPM-S	M1(F)	T1(1R)	DF01R	00:06
	R-NPM-S	M1(F)	T1(2R)	DF02R	03:35

■ End Event

The following table shows how to end an event from a mobile scanner.

Step	Operation
1	After [Warning Parts Empty] detected on a machine, an event message appears at [Chip Supply] control area. 
2	After you end an event from a mobile scanner's PMD screen, the event will disappear from [Chip Supply Event] control. 

Parts Supply Events — Tape Splice: 2 — Parts Exhaust: 0

Operator	Route Name	M/C Name	Location	Part Number	Time
	R-NPM-S	M1(F)	T1(2L)	DF02L	04:51
	R-NPM-S	M1(F)	T1(2R)	DF02R	04:51

5. FUNCTION – Template Designer

This chapter explains each operation of the template designer related to the PMD.

This chapter explains following PanaCIM application.

- Template Designer

5.1 Overview

PMD template designer lays out controls (information) to be displayed on Visual Control Panel (VCP) and save it as a template.
You can open a saved template and save it with another name.

5.1.1 Each Control

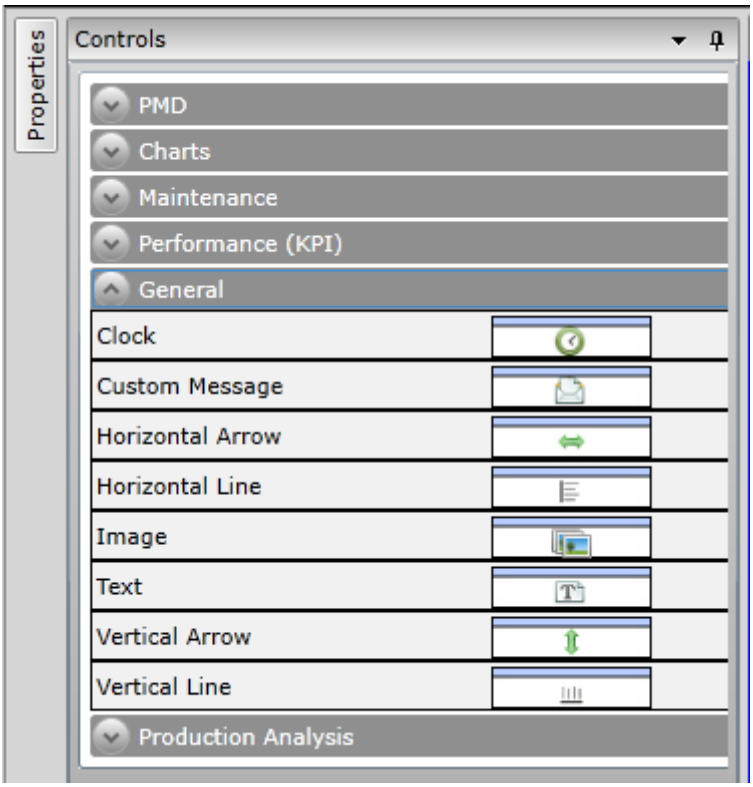
The following six controls are set on Visual Control Panel (CVP) as the operating monitor.
- General control
- Performance(KPI) control
- PMD control
- Maintenance control
- Production Analysis control
- Chart control

The functions of each of these six controls are described below.

■ General Control

The following is a control which enables you to set on Visual Control Panel (VCP) as the operating monitor.

<Visual Control Panel (VCP)>



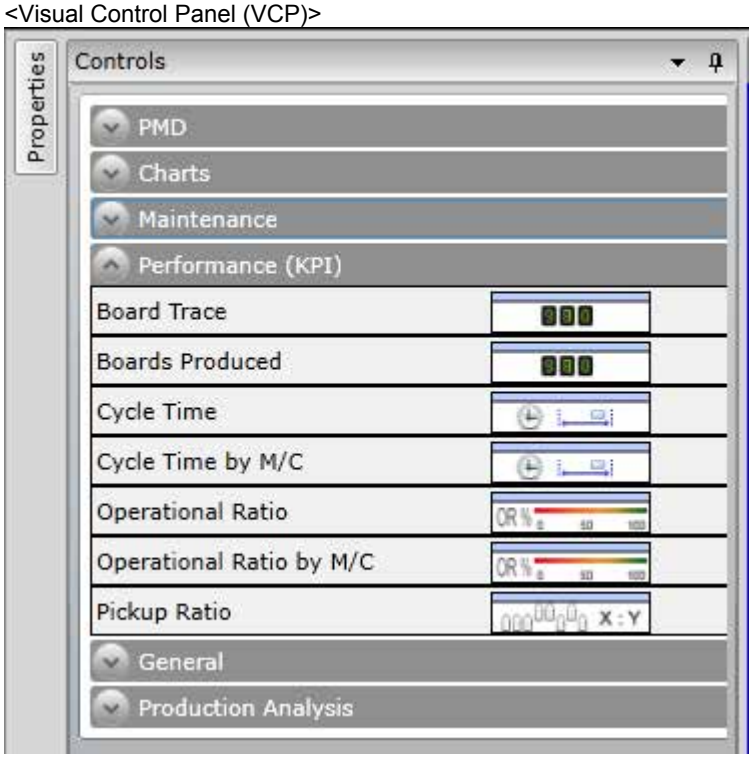
<Description of controls>

No.	Control name	Description
1	Custom Message	Text-base custom message
2	Text	Fixed text.
3	Image	Fixed original image <Image file type> ・bmp ・gif ・jpg ・png
4	Clock	Actual time
5	Vertical Line	Vertical lines
6	Vertical Arrow	Vertical up/down arrows
7	Horizontal Line	Horizontal lines
8	Horizontal Arrow	Horizontal right/left arrows

■ Performance (KPI) Control

The following is a control which enables you to set on Visual Control Panel (VCP) as the operating monitor.

<Visual Control Panel (VCP)>



<Description of controls>

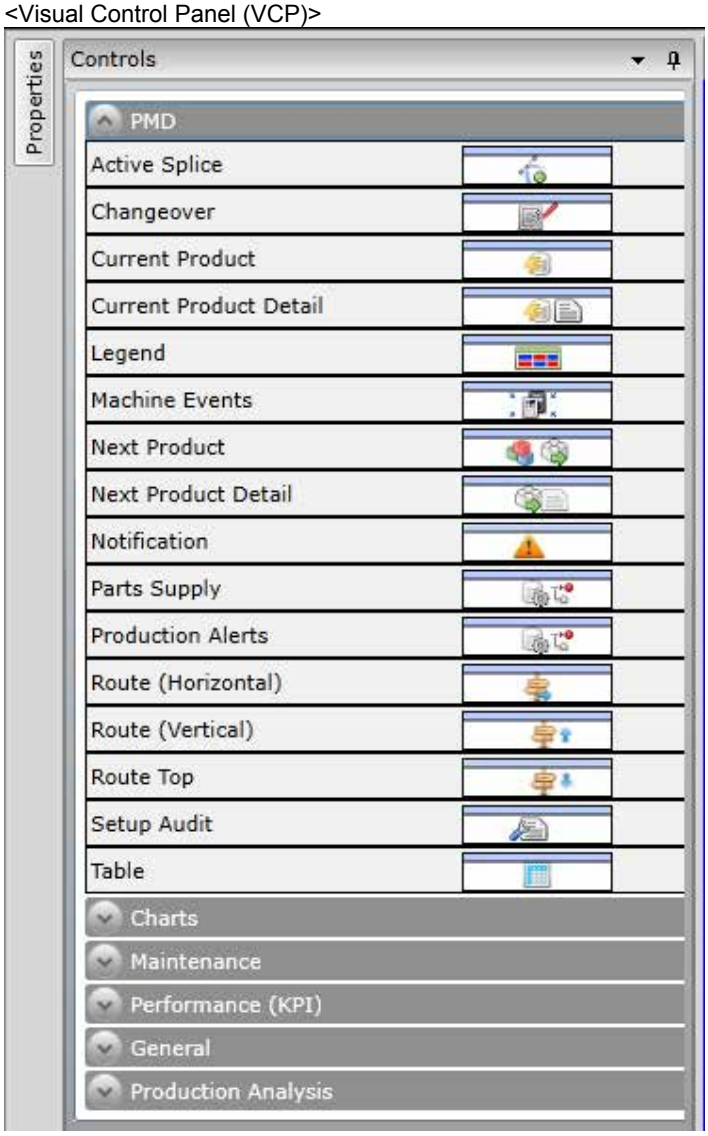
No.	Control name	Description
1	Cycle Time	Production cycle time of the current route. When iLNB is used, the following file is created each time one board is produced. File name: Event_Info.inf Example of description: <Parameter> CycleTime = 18.1 </Parameter> When iLNB is used, it also supports to the third party machines (SPI/AOI/Reflow furnace).
2	Operational Ratio	Shows the following operational ratio of lines. <Actual> Show operational ration from the 0 min of the actual hour (1:00/2:00) to the actual time <Previous Hour> Shows operational ratio of previous one hour Example: When actual time is 17:40, show the operational ratio of 16:00～17:00 <Day> Shows the ratio from (0:00)～actual time. When iLNB is used, the following file is created each time one board is produced. File name: Product_Info.inf Example of description: <EquipmentSpecific> OperationRatio = 88.1 </EquipmentSpecific> When iLNB is used, it also supports to the third party machines (SPI/AOI/Reflow furnace).
3	Panel Trace	Barcode of produced panels
4	Board Quality Ratio	When iLNB is used, it also supports to the third party machines (SPI/AOI).
5	Pickup Ratio	Pickup ratio of feeders as performance data
6	Produced Panel(s)	Shown as the numerator of a fraction with the planned number specified in the work order as denominator. Only hyphen appears when no work order is used.
7	Cycle Time by M/C	Shows the current production cycle time by machine. When iLNB is used, the following file is created each time one board is produced. File name: Event_Info.inf Example of description: <Parameter> CycleTime = 18.1 </Parameter> When iLNB is used, it also supports to the third party machines (SPI/AOI/Reflow furnace).

No.	Control name	Description
8	Operational Ratio by M/C	Shows the current operational ratio by machine. When iLNB is used, the following file is created each time one board is produced. File name: Product_Info.inf Example of description: <EquipmentSpecific> OperationRatio = 88.1 </EquipmentSpecific> When iLNB is used, it also supports to the third party machines (SPI/AOI/Reflow furnace).

■ PMD Control

The following is a control which enables you to set on Visual Control Panel (VCP) as the operating monitor.

<Visual Control Panel (VCP)>



<Description of controls>

No.	Control name	Description
1	Splicing	Splice information if spliced during production
2	Table	Shows the selectable route/machine table boxes. Machine table numbers are displayed in these boxes. When splice warning occurs, it turns yellow, and if

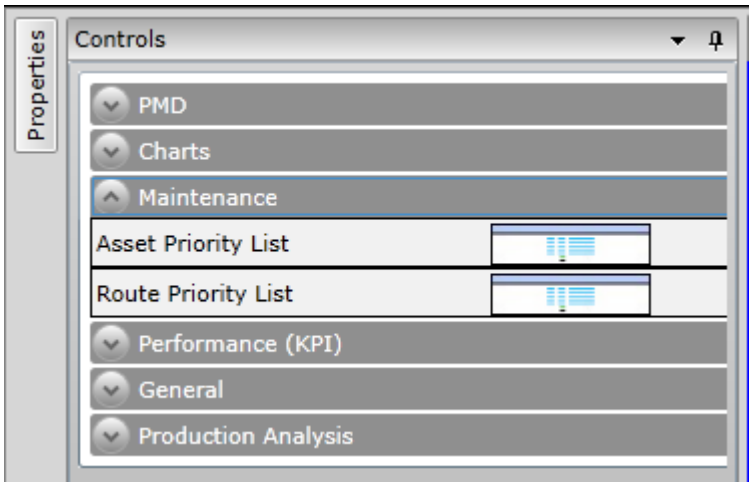
No.	Control name	Description
		parts empty occurs, it turns red.
3	Route(Vertical)	Aligns and shows machines vertically on the route
4	Route(Horizontal)	Aligns and shows machines horizontally on the route. Once an error or warning occurs on the machine, a target machine icon with the designated color is shown. Machine error: Indicate a production failure on the machine side Upon reception of an event (Cover open / PCB transfer Error / Emergency stop, etc.) Machine warning: When a routine check is started from the scanner Part error: When parts are exhausted Part warning: When part exhaustion warning has been issued. When iLNB is used, you can display information including the third party machines (SPI/AOI/Reflow furnace).
5	Route Top	Shows machines on the route in a list. Show tables of the machines. If splice warning detected, display the entire table in yellow. Display the entire table in red in case of parts exhausted. If you use two machines configuring single lane as dual lane, status of each machine can be displayed. When iLNB is used, you can display information including the third party machines (SPI/AOI/Reflow furnace).
6	Current Product	The product being produced. This can be displayed for multiple routes.
7	Current Product Details	Shows route name, product name being produced, planned panel count, planned production end date and time. This can be displayed for multiple routes.
8	Next Product	Next program name to be produced. This can be displayed for multiple routes.
9	Next Product Details	Shows route name, program name to be produced in the route, planed panel count, planned production start date and time. This can be displayed for multiple routes.
10	Production Alert	Shows information in case an event causing PanaCIM system trouble occurs.
11	Machine Event	Shows Splice warning or other events that stop production (parts empty, emergency stop etc.) on real time basis
12	Changeover	Shows produced/planned product count of the current one, as well as the planned count and production start date/time of the next product. This can be set for multiple routes.
13	Notification	Show a "Production Complete" message together with the production data in case the planned product count has been completed.
14	Setup Audit	Shows setup audit information. Only scheduled setup audits are supported. Show the next audit time including machines. Once a machine is scanned, the schedule will be reset.
15	Parts Supply	Shows splicing and parts empty. This can also be displayed for multiple routes.

No.	Control name	Description
16	Legend	Shows machine status in different colors. ・In production (Green) ・Standby (Yellow) ・Error (Red) ・Down (Pink) ・Changeover (Blue) ・Disabled (Gray)

■ Maintenance Control

The following is a control which enables you to set on Visual Control Panel (VCP) as the operating monitor.

＜Visual Control Panel (VCP)＞



＜Description of controls＞

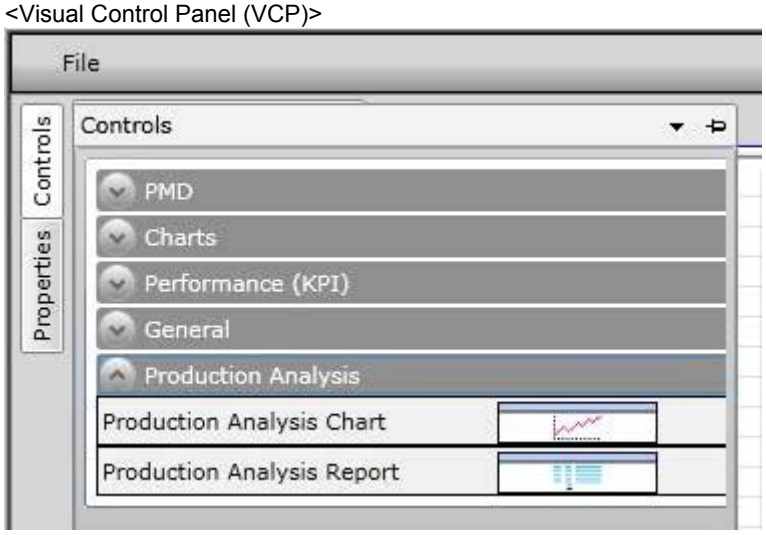
No.	Control name	Description
1	Maintenance warning list per line	Shows the list of maintenance warning per line.
2	Maintenance warning list per asset	Shows a list of maintenance warning per asset.

■ Production Analysis Control

The following is a control which enables you to set on Visual Control Panel (VCP) as the operating monitor.

* Production Analysis control can be registered, but its charts or reports are not shown on Visual Control Panel (VCP).

<Visual Control Panel (VCP)>



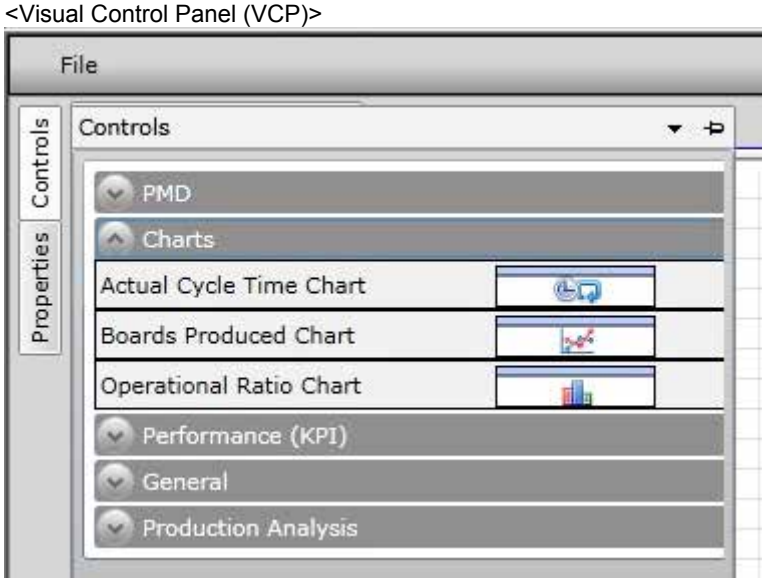
<Description of Controls>

№	Control name	Description
1	Production Analysis Chart	Shows the production analysis chart.
2	Production Analysis Report	Shows the production analysis report.

■ Chart Control

PMD shows controls on Visual Control Panel (VCP) as follows.

<Visual Control Panel (VCP)>

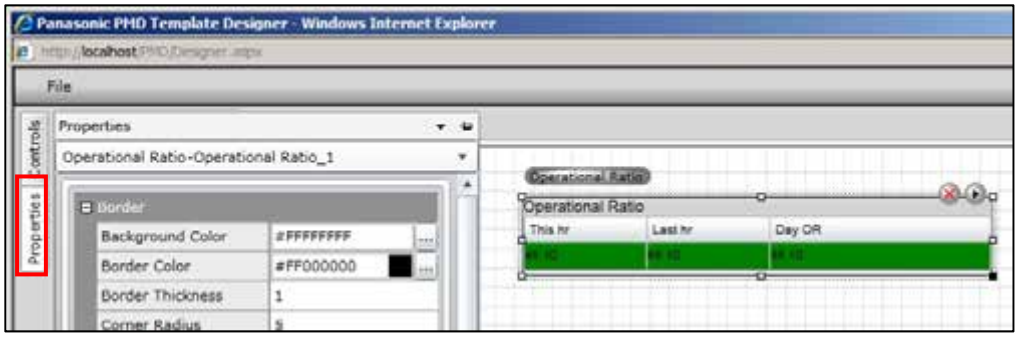


<Description of Controls>

№	Control name	Description
1	Operational Ratio Chart	Shows operational ratio graph per line.
2	Actual Cycle Time Chart	Shows actual cycle time graph per line.
3	Boards Produced Chart	Shows produced board graph per line.

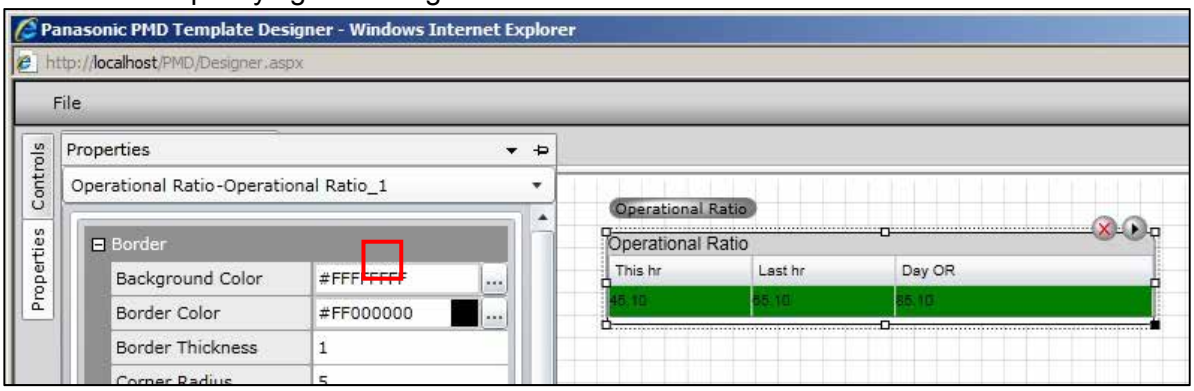
5.2 Settings

The following properties are used in many controls.
Select a desired control (Operational ratio in the below example) and press the Property button to set properties.



5.2.1 Color Selection

Use color property to set the color for each control.
Select the button for specifying the background color on the below screen.

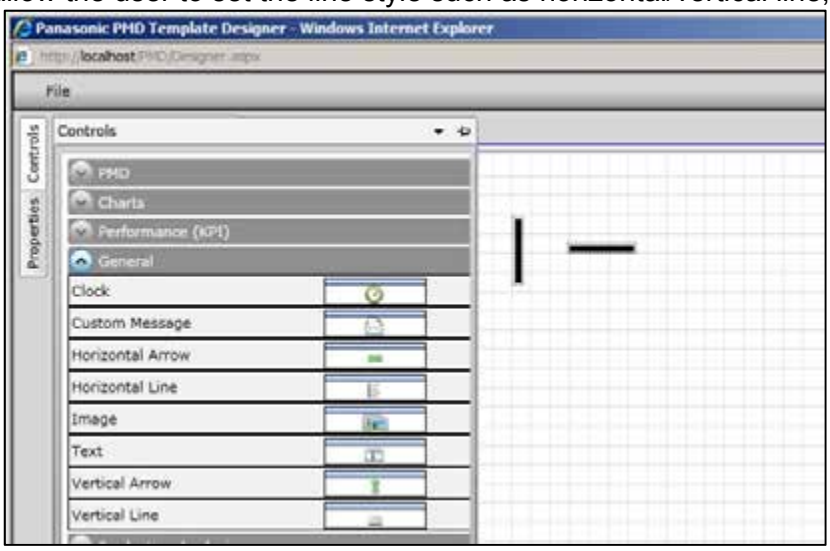


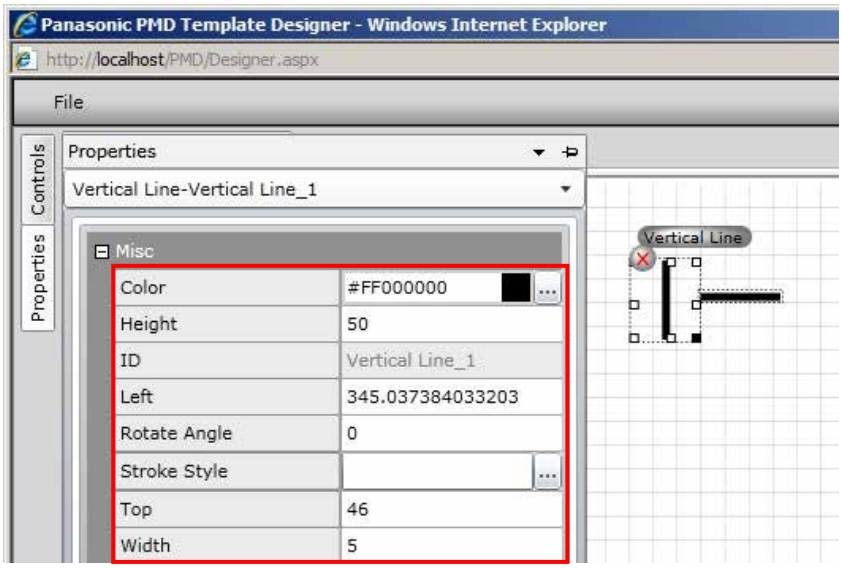
The following dialog appears.



5.2.2 Line Style Selection

The system shall allow the user to set the line style such as horizontal/vertical line, line width, etc.





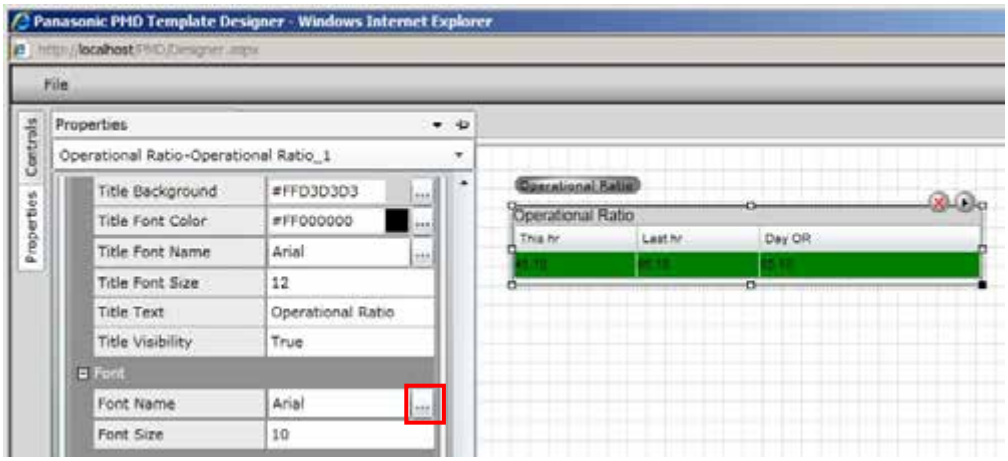
Use the above Stroke style button to set solid or dotted line.

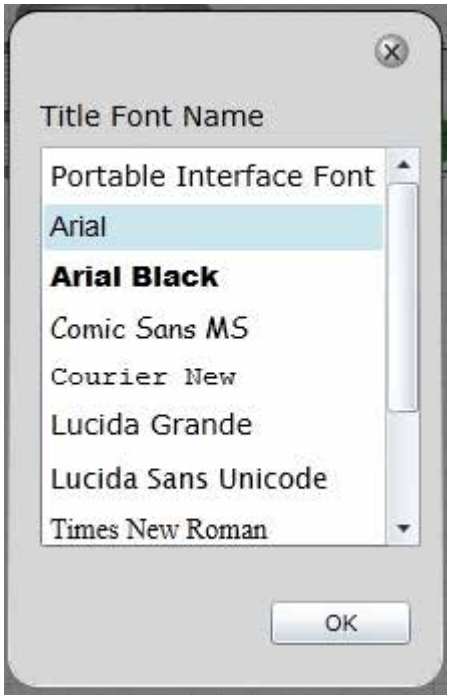


5.2.3 Font Selection

The system shall allow the user to select the font style/color. It shall display the standard font selection dialog.
Pressing the Font button enables the user to set the following:

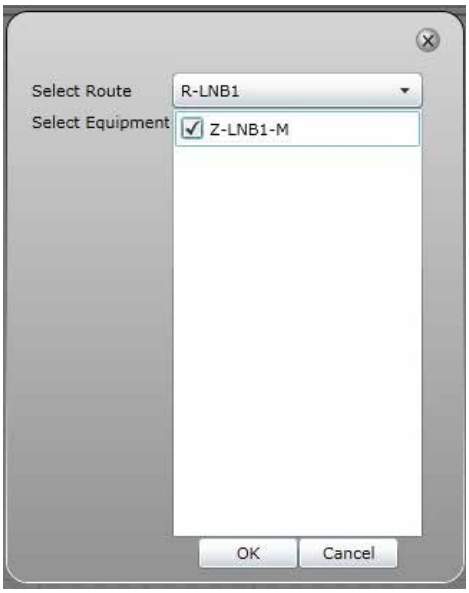
- Font Color
- Font size
- Font type





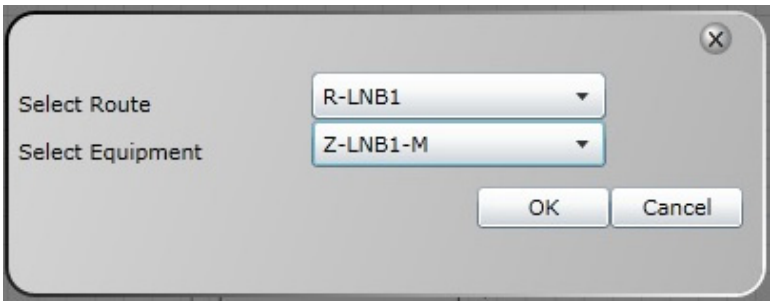
5.2.4 Equipment Selection

The system shall allow the user to select the equipment to be displayed for the given line.



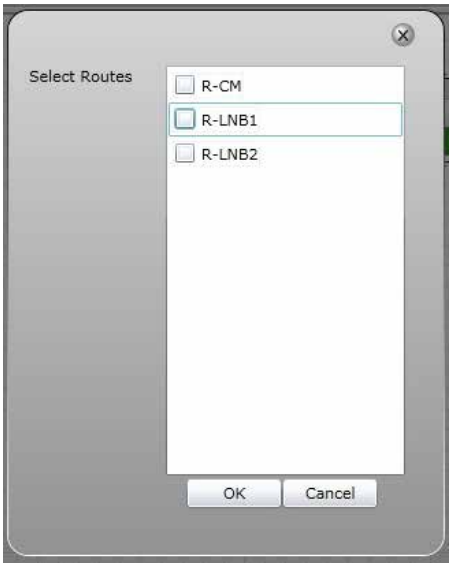
5.2.5 Single - Routes Selection

The system shall allow the user to select a single route.



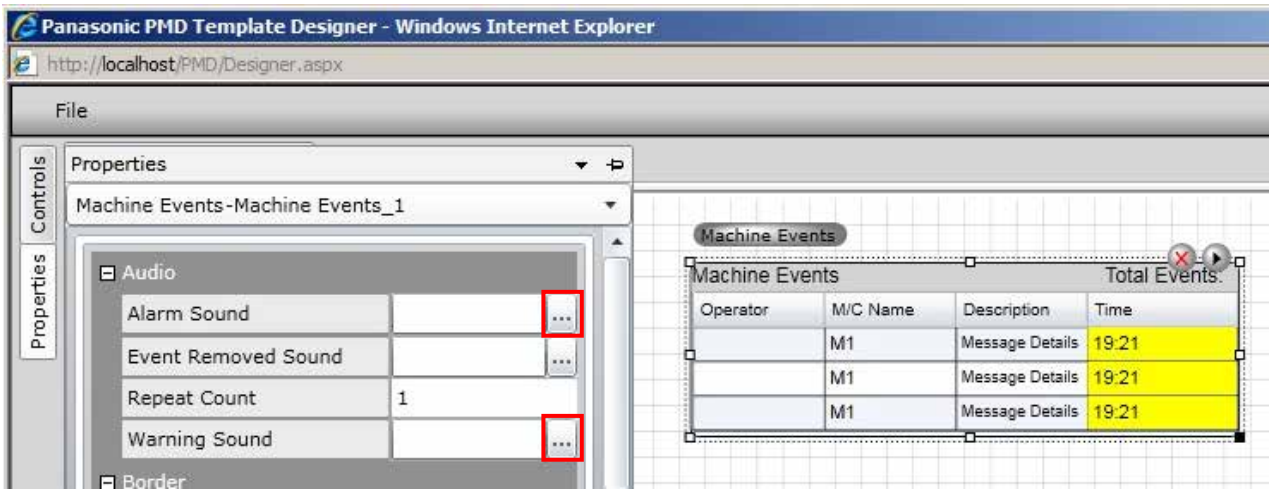
5.2.6 Multi – Routes Selection

The system shall allow the user to select one or more routes.



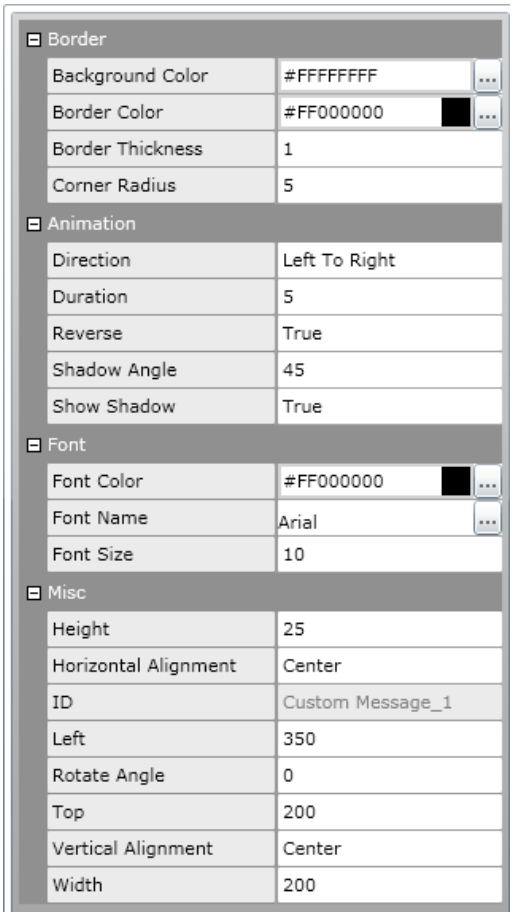
5.2.7 Audio Selection

The System shall allow the user to select an audio file. The file shall be a .wav file. The standard file selection dialog shall be displayed.



5.2.8 Custom Message

The following are the default settings for Custom Message Control.

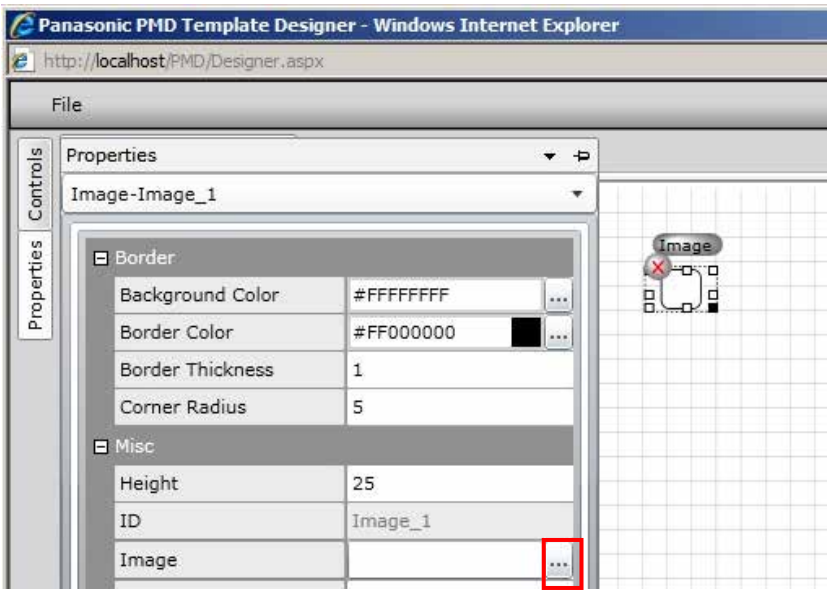


5.2.9 Image Selection

The System shall allow the user to select the image file. The file shall be one of the following formats:

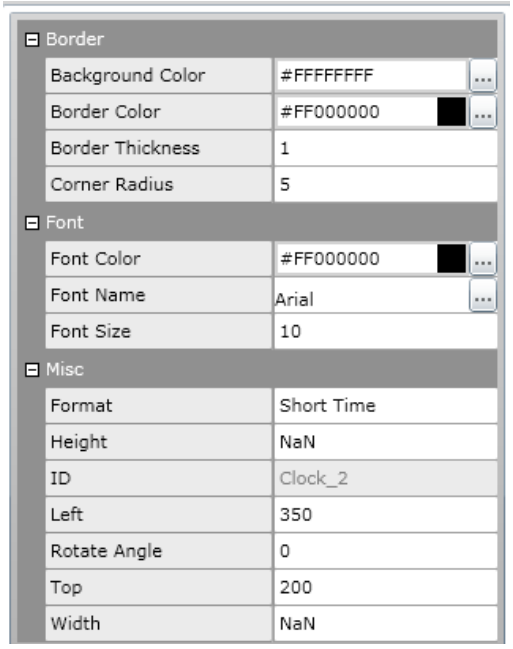
- Bitmaps
- Gifs
- Jpeg
- Png

Pressing the Image Selection button displays the file select screen.



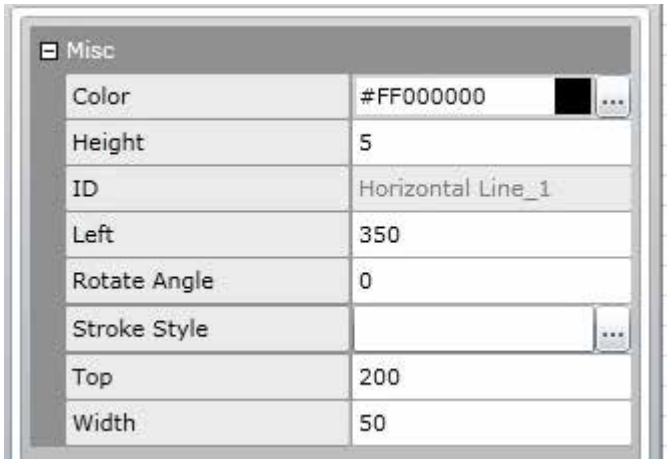
5.2.10 Clock

The following are the default settings for Clock Control



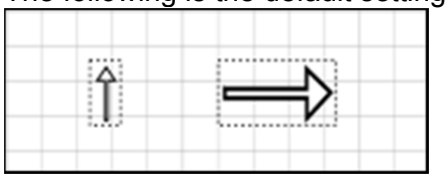
5.2.11 Historical/Vertical Lines

The following is the default settings for Horizontal/Vertical Lines.



5.2.12 Historical/Vertical Arrows

The following is the default settings for Arrows (Vertical and horizontal arrows).





5.2.13 Route Cycle Time

The following is the default settings for the Route Cycle Time.

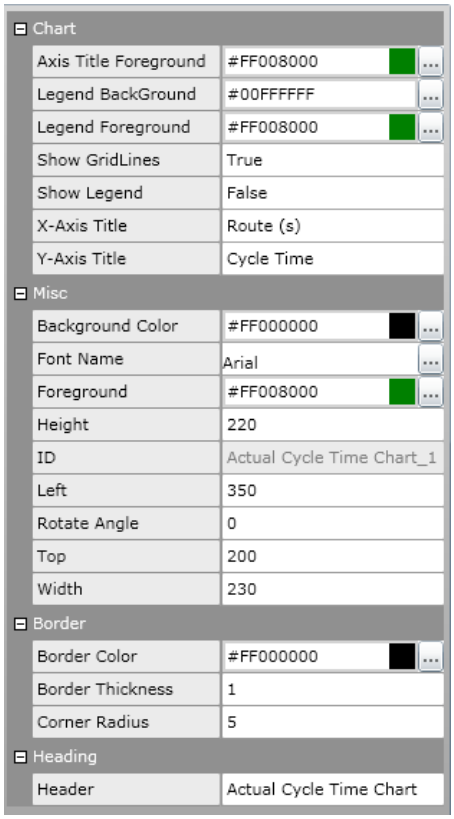


Chart	
Axis Title Foreground	#FF008000
Legend BackGround	#00FFFFFF
Legend Foreground	#FF008000
Show GridLines	True
Show Legend	False
X-Axis Title	Route (s)
Y-Axis Title	Cycle Time
Misc	
Background Color	#FF000000
Font Name	Arial
Foreground	#FF008000
Height	220
ID	Actual Cycle Time Chart_1
Left	350
Rotate Angle	0
Top	200
Width	230
Border	
Border Color	#FF000000
Border Thickness	1
Corner Radius	5
Heading	
Header	Actual Cycle Time Chart

5.2.14 Pickup Ratio

The default setting of the pickup ratio control is displayed.

Border	
Background Color	#FFFFFFFF
Border Color	#FF000000
Border Thickness	1
Corner Radius	5
Heading	
Column1 Heading	Route Name
Column1 Visibility	False
Column1 Width	0
Column2 Heading	Equipment Name
Column2 Visibility	True
Column2 Width	101
Column3 Heading	Feeder
Column3 Visibility	True
Column3 Width	101
Column4 Heading	Pickup Ratio
Column4 Visibility	True
Column4 Width	183
Gridlines Visibility	False
Header Visibility	True
Title	
Title Background	#FFD3D3D3
Title Font Color	#FF000000
Title Font Name	Arial
Title Font Size	12
Title Text	Pickup Ratio
Title Visibility	True
Font	
Font Color	#FF000000
Font Name	Arial
Font Size	10
Misc	
Data Grid Row Height	20
Height	104
ID	Pickup Ratio_1
Left	350
Rotate Angle	0
Row Count	3
Show Current Product Only	False
Top	200
Width	390

You can change the display contents by changing the setting.

・Show Current Product Only

False: The value including lot before changeover is displayed.

True: The value only currently producing lot is displayed.

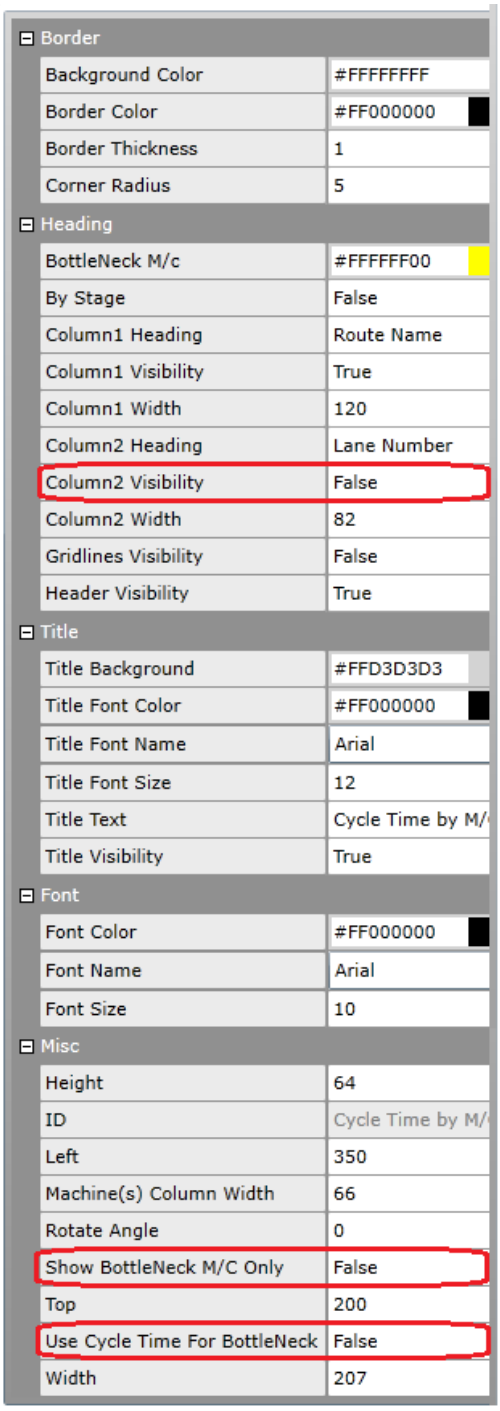
5.2.15 Boards Produced

The following is the default settings for the Boards Produced.

Border	
Background Color	#00FFFFFF
Border Color	#FF000000
Border Thickness	1
Corner Radius	5
Title	
Title Background	#FFD3D3D3
Title Font Color	#FF000000
Title Font Name	Portable User Interface
Title Font Size	12
Title Text	Board Produced
Title Visibility	True
Misc	
Gauge Type	Circle
Height	150
ID	Boards Produced Chart_1
Left	350
Rotate Angle	0
Top	200
Width	250

5.2.16 Machine Cycle Time

The following shows the default settings of machine cycle time control.



Border	
Background Color	#FFFFFFFF
Border Color	#FF000000
Border Thickness	1
Corner Radius	5
Heading	
BottleNeck M/c	#FFFFFF00
By Stage	False
Column1 Heading	Route Name
Column1 Visibility	True
Column1 Width	120
Column2 Heading	Lane Number
Column2 Visibility	False
Column2 Width	82
Gridlines Visibility	False
Header Visibility	True
Title	
Title Background	#FFD3D3D3
Title Font Color	#FF000000
Title Font Name	Arial
Title Font Size	12
Title Text	Cycle Time by M/
Title Visibility	True
Font	
Font Color	#FF000000
Font Name	Arial
Font Size	10
Misc	
Height	64
ID	Cycle Time by M/
Left	350
Machine(s) Column Width	66
Rotate Angle	0
Show BottleNeck M/C Only	False
Top	200
Use Cycle Time For BottleNeck	False
Width	207

You can change the display contents by changing the setting.

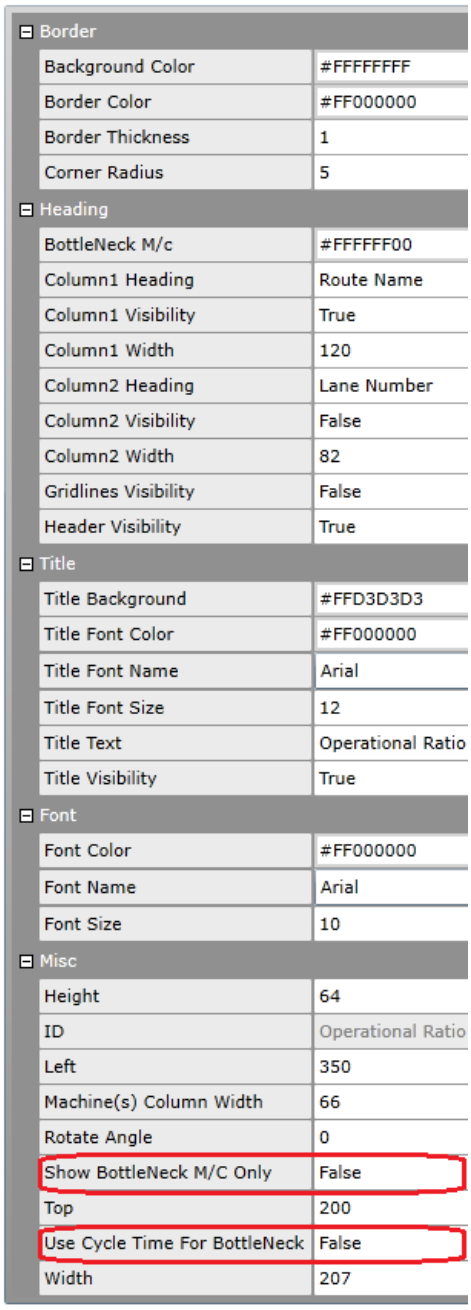
・Show BottleNeck M/C Only
  False: All machines are displayed.
  True: Only bottleneck machine is displayed.

・Use Cycle Time For BottleNeck
  False: A machine with the highest operation rate is displayed as bottleneck.
  True: A machine with the highest cycle time is displayed as bottleneck.

・Row 2 display (Lane No.)
  False: The count value is displayed for alternative placement and the bottleneck lane value is displayed for independent placement.
  True: Displays per lane.

5.2.17 Operating Rate per Machine

The default setting of the operating rate control per machine is displayed.



Border	
Background Color	#FFFFFFFF
Border Color	#FF000000
Border Thickness	1
Corner Radius	5
Heading	
BottleNeck M/c	#FFFFFF00
Column1 Heading	Route Name
Column1 Visibility	True
Column1 Width	120
Column2 Heading	Lane Number
Column2 Visibility	False
Column2 Width	82
Gridlines Visibility	False
Header Visibility	True
Title	
Title Background	#FFD3D3D3
Title Font Color	#FF000000
Title Font Name	Arial
Title Font Size	12
Title Text	Operational Ratio
Title Visibility	True
Font	
Font Color	#FF000000
Font Name	Arial
Font Size	10
Misc	
Height	64
ID	Operational Ratio
Left	350
Machine(s) Column Width	66
Rotate Angle	0
Show BottleNeck M/C Only	False
Top	200
Use Cycle Time For BottleNeck	False
Width	207

You can change the display contents by changing the setting.

・Show BottleNeck M/C Only

False: All machines are displayed.

True: Only bottleneck machine is displayed.

・Use Cycle Time For BottleNeck

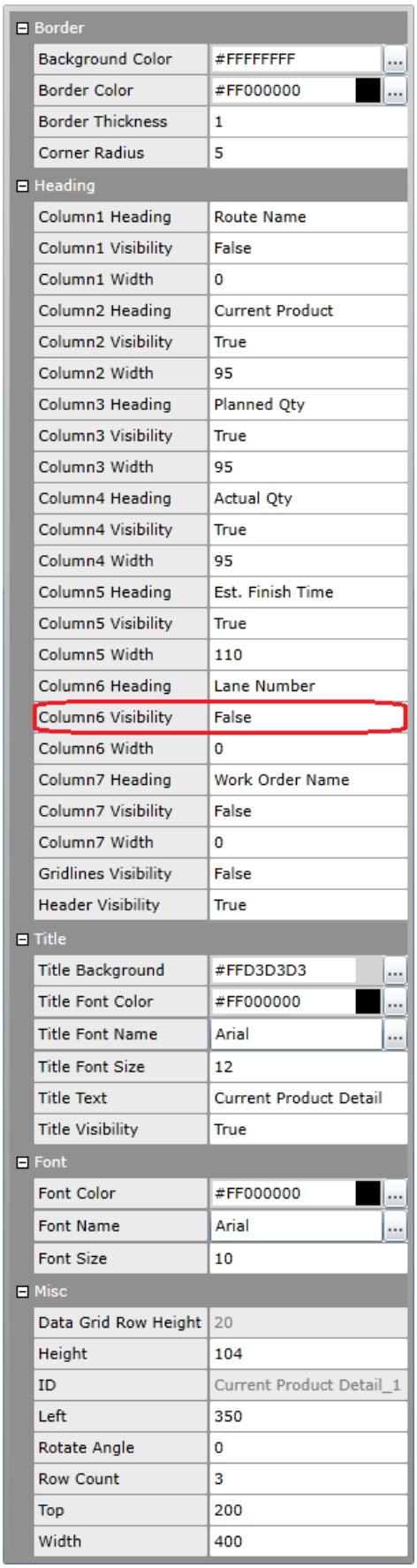
False: A machine with the highest operation rate is displayed as bottleneck.

True: A machine with the highest cycle time is displayed as bottleneck.

* When iLNB is used, it also supports to the third party machines (SPI/AOI/Reflow furnace).

5.2.18 Details of Current PCB

The default setting of the current PCB detailed control is displayed.



Border	
Background Color	#FFFFFFFF
Border Color	#FF000000
Border Thickness	1
Corner Radius	5
Heading	
Column1 Heading	Route Name
Column1 Visibility	False
Column1 Width	0
Column2 Heading	Current Product
Column2 Visibility	True
Column2 Width	95
Column3 Heading	Planned Qty
Column3 Visibility	True
Column3 Width	95
Column4 Heading	Actual Qty
Column4 Visibility	True
Column4 Width	95
Column5 Heading	Est. Finish Time
Column5 Visibility	True
Column5 Width	110
Column6 Heading	Lane Number
Column6 Visibility	False
Column6 Width	0
Column7 Heading	Work Order Name
Column7 Visibility	False
Column7 Width	0
Gridlines Visibility	False
Header Visibility	True
Title	
Title Background	#FFD3D3D3
Title Font Color	#FF000000
Title Font Name	Arial
Title Font Size	12
Title Text	Current Product Detail
Title Visibility	True
Font	
Font Color	#FF000000
Font Name	Arial
Font Size	10
Misc	
Data Grid Row Height	20
Height	104
ID	Current Product Detail_1
Left	350
Rotate Angle	0
Row Count	3
Top	200
Width	400

You can change the display contents by changing the setting.

・Row 6 display (Lane No.)

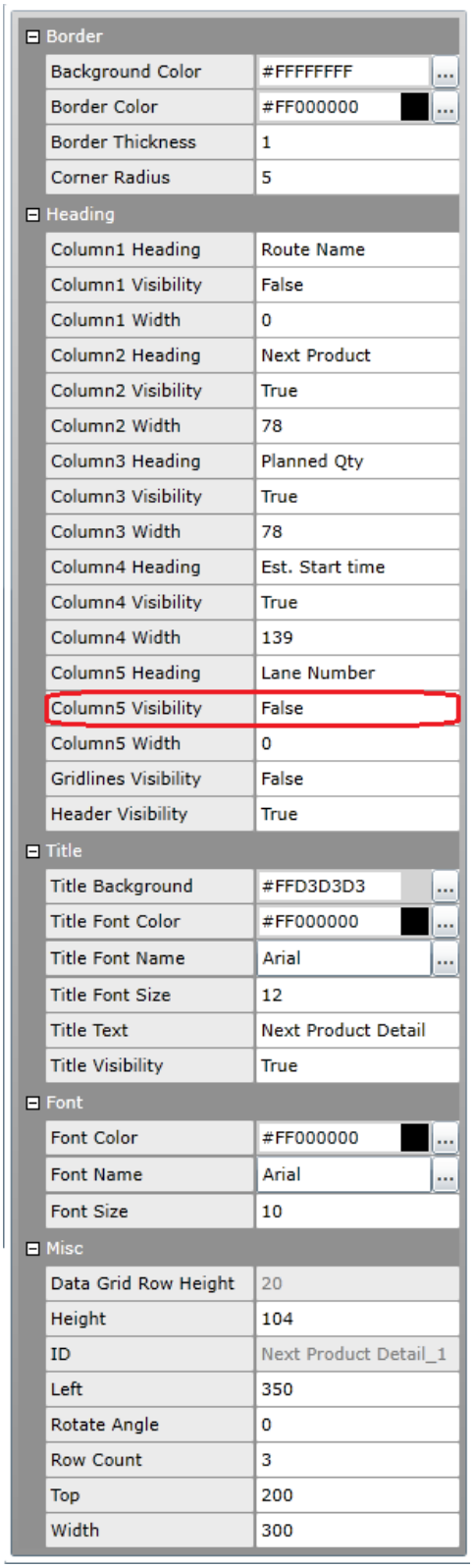
False: The lane No. is not displayed.

True: The lane No. selected in work order is displayed.

If both lanes are selected, [Both Lane] is displayed.

5.2.19 Details of Next PCB

The following shows the default settings of next PCB detailed control..



Border	
Background Color	#FFFFFFFF
Border Color	#FF000000
Border Thickness	1
Corner Radius	5
Heading	
Column1 Heading	Route Name
Column1 Visibility	False
Column1 Width	0
Column2 Heading	Next Product
Column2 Visibility	True
Column2 Width	78
Column3 Heading	Planned Qty
Column3 Visibility	True
Column3 Width	78
Column4 Heading	Est. Start time
Column4 Visibility	True
Column4 Width	139
Column5 Heading	Lane Number
Column5 Visibility	False
Column5 Width	0
Gridlines Visibility	False
Header Visibility	True
Title	
Title Background	#FFD3D3D3
Title Font Color	#FF000000
Title Font Name	Arial
Title Font Size	12
Title Text	Next Product Detail
Title Visibility	True
Font	
Font Color	#FF000000
Font Name	Arial
Font Size	10
Misc	
Data Grid Row Height	20
Height	104
ID	Next Product Detail_1
Left	350
Rotate Angle	0
Row Count	3
Top	200
Width	300

You can change the display contents by changing the setting.

・Row 5 display (Lane No.)

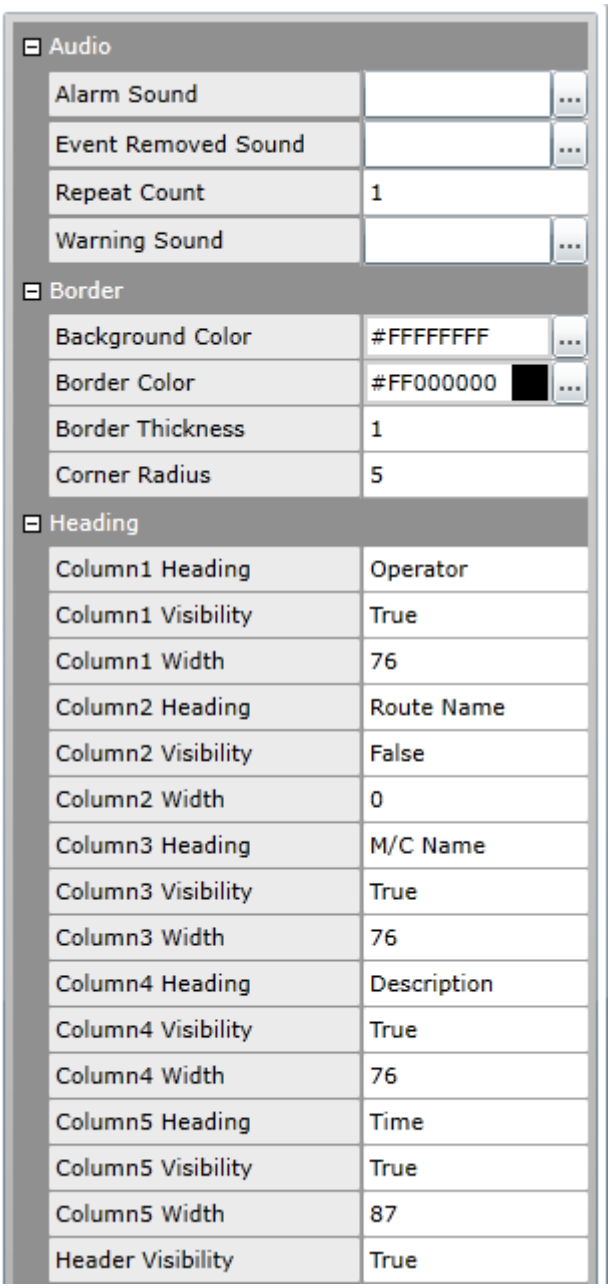
False: The lane No. is not displayed.

True: The lane No. selected in work order is displayed.

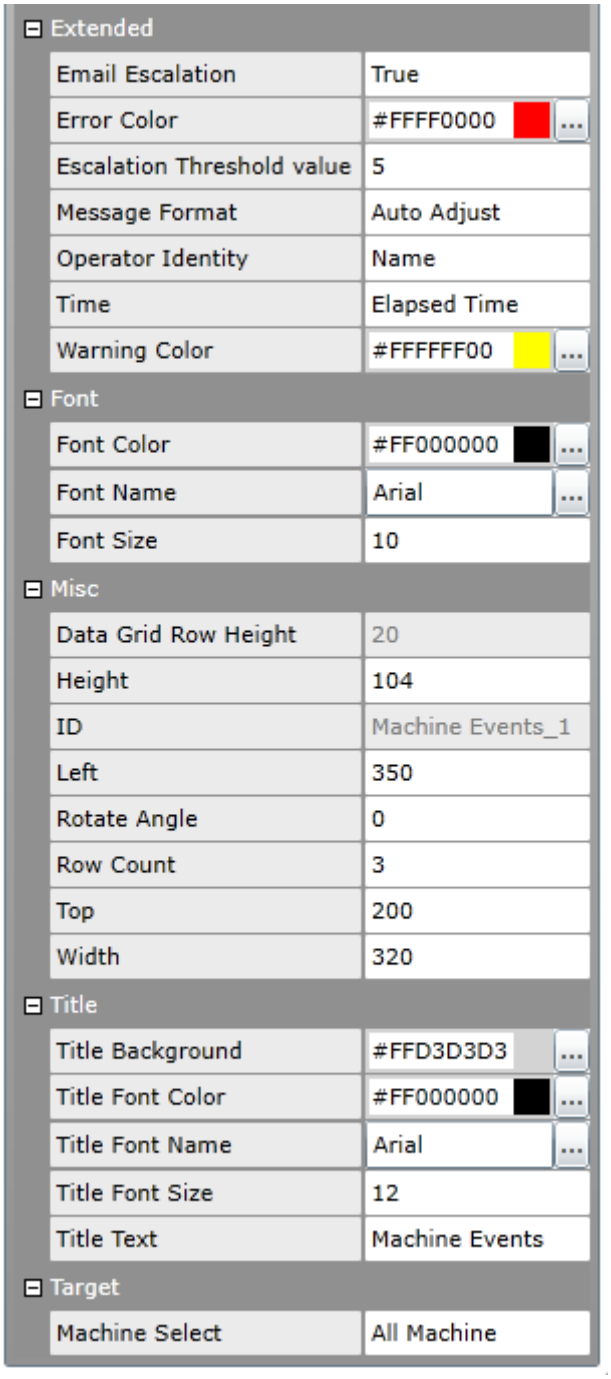
If both lanes are selected, [Both Lane] is displayed.

5.2.20 Machine Events

The following is the default settings for Machine Events.



Audio	
Alarm Sound	
Event Removed Sound	
Repeat Count	1
Warning Sound	
Border	
Background Color	#FFFFFFFF
Border Color	#FF000000
Border Thickness	1
Corner Radius	5
Heading	
Column1 Heading	Operator
Column1 Visibility	True
Column1 Width	76
Column2 Heading	Route Name
Column2 Visibility	False
Column2 Width	0
Column3 Heading	M/C Name
Column3 Visibility	True
Column3 Width	76
Column4 Heading	Description
Column4 Visibility	True
Column4 Width	76
Column5 Heading	Time
Column5 Visibility	True
Column5 Width	87
Header Visibility	True



Extended	
Email Escalation	True
Error Color	#FFFF0000
Escalation Threshold value	5
Message Format	Auto Adjust
Operator Identity	Name
Time	Elapsed Time
Warning Color	#FFFFFF00
Font	
Font Color	#FF000000
Font Name	Arial
Font Size	10
Misc	
Data Grid Row Height	20
Height	104
ID	Machine Events_1
Left	350
Rotate Angle	0
Row Count	3
Top	200
Width	320
Title	
Title Background	#FFD3D3D3
Title Font Color	#FF000000
Title Font Name	Arial
Title Font Size	12
Title Text	Machine Events
Target	
Machine Select	All Machine

You can change the display contents by changing the setting.

・Maintenance target / Machine selection

All machines: All machine events are displayed.
Chip mounter: Only machine events of the placement machine are displayed.
Screen printer: Only machine events of the printer are displayed.
Third party machine: Only machine events of the third party machine are displayed.

5.2.21 Part Supply

The following is the default settings for parts supply event control.

Audio	
Alarm Sound	
Event Removed Sound	
Repeat Count	1
Warning Sound	
Border	
Background Color	#FFFFFFFF
Border Color	#FF000000
Border Thickness	1
Corner Radius	5
Heading	
Column1 Heading	Operator
Column1 Visibility	True
Column1 Width	98
Column2 Heading	Route Name
Column2 Visibility	False
Column2 Width	0
Column3 Heading	M/C Name
Column3 Visibility	True
Column3 Width	98
Column4 Heading	Location
Column4 Visibility	True
Column4 Width	98
Column5 Heading	Part Number
Column5 Visibility	True
Column5 Width	98
Column6 Heading	Time
Column6 Visibility	True
Column6 Width	98
Column7 Heading	Time To Exhaust
Column7 Visibility	True
Column7 Width	105
Header Visibility	True

Extended	
Email Escalation	True
Escalation Threshold value	5
Message Format	Auto Adjust
Operator Identity	Name
Parts Error Color	#FFFF0000
Parts Warning Color	#FFFFFF00
Time	Elapsed Time
Title	
Exhaust Header	Parts Exhaust:
Splice Header	Tape Splice:
Title Background	#FFD3D3D3
Title Font Color	#FF000000
Title Font Name	Arial
Title Font Size	12
Title Text	Parts Supply Events
Title Visibility	True
Font	
Font Color	#FF000000
Font Name	Arial
Font Size	10
Misc	
Data Grid Row Height	20
Height	104
ID	Parts Supply_1
Left	350
Rotate Angle	0
Row Count	3
Top	200
Width	600
Target	
Machine Select	All Machine

You can change the display contents by changing the setting.

・Maintenance target / Machine selection

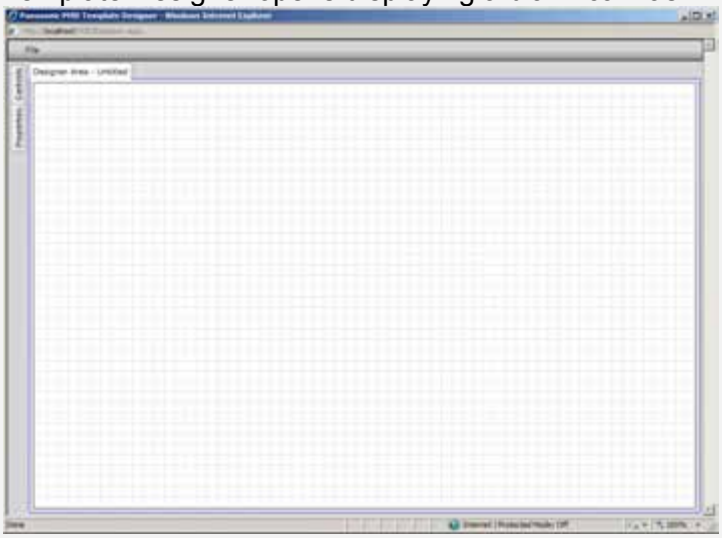
All machines: All part supply events are displayed.
Chip mounter: Only part supply events of the placement machine are displayed.
Screen printer: Only machine events of the printer are displayed.
Third party machine: Only machine events of the third party machine are displayed.

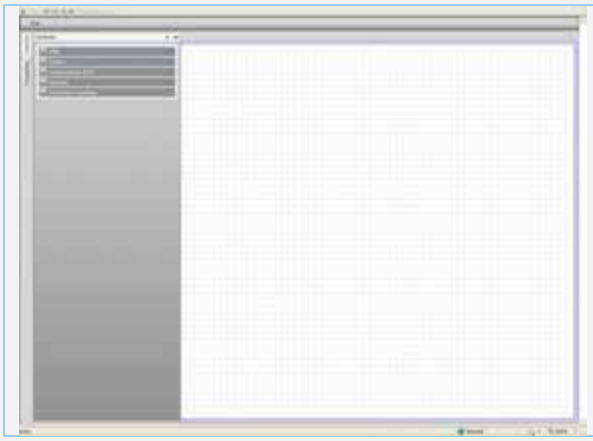
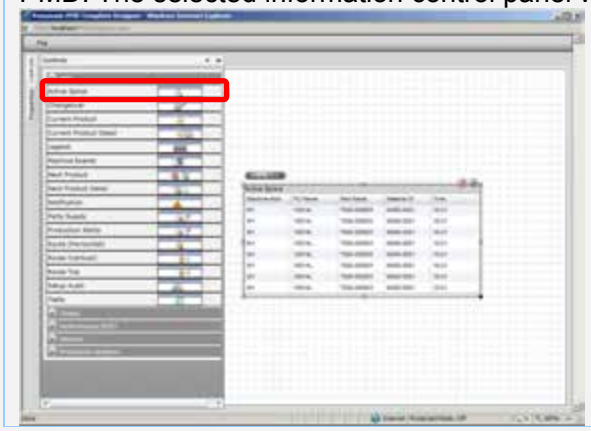
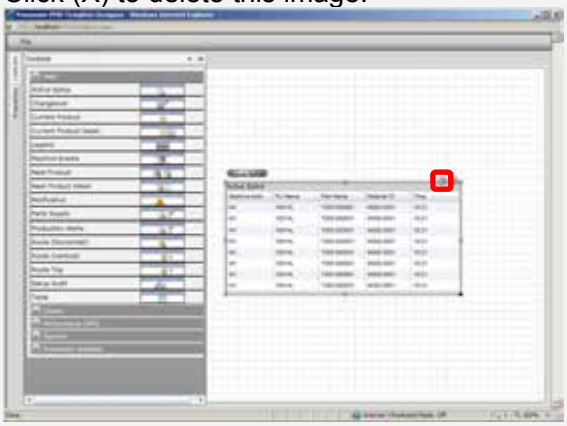
5.3 Operation

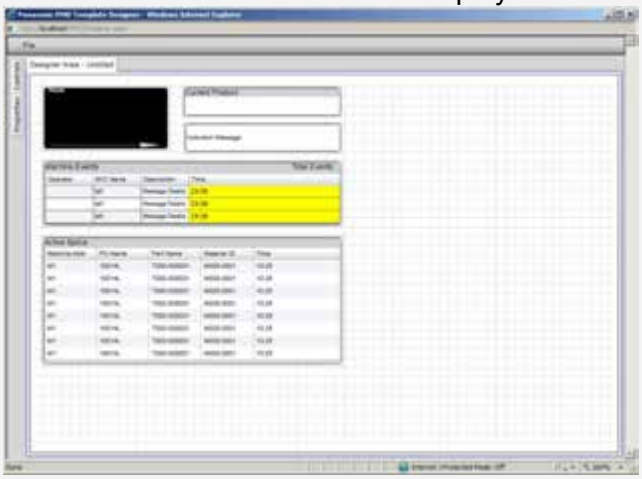
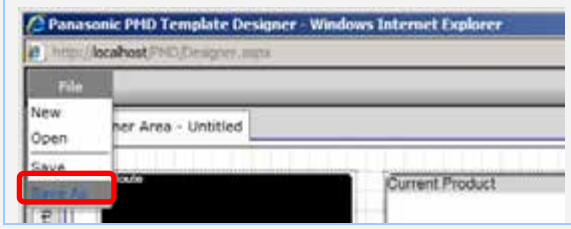
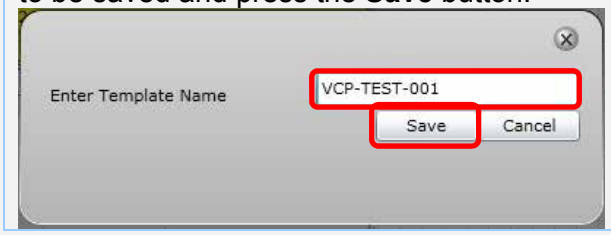
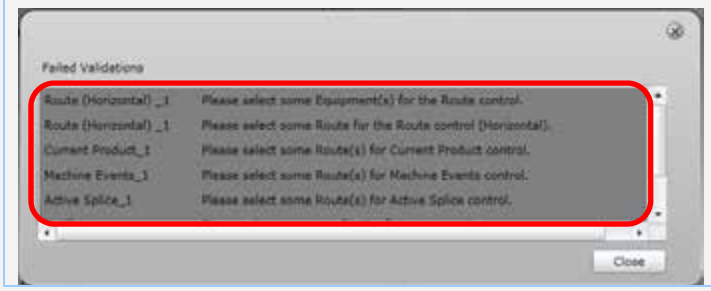
PMD operation runs using various applications and utilities.

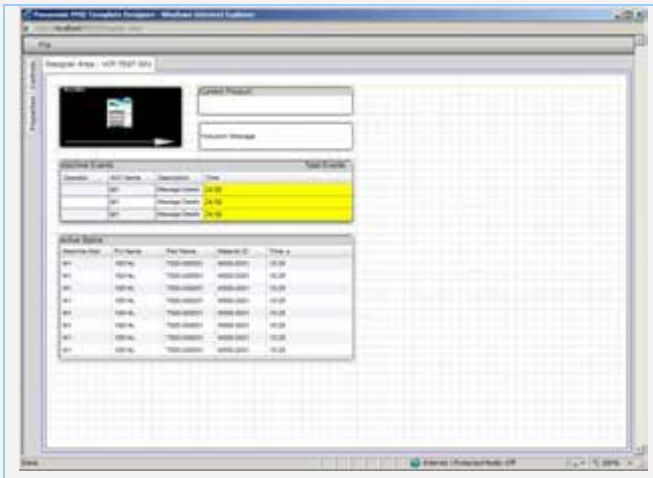
5.3.1 Create Template

Screen layout on Visual Control Panel can be customized using template function. You can create as many templates as you need.

Step	Procedure
1	Select **Production Monitoring & Dispatch** from the Launchpad and click the **Designer** link. 
2	Template Designer opens displaying a blank canvas. 
3	At this time, you may begin selecting Controls and their options to customize your template (or further customize existing templates). Control options include **PMD, Charts, Performance (KPI), General, and Production Analysis**. Once a main Control is selected, related options display that you can incorporate to your template design.

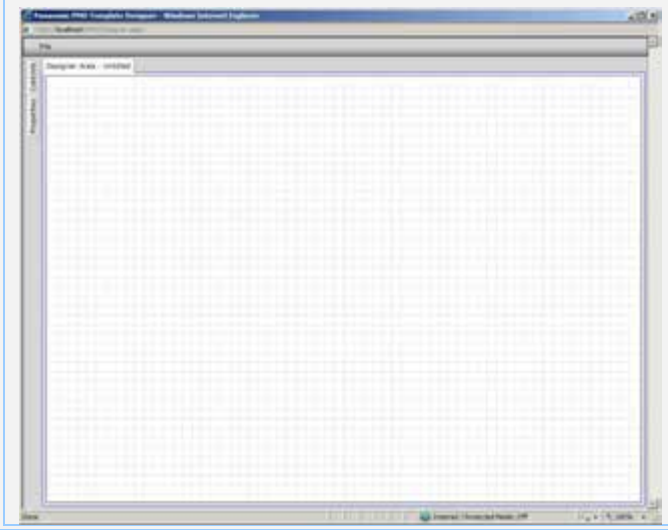
Step	Procedure
	
4	From the Control tab, select the information control panel to be displayed on the PMD. The selected information control panel will be shown on the display area. 
5	A control corresponding to the selected item will be displayed on the canvas. Click (X) to delete this image.  * Select the top left corner to move the control.

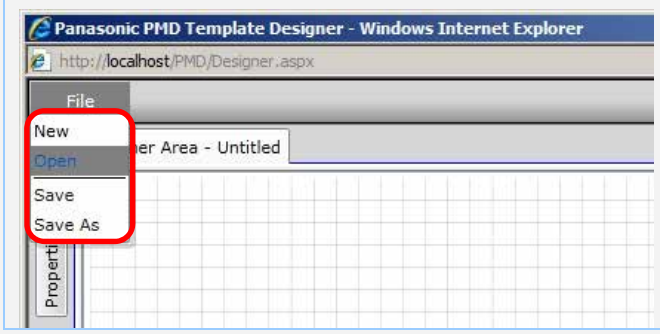
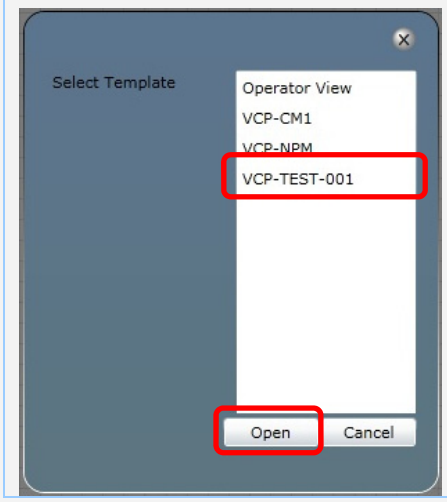
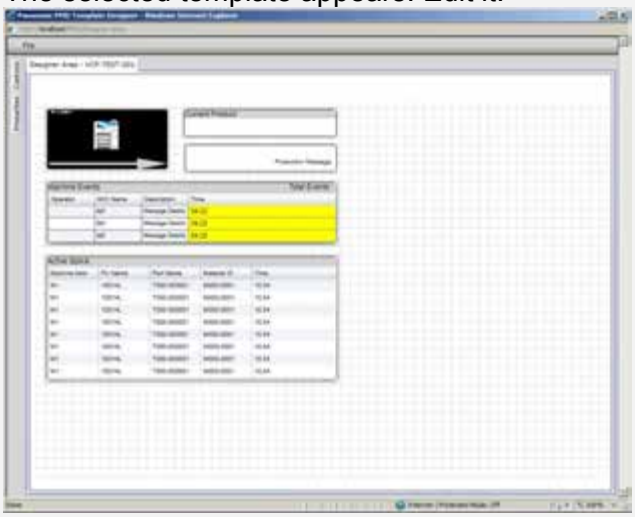
Step	Procedure
6	Add the information items to be displayed on the PMD and adjust the layout. 
7	Select **Save As** from [File] menu. 
8	Pop-up screen for [Template name input] appears. Type in the desired name to be saved and press the **Save** button. 
9	When saving the template, if any information items are yet to be set, the following dialog will appear. Confirm the message and set the required items. 
10	After the template has been saved, the template designer screen for PMD appears. Press the  button to close the screen.

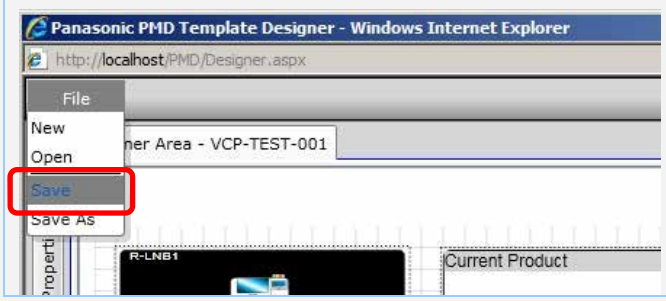
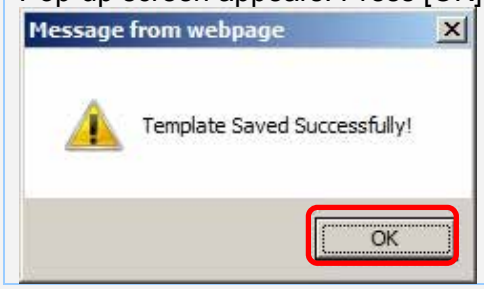
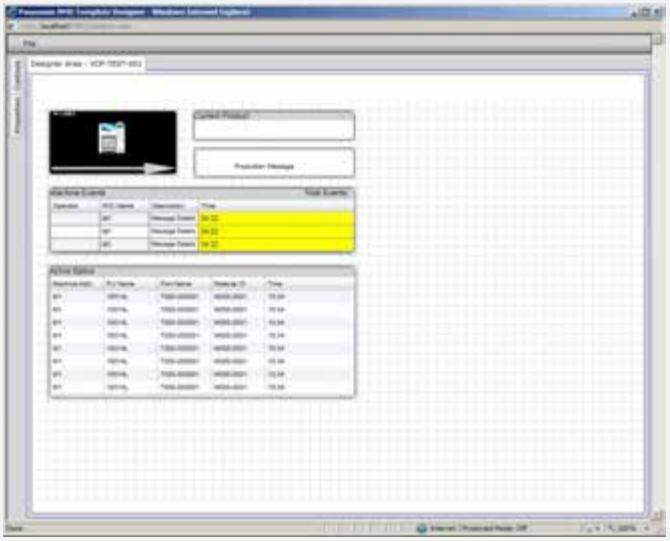
Step	Procedure
	

5.3.2 Edit Template

Screen layout on Visual Control Panel can be customized using template function. You can create as many template as you need.

Step	Operation
1	Launch WebApplication and select <Design> from <PMD> menu. 
2	Template designer screen for PMD opens. 

Step	Operation
3	Go to [File] menu and select [Open]. 
4	[Select Template] screen appears. Select a template and click [Open] button. 
5	The selected template appears. Edit it. 

Step	Operation
6	Go to [File] menu and select [Save]. 
7	Pop-up screen appears. Press [OK] button. 
8	Template designer screen for PMD appears. Click  button to end. 

= MEMO =

6. Function – Production Information

This chapter explains the operations for displaying items on the Visual Control Panel (VCP) of the PMD, with production information as an example.

Use the following PanaCIM applications to configure the production information and to operate the PMD.

- Template Designer
- Visual Control Panel (VCP)

6.1 Overview

"Production Information" of PMD displays different items of machine information on the Visual Control Panel (VCP) and visually informs the user of the machine condition as well as the production status.

6.2 Settings

"Production Information" of PMD requires the following settings.

- Add various controls that display production information onto the Visual Control Panel (VCP).

For settings and operation methods, refer to "5 Function – Template Designer".

6.3 Operation

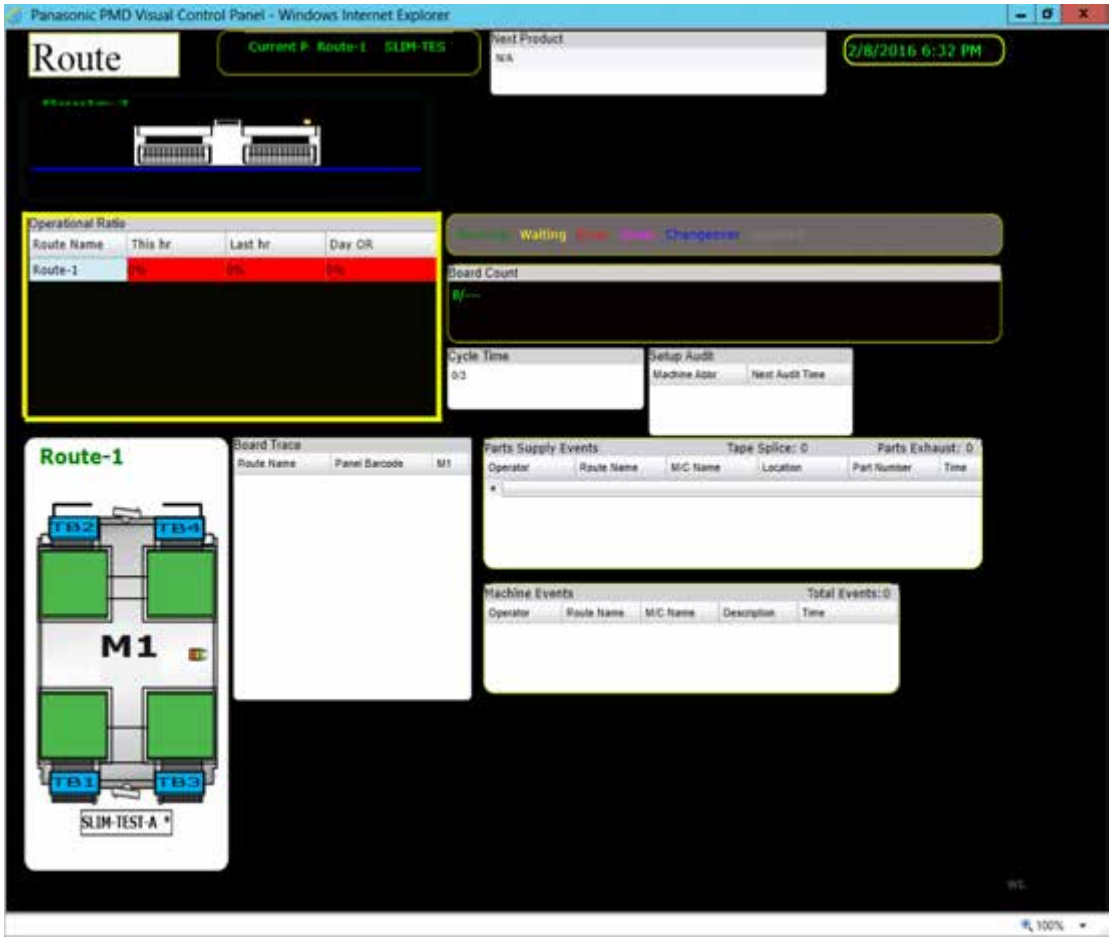
6.3.1 Display on Visual Control Panel (VCP)

As a setting example of [Production information] of the production monitor, the display screen is shown in case the following controls are arranged.

<Configured control>

- General control:
  Text/Time
- PMD control:
  Route (Horizontal)/Legend/Current Product/Next Product/Route Top/Setup Audit/
  Machine Event/Parts Supply
- Performance (KPI) control:
  Operational Ratio/Cycle Time/Panel Trace/Produced Panel(s)

<Display example of visual control panel (VCP)



6.3.2 Change of Settings for Route (Horizontal) Control

In the route (horizontal) control, the display color of the machine number displayed on the machine icon can be set to classify the types of events according to the event message received from the machine.

<Types of event messages to be classified>

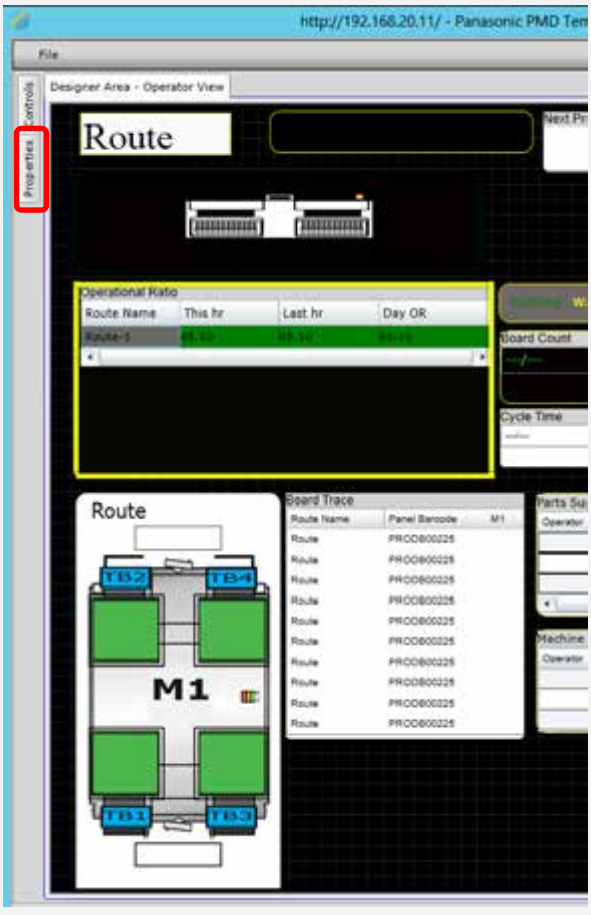
- Machine Error
- Machine Warning
- Part Error
- Part Warning

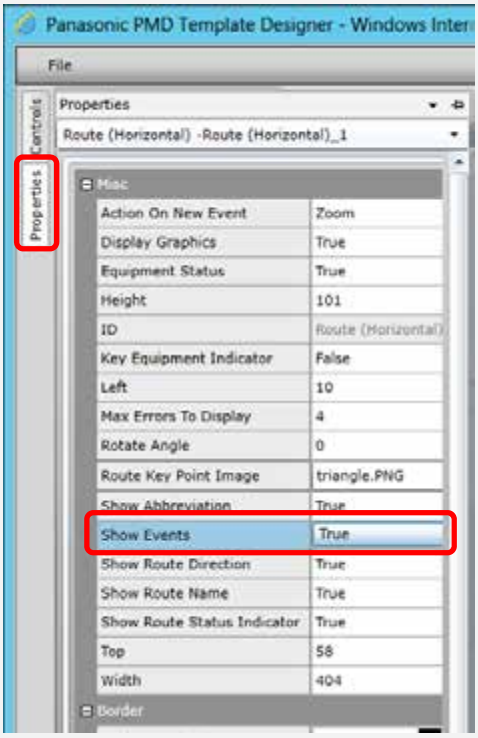
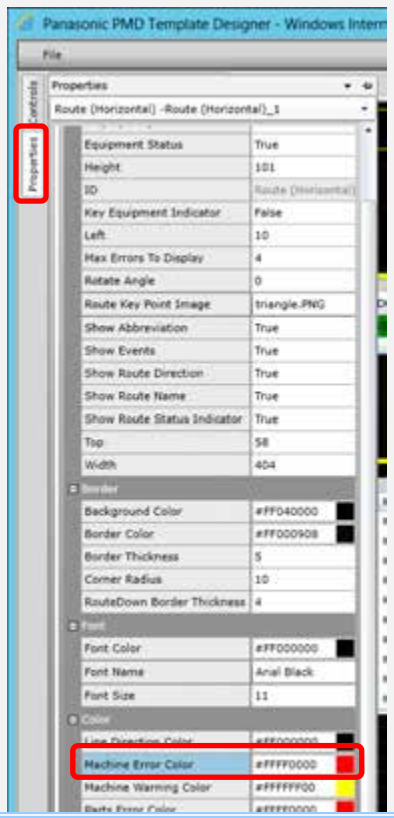
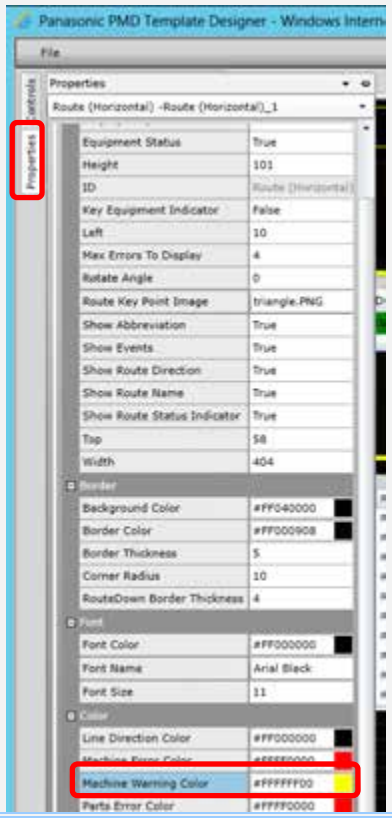
Also, if you receive more than one event message at the same time, priority order to display on the machine is decided in advance. The types of event messages are displayed in the following order of decreasing precedence.

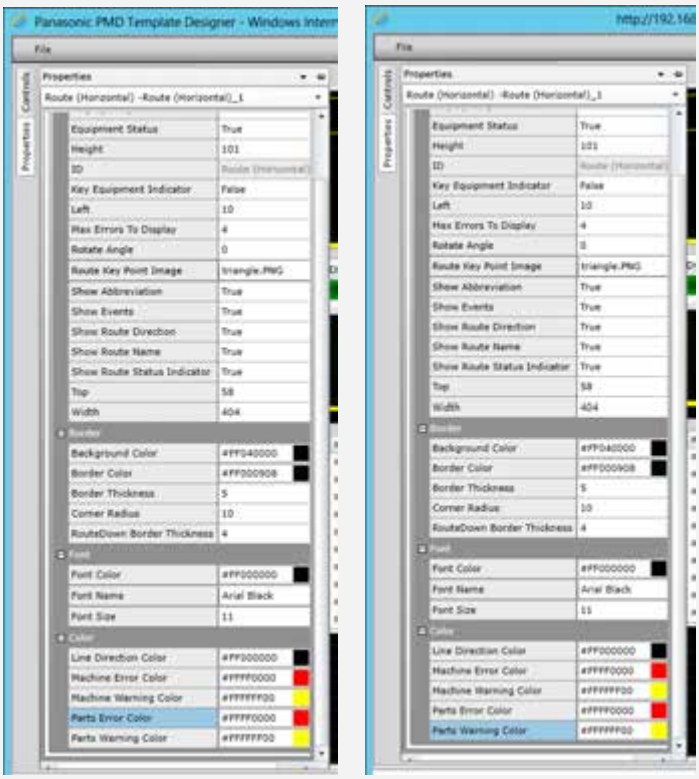
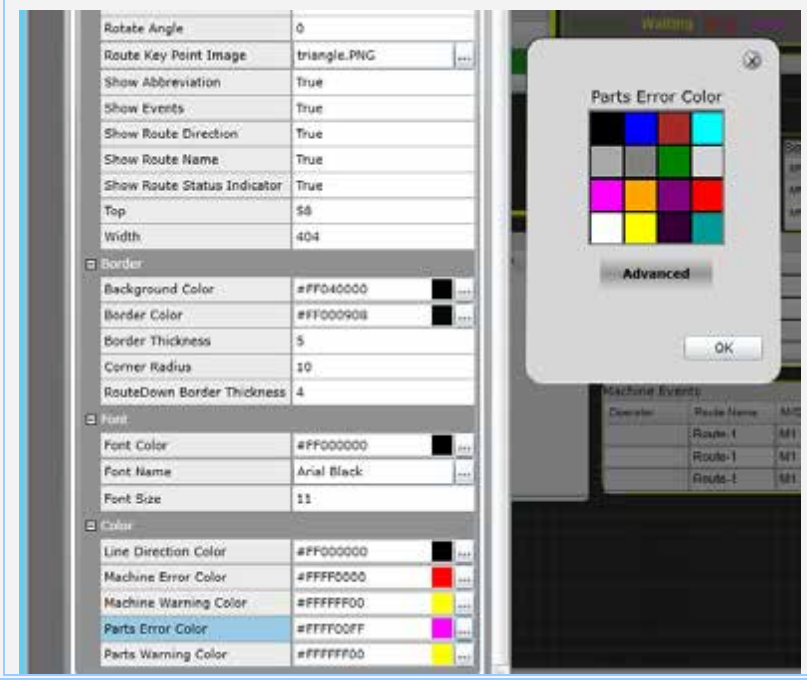
<Priority level of event message>

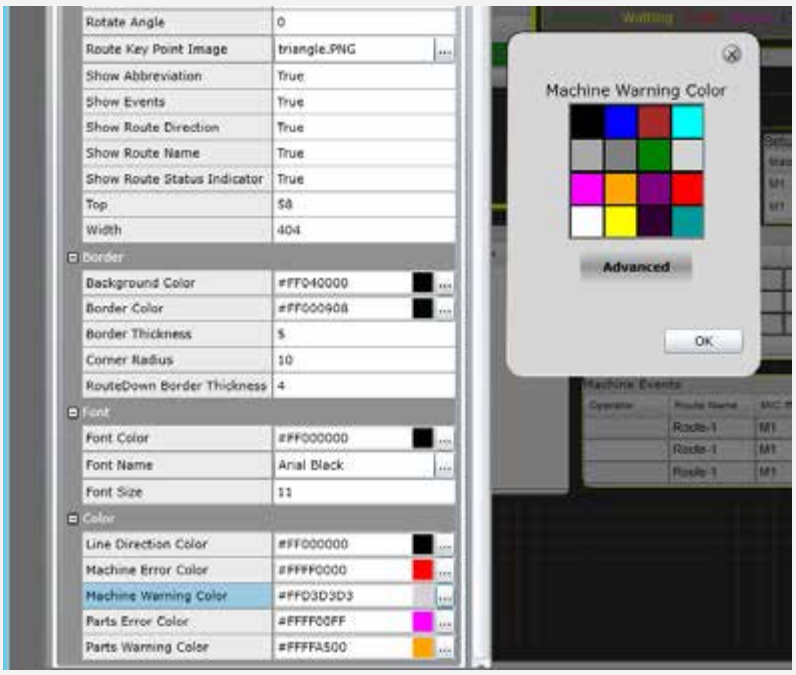
1) Machine Error
2) Part Error
3) Machine Warning
4) Part Warning

The following shows the procedure of how to change the setting of the route (horizontal) control.

Step	Procedure
1	Select [Route (Horizontal)] on the [PMD Template Designer] screen and click [Properties] tab on the left. 

Step	Procedure
2	Change [Show Events] to [True] on the [Properties] tab. ***Make sure to configure [Show Events] to [True]. Otherwise it does not display.** 
3	Confirm the color setting of the following items on the [Properties] tab. ・Machine Error Color: Red ・Machine Warning Color: Yellow  

Step	Procedure
4	Confirm the color setting of the following items on the [Properties] tab. ・Parts Error Color: **Red (Same as the machine color as default)** ・Parts Warning Color**: Yellow (Same as the machine error as default)** *Change the color setting because it is difficult to distinguish between machine error and machine warning by the default setting. 
5	Click the [...] button of [Parts Error Color] on the [Properties] tab and select [Pink] from the [Color] pop-up screen. 

Step	Procedure
6	Click the [...] button of [Parts Warning Color] on the [Properties] tab and select [Yellowish brown] from the [Color] pop-up screen. 
7	Save the changes on the [PMD Template Designer] screen. *The colors of the machine number displayed on the machine icon are changed by the above procedure and it will be easier to distinguish the types of events occurred such as error or warning by the color. ・**Machine Error Color:** Red ・**Machine Warning Color:** Yellow ・**Parts Error Color:** Pink ・**Parts Warning Color:** Yellowish brown

= MEMO =

7. Function – Setup Audit

This chapter explains operation for displaying items on the Visual Control Panel (VCP) of the PMD, with Setup Audit as an example.

Use the following PanaCIM applications to configure the Setup Audit information and to operate the PMD.

- CIMC controller
- Template (VCP)
- Mobile Scanner
- VCP

7.1 Overview

PMD [Setup Audit] is a function to make sure no wrong parts are used in production. Production stops at the specified time and Setup Audit mode starts. Scan the reel barcode of the parts being used. When Setup Audit mode starts, a message appears on Visual Control Panel (CVP) and the scanner's PMD screen to inform operators.

7.2 Settings

PMD [Setup Audit] needs the following settings.

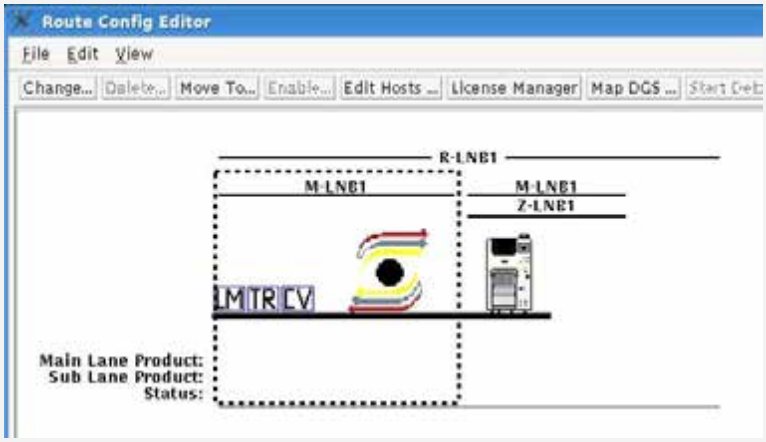
- Set setup audit action mode at CIMC controller
- Add controls to display setup audit information on Visual Control Panel (VCP)

7.2.1 Configure CIMC Controller

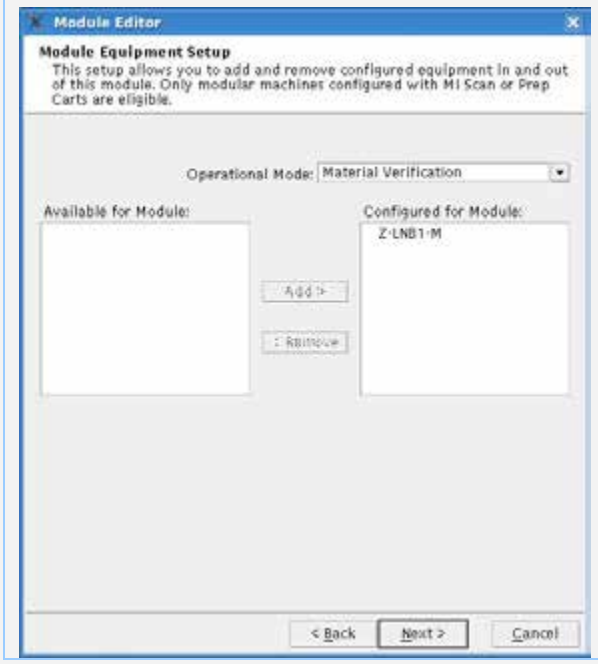
Set Setup Audit action mode from CIMC Controller's [Route Setting Editor」 screen.

- Program change check: Select this to check at changeover
- Setup Audit Time: Check regularly (at specified interval)

* Make sure to finish setup audit within the specified time otherwise interlock will be activated and production stops.

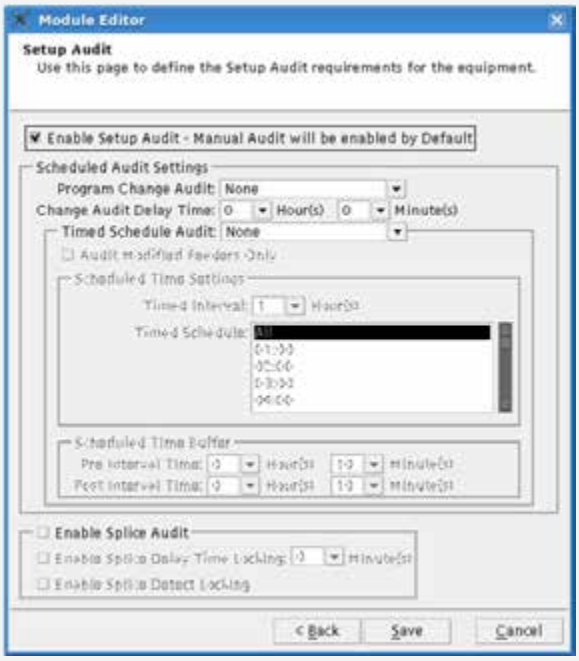
Step	Operation
1	From CIMC Controller, click [Route Editor] icon at application launcher. 
2	[**Route Setting Editor**] screen appears, click the route's [**Module**]. 
3	On [**Module Editor**]-[**GeneralSetting**] screen, click [**Next**] button**.** 

On [**Module Editor**]-[**Module Machine Setting**] screen, click [**Next**] button**.**



On [**Module Editor**]-[**Setup Audit**] screen, set the action mode and time to activate interlock. Click [**Save**] button.

<Settings>
- Enable Setup Audit: Check
- Program Change Audit: :
  Select here to check at changeover.
  * Set interlock at [**Change Audit Delay Time**].
- Timed Schedule Audit:
  Select this to check regularly (at specified interval).
  * Set interlock at [**Scheduled Time Buffer**].

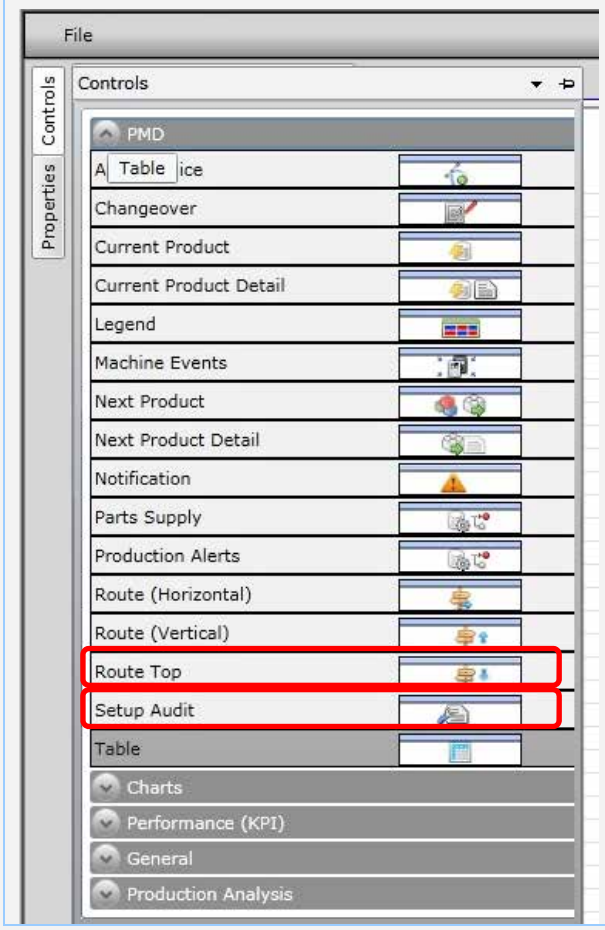


* The settings above shows the example of hourly setup audit.

7.2.2 Template Designer(Configure Template)

The following table shows how to add controls to show information on Visual Control Panel (CVP) in coordination with work order.

* See 5.3 Operation for how to open or save templates.

Step	Operation
1	On [PMD Template Designer] screen click [Controls] tab on the left to open [PMD]. 
2	Add the following controls to show setup audit information on Operation Monitor Control. • Setup Audit: As setup audit information, the next audit time and machines are displayed. Once setup audit is done by a scanner (parts scanned), the schedule will be reset. * Only scheduled setup audits are supported. • Top table: Shows routes/machines in a list to select for setup audit. Machines are displayed with their table numbers. <Icons> Icons on tables show various colors depending on the setup audit status. <Tables> In case of splice warning, the entire table becomes yellow. For parts empty, it is displayed with red.

7.3 Operation

Use a mobile scanner to clear setup audit warning display from PMD. Click the setup audit icon on the mobile scanner to finish the audit.

- **Program Change Audit:**
  Start setup audit at changeover (when different recipe downloaded) to check necessary parts.

- **Timed Schedule Audit:**
  Start setup audit at a specified time (interval) to check necessary parts.

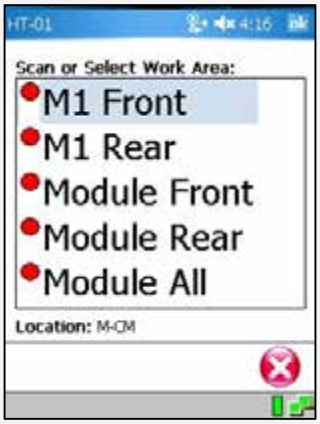
7.3.1 Display Warning message

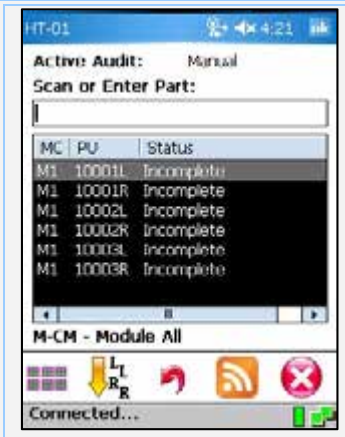
Setup audit warning message appears on a scanner's Operational Monitor-[PMD] screen.

Step	Operation
1	Click  icon on Production main menu screen. 
2	Setup audit status (when interlock activated) appear on PMD [PMD] screen.  *** The screen above shows the following setup audit status.** **・Red: Interlock activated during setup audit.** **(When setup audit did not finish within the specified time.)** **・Yellow: It is at setup audit start time.**

7.3.2 Execute Setup Audit

The following steps shows how to perform setup audit with a scanner.

Step	Operation
1	Click  icon on production main menu screen. 
2	Select [Module All] on [Work Area] screen. 
3	Scan or enter the parts at the location with cursor on [Scan or Enter Material ID] screen. <Interlock activated>  <Interlock inactive>

	 * See [PanaCIM-EE_V8.6.0_Material Verification」 for more details on setup audit.
4	Scan the parts at the location pointed with the cursor and all other parts on the front table side. 
5	Then scan the parts at the location pointed with the cursor and all other parts on the front table side. 
6	After scanned or entered all parts, the following message appears. Click  icon to finish.



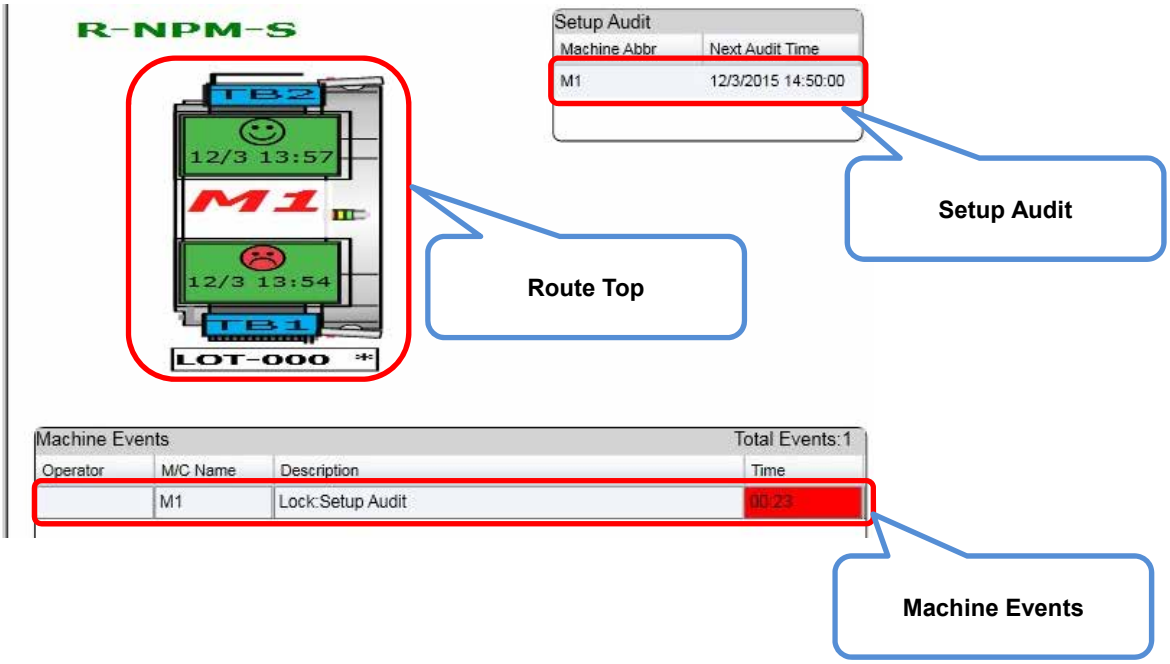
7.3.3 Display on Visual Control Panel(VCP)

Visual Control Panel (CVP) shows the status of setup audit done by a scanner to inform operators.

■ Display Setup Audit Warning

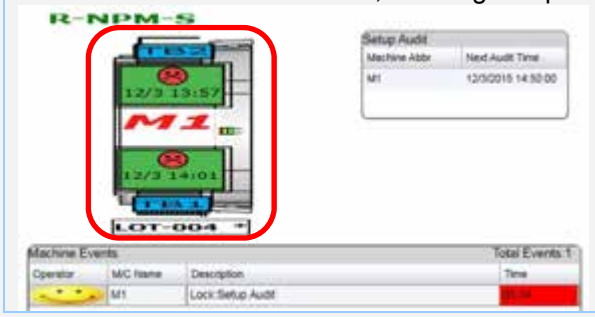
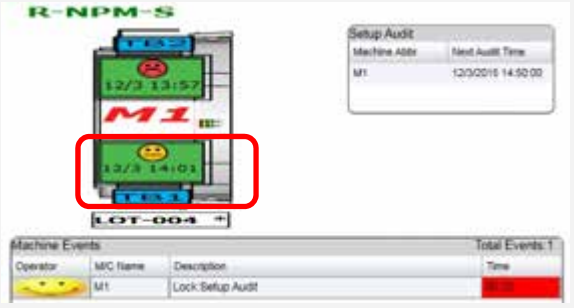
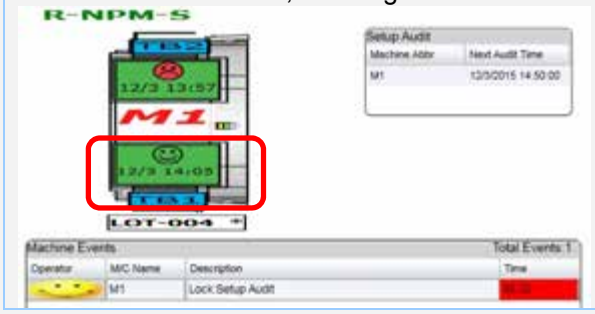
Visual Control Panel (CVP) shows various information as setup audit warning with route top.

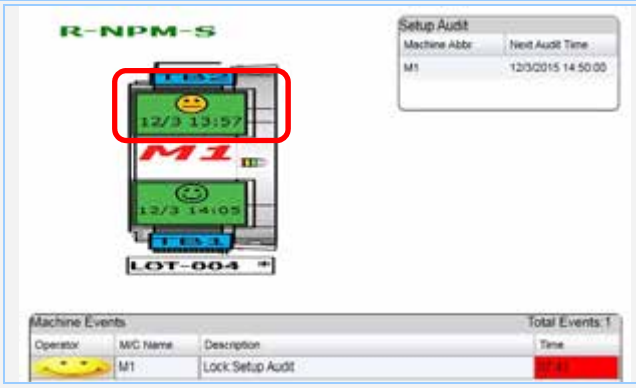
<Route Top>
・Setup audit status can be checked per table.
<Setup Audit>
・Shows the time to start setup audit.
<Machine Events>
・Check if interlock is activated because the audit did not finish.



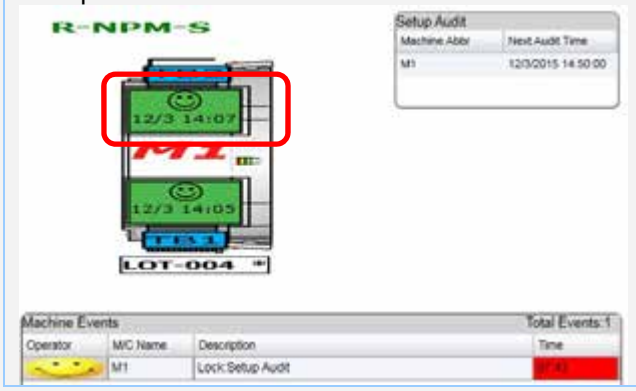
■ Display Setup Audit Status

After setup audit was done by a scanner, the contents at Route Top on Visual Control Panel (CVP) are changed as follows.

Step	Operation
1	The screen is as follows when setup audit is stared with a scanner. <Route Top> ・ The Front/Rear icon are red, showing setup audit has not been started. 
2	After the audit started on the front side, the screen changes as follows. <Route Top> ・ The Front icon is yellow, showing the audit running for the front parts. ・ The Rear icon is red, showing the audit has not started for the front parts. 
3	After the audit finished on the front side, the screen changes as follows. <Route Top> ・ The Front icon is green with date/time, showing the audit has finished for the front parts. ・ The Rear icon is red, showing audit has not done for the rear parts. 
4	After the audit started for the rear side, the screen changes as follows. <Route Top> ・The Front icon is green with date/time, showing the audit has finished for the front parts. ・The Rear icon is yellow, showing the audit running for the rear parts.



5 After setup audit finished (checked all parts), the screen changes as follows.
<Route Top>
・The Front/Rear icon are green with date/time, showing the audit has finished for the front parts.



8. Function - Link to Work Order

This chapter explains operations for displaying items on the Visual Control Panel (VCP) of the PMD, with Work Order information as an example.

Use the following PanaCIM applications to configure the Work Order information and to operate the PMD.

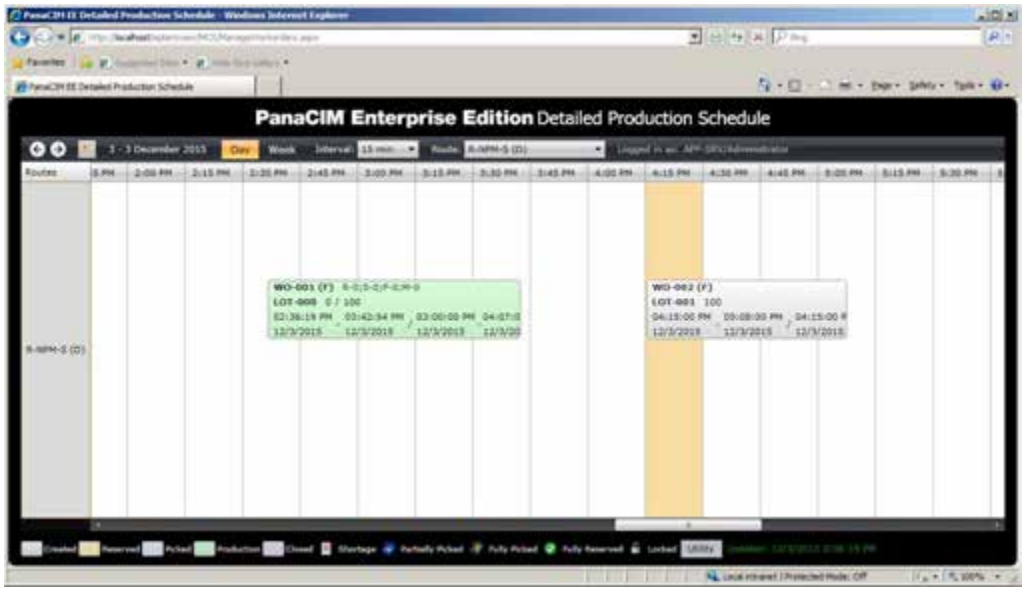
- Template Designer
- Detailed Production Schedules (Work Order)
- Visual Control Panel

8.1 Overview

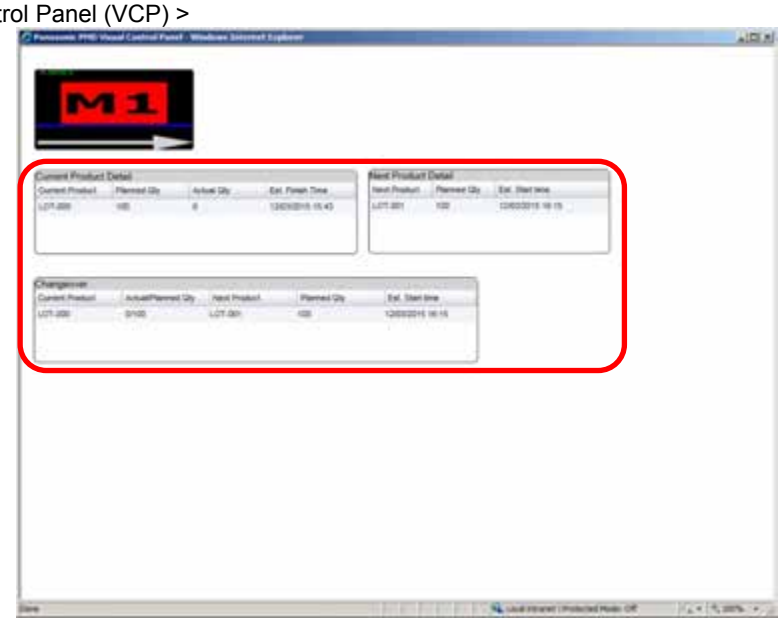
PMD [Coordination with Work Order] works with the status displayed on [Detailed Production Schedules] screen, and displays the information at the following controls on Visual Control Panel (CVP).

<Allocated controls>
- Current Product: General information on the product being produced.
- Current Product Details: Detailed information of the product being produced.
- Next Product: General information on the product to be produced next.
- Next Product Details: Detailed information of the product to be produced next.
- Changeover: Detailed information on the products that need changeover.

<Work Order- Detailed Production Schedules>



<Visual Control Panel (VCP) >



8.2 Settings

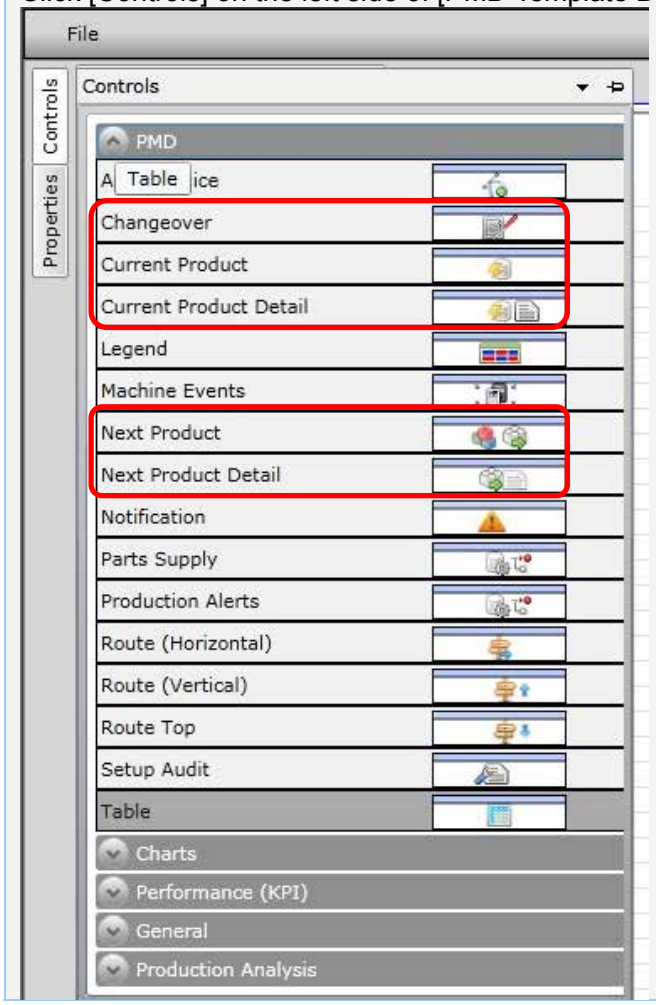
Add controls to display work order information on Visual Control Panel (CVP) by PMD [Coordination with work order].

Setting this will enable to omit steps to check work order status or board count on work order [Detailed Production Schedules].

8.2.1 Template Designer(Configure Template)

The following table shows how to add controls to display information on Visual Control Panel (CVP) in coordination with work order.

* See [5.3 Operation」 for how to open or save templates.

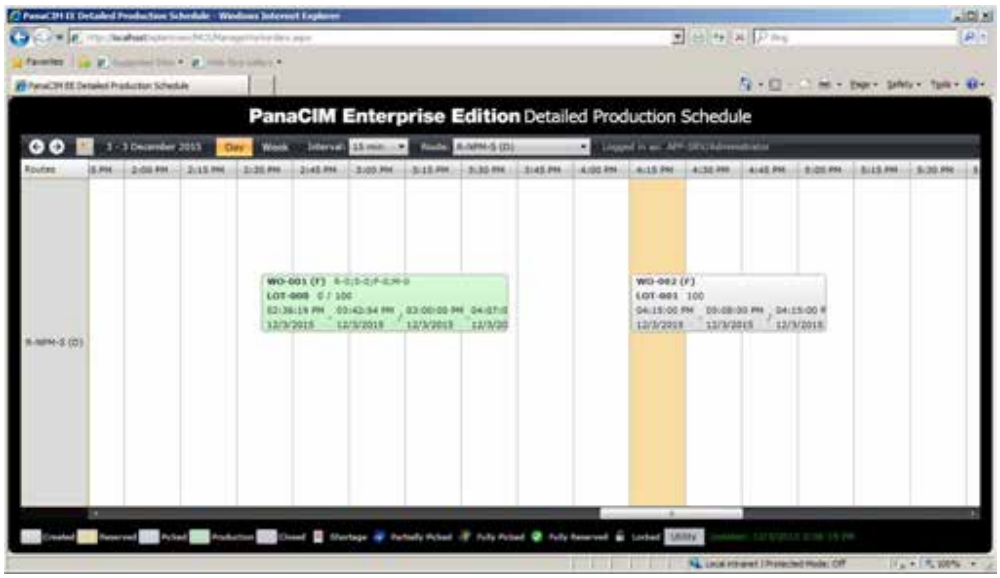
Step	Operation
1	Click [Controls] on the left side of [PMD Template Designer] screen to open [PMD]. 
2	Add the following controls to display work order information on Operation Monitor Control. • **Current Product** : Shows the product being produced. • **Current Product Details**: Shows route name, product name being produced, planned panel count, planned production end date and time. This can be displayed for multiple routes. • **Next Product:**

	Shows the name of the program to be produced next. • **Next Product Details:** Shows route name, program name to be produced next, planed panel count, planned production start date and time. • **Changeover:** Shows produced/planned count of the current product, and the next planned product count with production start date/time. * The above controls can be displayed for multiple routes.

8.3 Operation

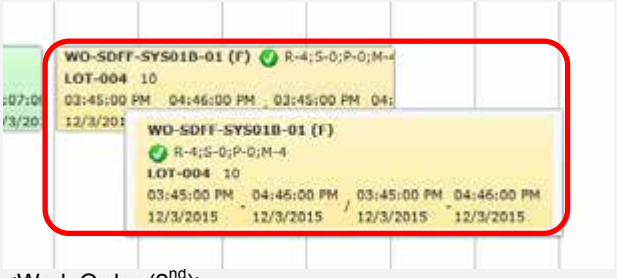
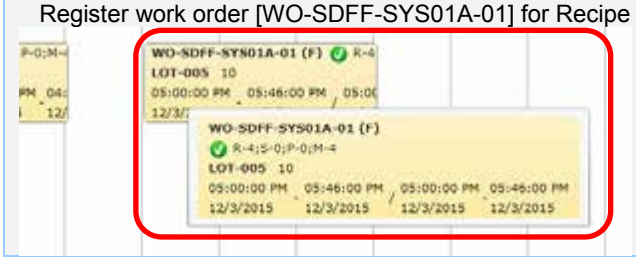
Register or reserve work orders from [Detailed Production Schedules] screen.

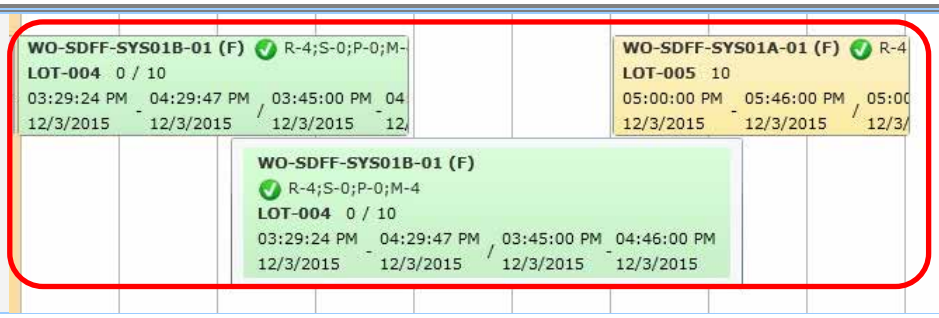
The status of registered work order changes or various information appears after you downloaded a recipe based on the work order or started production.



8.3.1 Create & Reserve Work Order

Work order [Detailed Production Schedules] screen has the following information.
See [9.4.1 Coordination with Work Order] for Visual Control Panel that corresponds with the following screen.

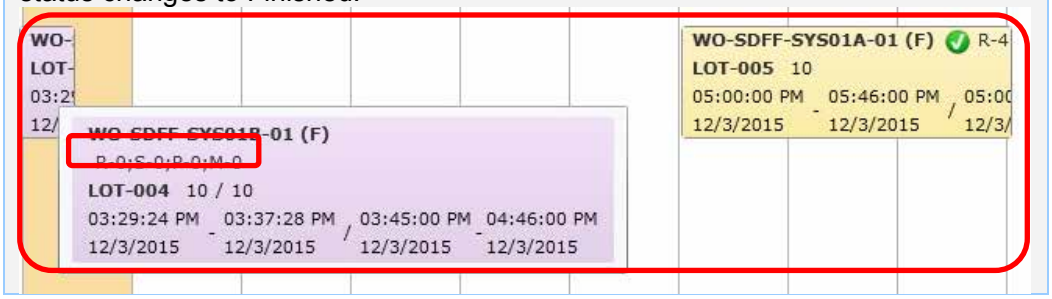
1	On [Detailed Production Schedules] screen, register two work orders. <Work Order (First)> Register Work Order [WO-SDFF-SYS01B-01] for Recipe [LSDFF-SYS01-B] and finish reservation.  <Work Order (2nd)> Register work order [WO-SDFF-SYS01A-01] for Recipe [LSDFF-SYS01-A]. 
2	After you downloaded a recipe that is used in the first work order to a machine, the work order status changes to In Production.



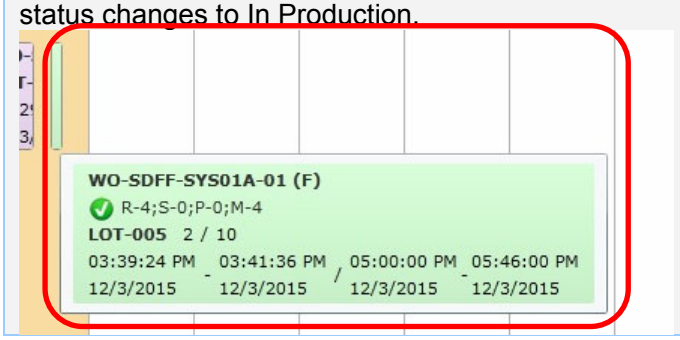
3 After produced two boards, the board count on the work order is updated.



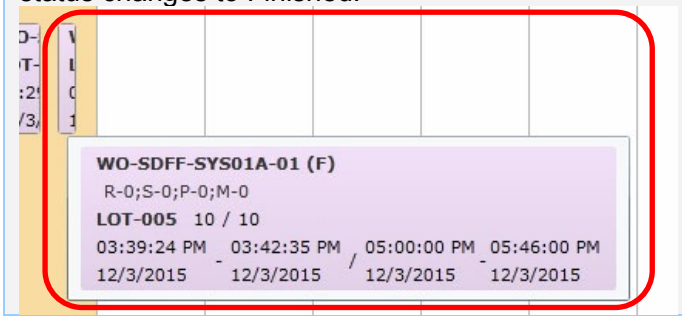
4 After the number of boards were produced as planned in the first work order, its status changes to Finished.



5 After changed to a recipe that is used in the 2nd work order from the machine, its status changes to In Production.

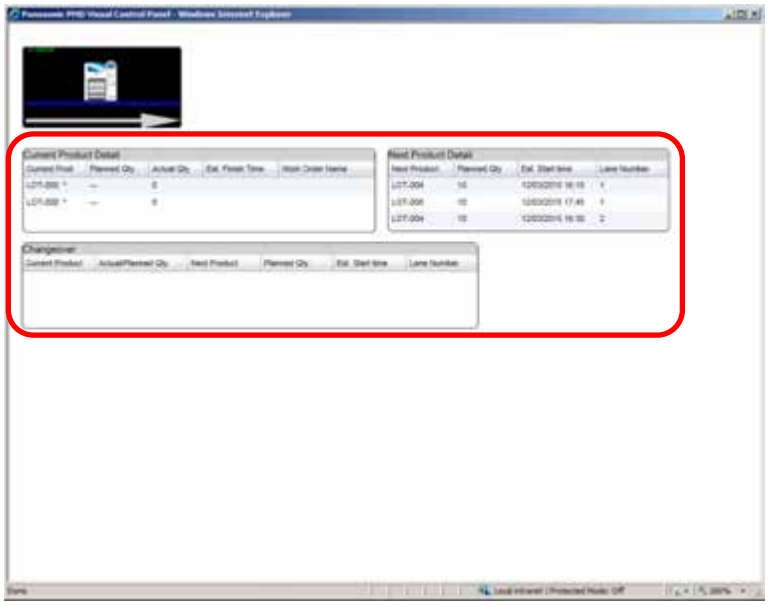


6 After the number of boards were produced as planned in the 2nd work order, its status changes to Finished.



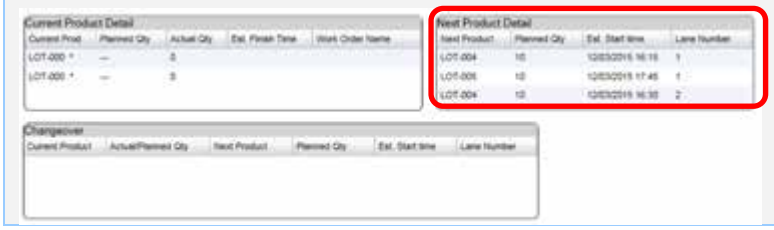
8.3.2 Display on Visual Control Panel(VCP)

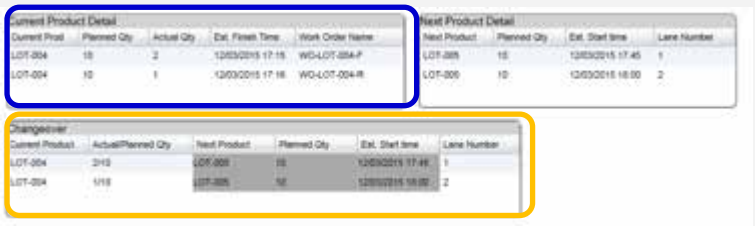
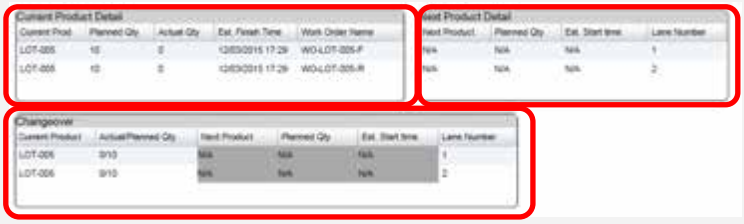
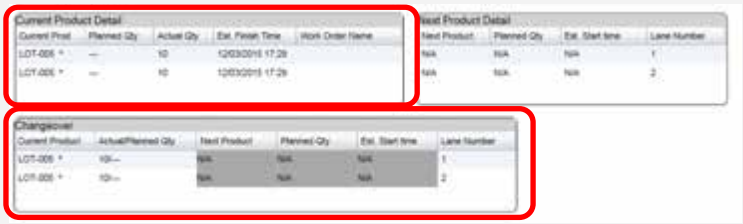
PMD Operation works with various applications and utilities.



■ Coordination with Work Order

Visual Control Panel (VCP) shows the following work order information.

1	After you register two work orders and finish reservation on [Detailed Production Schedules] screen, the first work order information appears at Next Product information. <Next Product Information> ・Next Product: Recipe name ・Plan: Planned board count ・Estimated start time: Estimated time to start production ・Lane: Lane for production 
2	Download the recipe that is used in the first work order from the machine. The following information will be changed. <Current Product Information (Blue)> ・Shows the first work order information. <Next Product Information (Red)> ・Shows the second work order information. <Changeover (Orange)> ・Shows the first work order information. 

3	When two boards are produced on the machine, the number of boards on the work order will be updated. <Current Product Information (Blue)> ・First work order information updated. <Changeover (Orange)> ・First work order information updated and displayed 
4	Produce as many number of boards as registered to the first work order. <Current Product Information (Blue)> ・First work order information updated. <Changeover (Orange)> ・First work order information updated and displayed. 
5	Changeover to the recipe that is used in the second work order from the machine and start production. 
6	Produce as many number of boards as registered to the second work order. 

9. Function – Route Downtime

This chapter explains operations for displaying items on the Visual Control Panel (VCP) of the PMD, with Route Downtime as an example.

Use the following PanaCIM applications to configure the Route Downtime information and to operate the PMD.

- Application Server
- Visual Control Panel
- Mobile Scanner

9.1 **Overview**

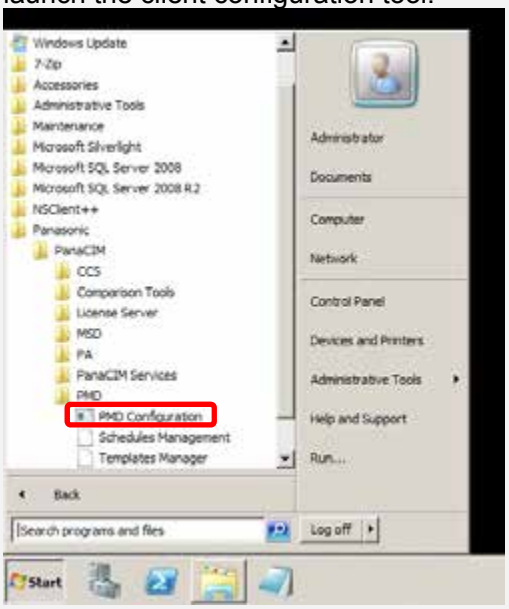
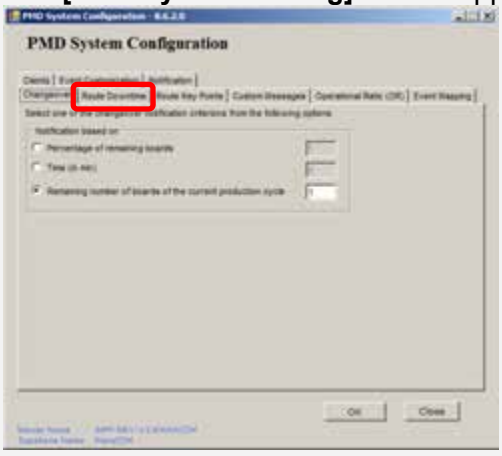
Users set downtime reasons in advance from a mobile scanner so PMD [Route Downtime] displays the target route in down status on Visual Control Panel (CVP) and visually notifies users of such condition.

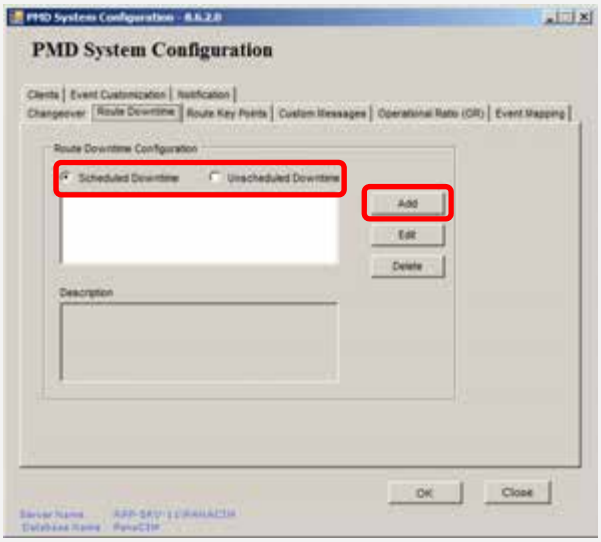
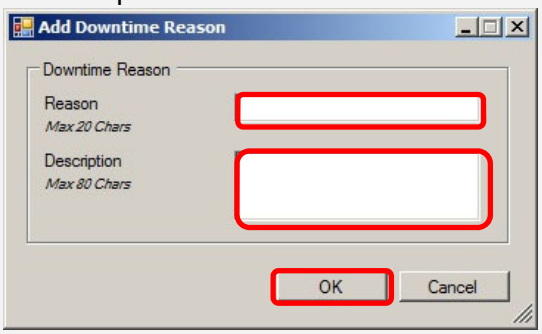
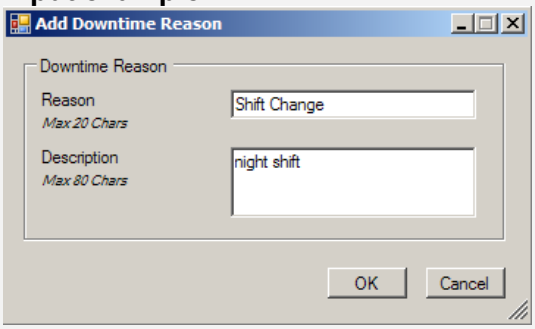
9.2 **Settings**

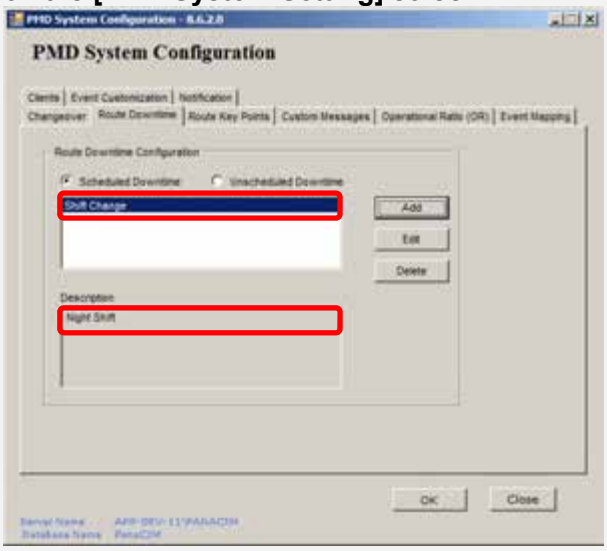
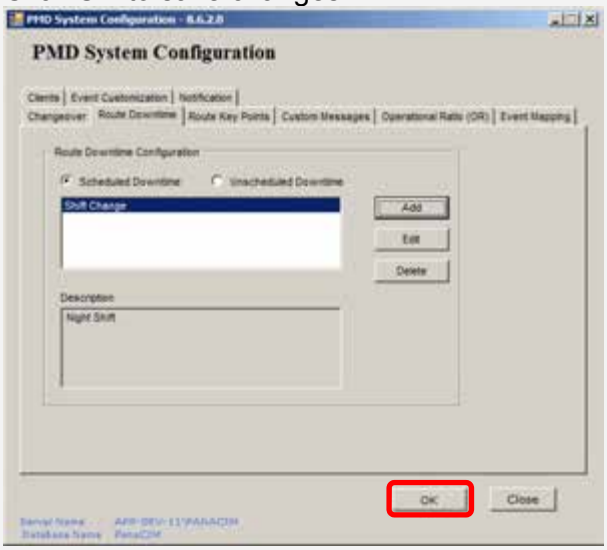
Take the following steps on the application server side to register the name (reason) and for route downtime beforehand. The setting information can include the following:

- Downtime kind: Scheduled or Unscheduled downtime.
- Scheduled downtime: Reason and description for scheduled downtime.
- Unscheduled downtime: Reason and description for unscheduled downtime.

* Select the reasons added here to initiate a route downtime via the Mobile scanner based acknowledgement system.

Step	Procedure
1	Select Start→Programs→Panasonic→PMD→PMD Configuration in order to launch the client configuration tool. 
2	The **[PMD System Setting]** screen appears. Click the **Route Downtime** tab. 

3	Select **Scheduled** or **Unscheduled** as the route downtime option. Click **Add** to add a reason for the selected downtime. 
4	The **Add Downtime Reason** window appears. Enter the Reason and the Description. Then click **OK**. **Downtime Reason** ・Reason: Enter the reason for the downtime. ・Description Describe the details of the downtime.  **Note:** This window does not accept any two-byte characters. **Input example** 

5	The registered downtime can be confirmed under the **Route Downtime** tab on the **[PMD System Setting]** screen.  To edit a reason, select it and click **Edit**. The **[Edit down time reason]** dialog appears. Modify the fields in the dialog. Similarly, to delete a reason, select it and click **Delete.** Pop-up message **[Do you want to delete the selected message?]** appears. Click the **OK** button to delete the scheduled/unscheduled route downtime reason.
6	Click **OK** to save changes. 

9.3 **Operation**

Use the mobile scanner to start/stop the Route Downtime of PMD monitor. Select the target route from the Route List on the Route Downtime screen of the mobile scanner. Then, select the downtime name.

- **Unscheduled Down Time**. Indicates time spent attending impromptu meetings, drills, unexpected delays, etc.
- **Scheduled Down Time**. Indicates reasons for attending scheduled activities.

9.3.1 Start Route Downtime

When the Route is Down situation, Route Downtime can be started by mobile scanner operation if the Downtime reason is defined.

Step	Procedure
1	Access the **Production** main menu screen with the operator who has PMD rights. **Production** main menu screen is shown below. 
2	Click the  icon. Following screen appears: 

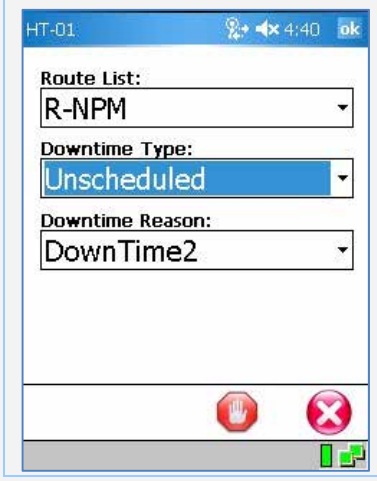
3

Click the  icon. Following screen appears from which Downtime kind and Downtime reason are chosen:

<Downtime kind: Scheduled>



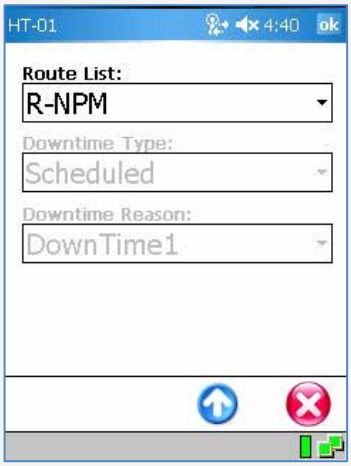
<Downtime kind: Unscheduled>



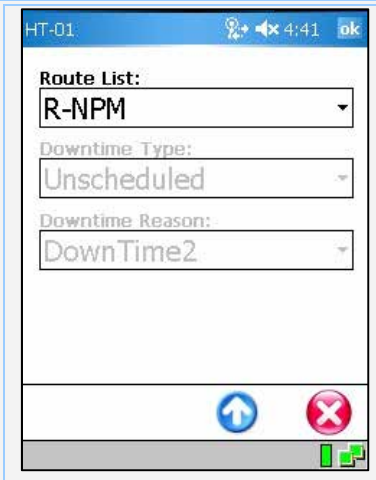
4

Click  to start the downtime.

<Downtime kind: Scheduled>



<Downtime kind: Unscheduled>



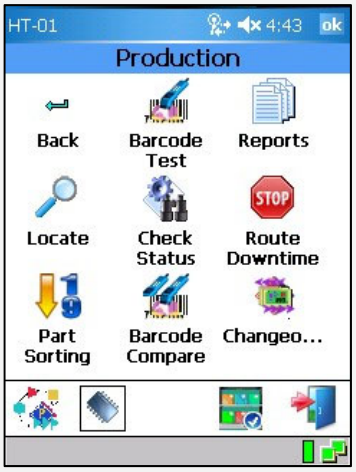
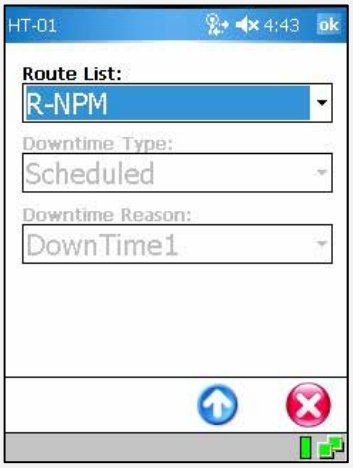
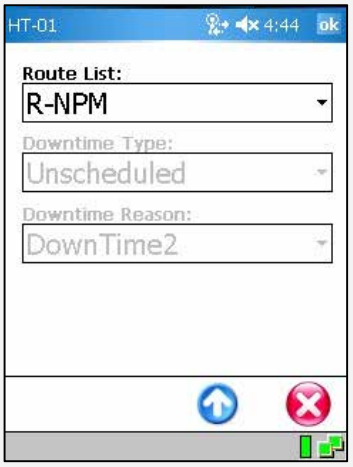
When the downtime is started and the **Downtime kind** and **Downtime reason** fields become inactive. The icon changes to .

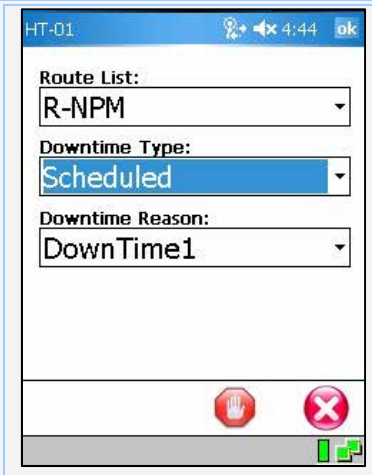
9.3.2 Stop Route Downtime

Stop Route Downtime via a scanner after the route (line) recovered from down.

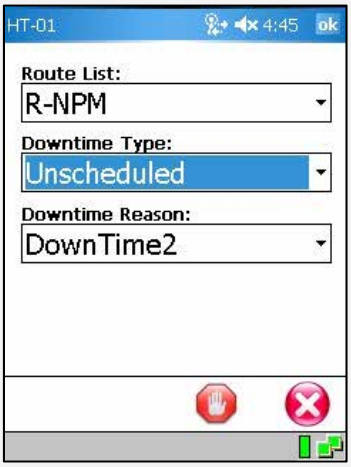
Step	Procedure
1	Access the **Production** main menu screen with the operator who has PMD rights. **Production** main menu screen is shown below.



2	Click the  icon. Following screen is displayed. 
3	Click the  icon. The screen below appears in which Downtime kind and reason are chosen: **<Downtime kind: Scheduled>**  **<Downtime kind: Unscheduled>** 
4	Click  to stop the downtime. **<Downtime kind: Scheduled>**



<Downtime kind: Unscheduled>



When the downtime is stopped and the **Downtime kind** and **Downtime reason** fields become active. The  icon changes to  .

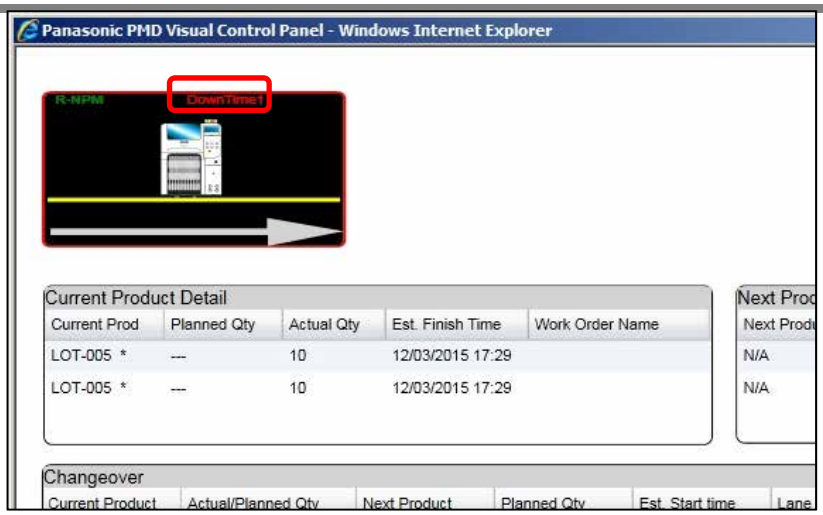
9.3.3 Display on Visual Control Panel (VCP)

Visual Control Panel (VCP) shows the selected reason and visually notifies users that route (line) stopped when Route Downtime was started from a mobile scanner. The reasons disappear after you stop Route Downtime, and the route's (line's) recovery will be notified.

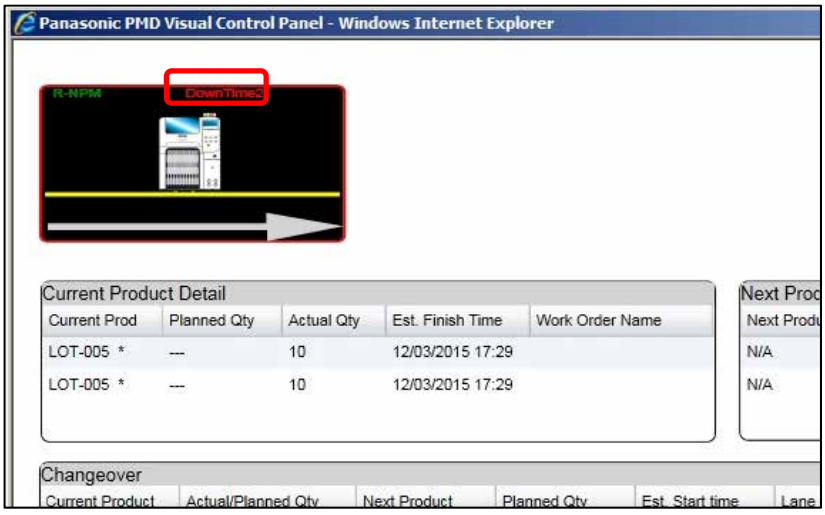
■ Start Route Downtime

Route (Horizontal) Control/Route (Vertical) Control on Visual Control Panel (VCP) show the reason of route downtime selected from the scanner.

<Downtime Kind: Scheduled>



<Downtime kind: Unscheduled>

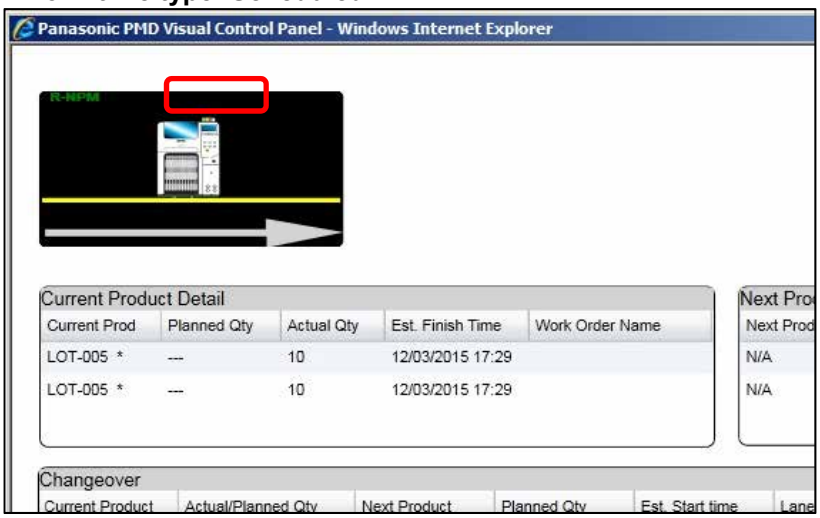


■ Stop Route Downtime

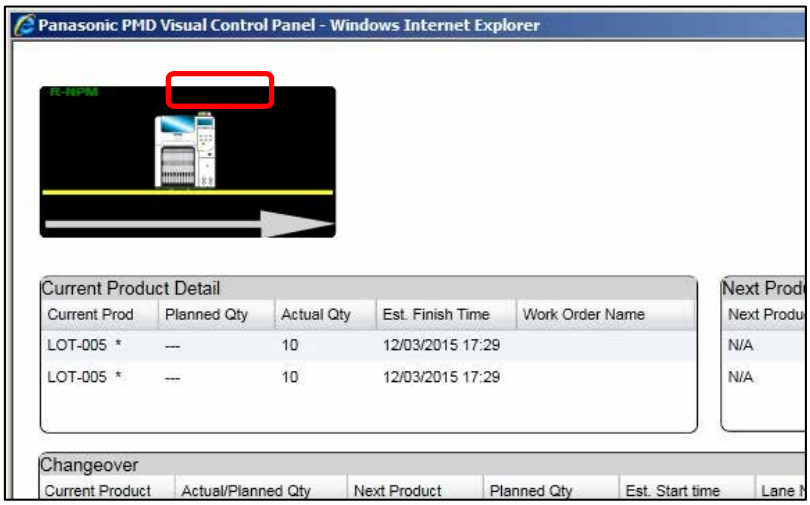
Route downtime reason displayed at Route (Horizontal) Control/Route (Vertical) Control on Visual Control Panel (VCP) disappears.

The reason disappears and the route (line) recovery notified after you stopped Route Downtime.

< Downtime type: Scheduled >



< Downtime kind: Unscheduled >



10. Function - Custom Message

This chapter explains Custom Message. This chapter explains operations for displaying items on the Visual Control Panel (VCP) of the PMD, with Custom Message as an example.

Use the following PanaCIM applications to configure the Custom Message information and to operate the PMD.

- Application Server
- Visual Control Panel

10.1 Overview

Users set custom message in advance from a mobile scanner so PMD [Route Downtime] displays the target route in down status on Visual Control Panel (CVP) and visually notifies users of such condition.

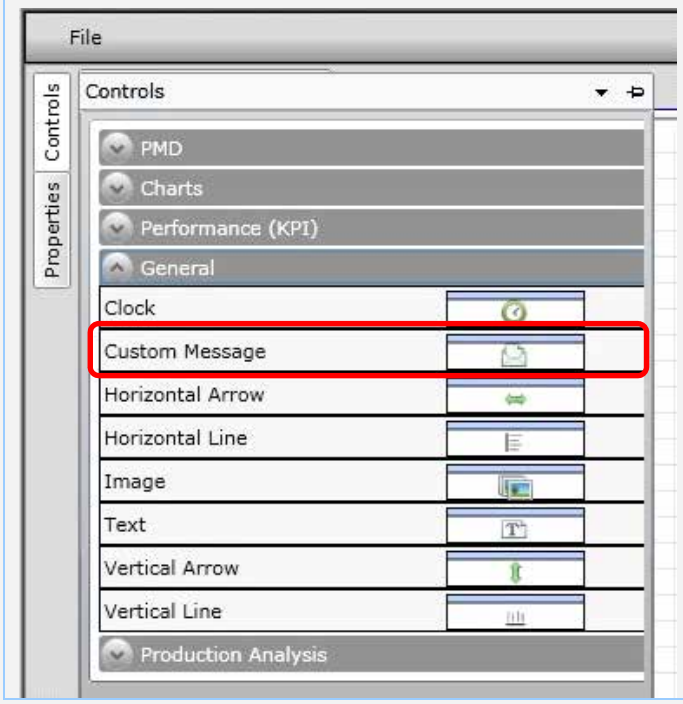
10.2 Settings

Take the following steps on the application server to register the custom messages beforehand.

10.2.1 Template Designer

Add controls to display information on Visual Control Panel (VCP) in coordination with work orders.

* See [5.3 Operation] for how to open or save templates.

Step	Operation
1	Click [Controls] tab on the left side of [PMD Template Designer] screen to open [General]. 
2	Add the following controls to display custom messages on Operation Monitor Control. • **Custom Message:** Shows text messages. Users can also specify time to show the text message. * This control can be displayed for multiple routes.

10.3 Operation

Take the following steps on the application server to display and clear the text messages on the visual control panel (VCP).

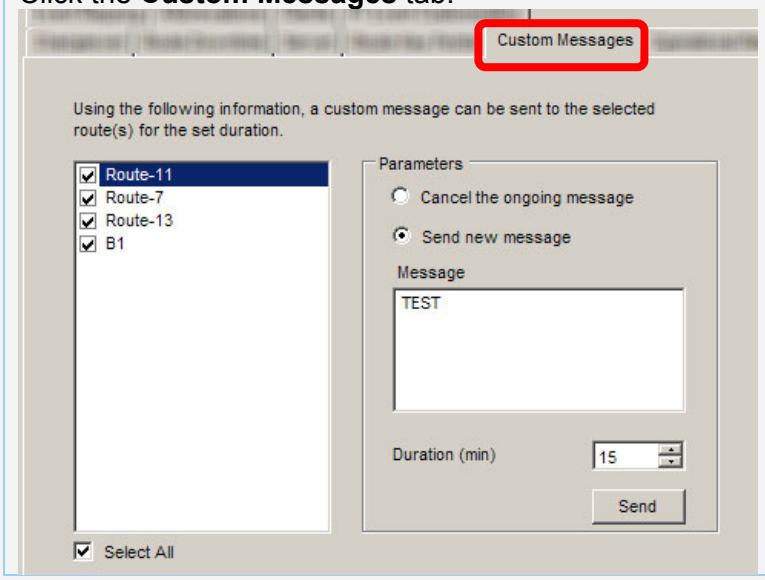
Select the Custom Messages tab to:

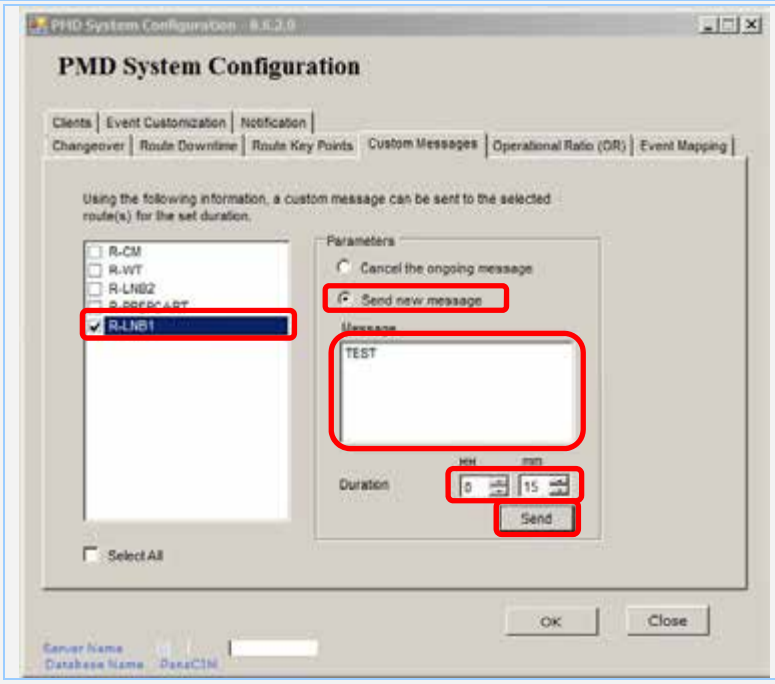
- Send the custom message to the selected route(s) for the set duration.
- Cancel the ongoing messages to the selected route(s).

Note: Custom messages work in the same way irrespective of whether these are sent from the server or the client.

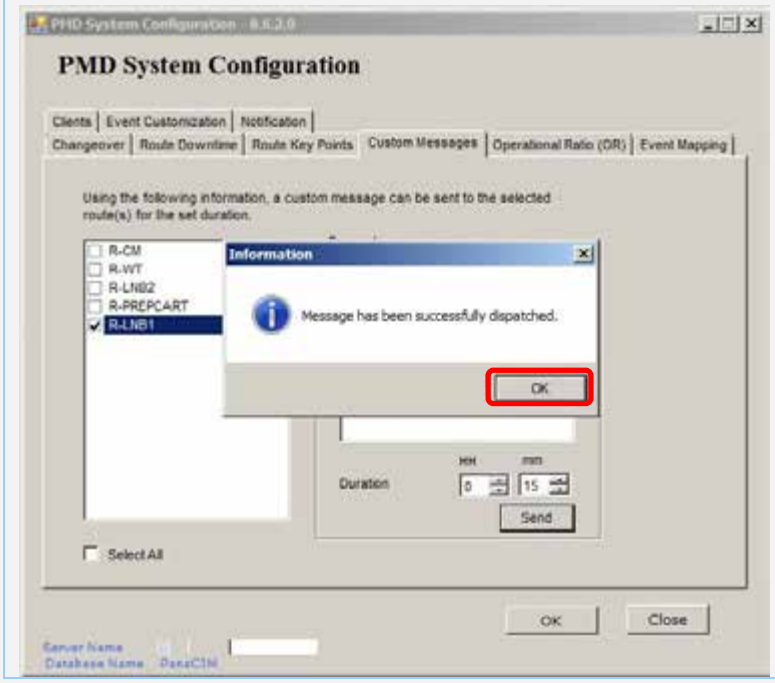
10.3.1 Send Custom Message

Add Custom Messages controls to display information on Visual Control Panel (VCP) in coordination with work orders.
Refer to the action table below to send custom messages.

Step	Procedure
1	Click the **Custom Messages** tab. 
2	Select the desires route(s), select the **[Send new message]** option, and type in the following items. Then, click the **Send** button. ・Message: In the **Message** field, enter a message to be displayed. ・Duration (m): In the **Duration** field, enter the time in minutes in which message will be sent.



3	Pop-up message [Message has been successfully dispatched] is displayed. Press the **OK** button.

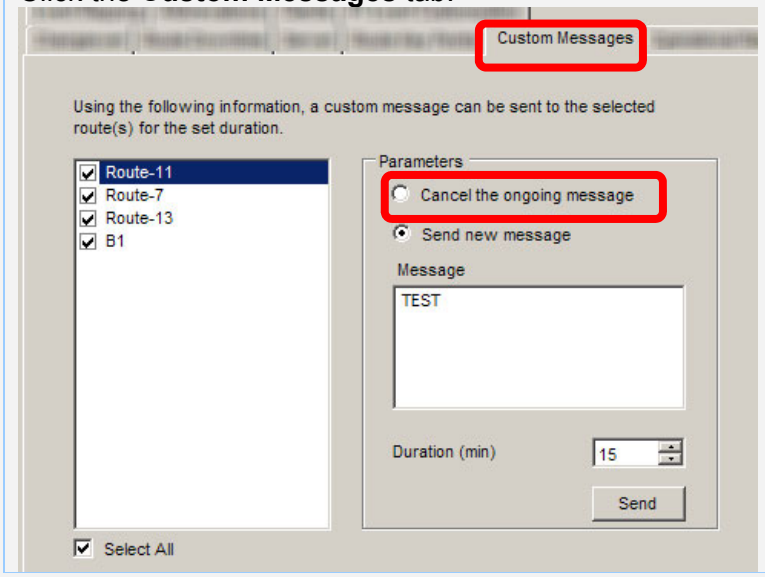
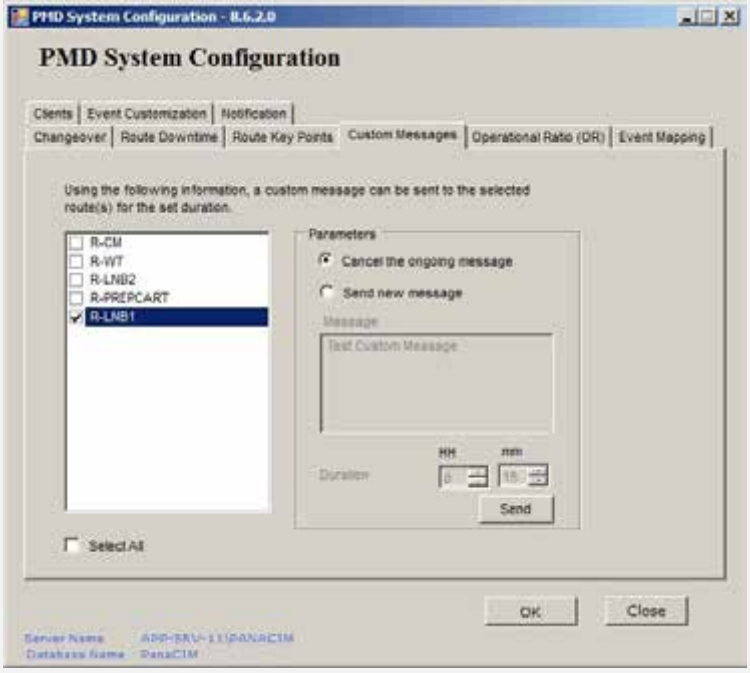


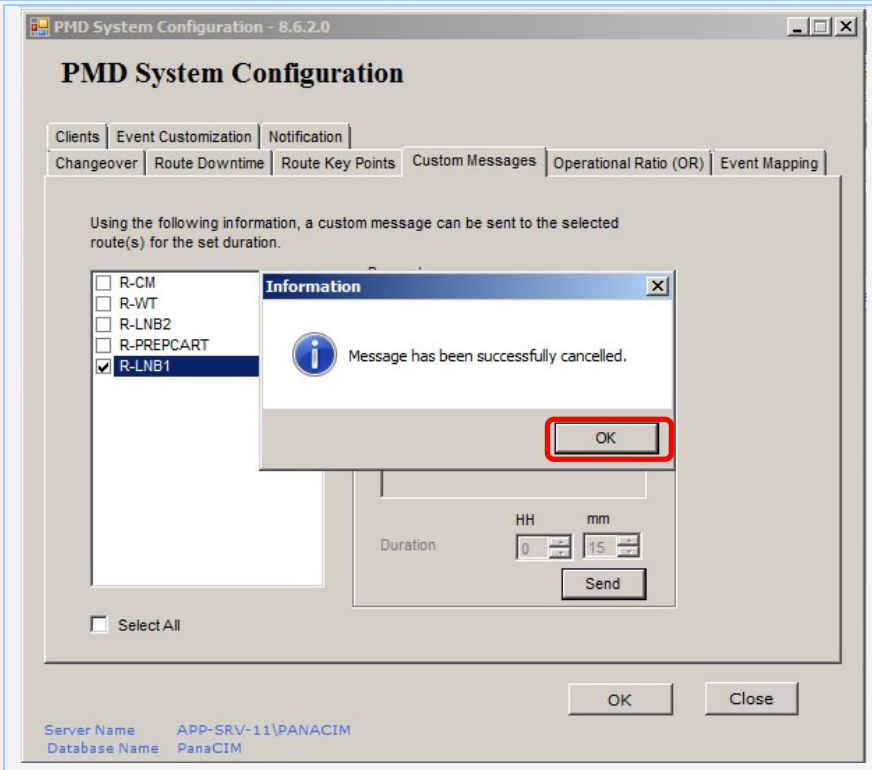
4	Click **OK** to save.

10.3.2 Cancel the Custom Message

The message displayed on the Custom Message control on the visual control panel (VCP) can be cleared.

Refer to the following table to cancel the custom message.

Step	Procedure
1	Click the **Custom Messages** tab. 
2	Select the **Cancel the ongoing message** option and click **Send**. 
3	The below dialog “Message has been successfully cancelled” is displayed. Click **OK.**



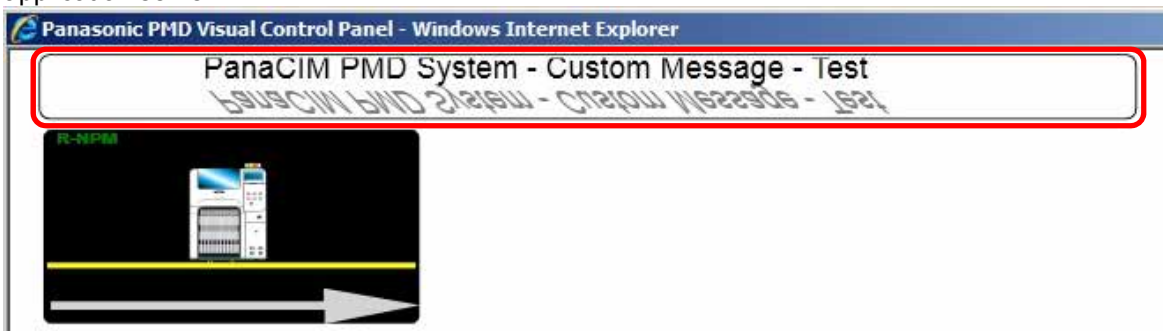
4	Click OK to save.

10.3.3 Display on Visual Control Panel(VCP)

Display/delete messages on/from [Custom Message] control area on Visual Control Panel (VCP) according to the operations down on the application server side.

■ Sending Custom Messages

Visual Control Panel (VCP) Custom Message Control shows the message sent from the application server.



The message disappears automatically after a certain time (time specified when the message sent).

■ Stopping Custom Messages

The message disappears from the custom message control on Visual Control Panel (VCP).



11. Function - Template Explorer

This chapter explains Template Explorer.

11.1 Overview

PMD operation runs using various applications and utilities.

11.2 Operation

PMD operation runs using various applications and utilities.

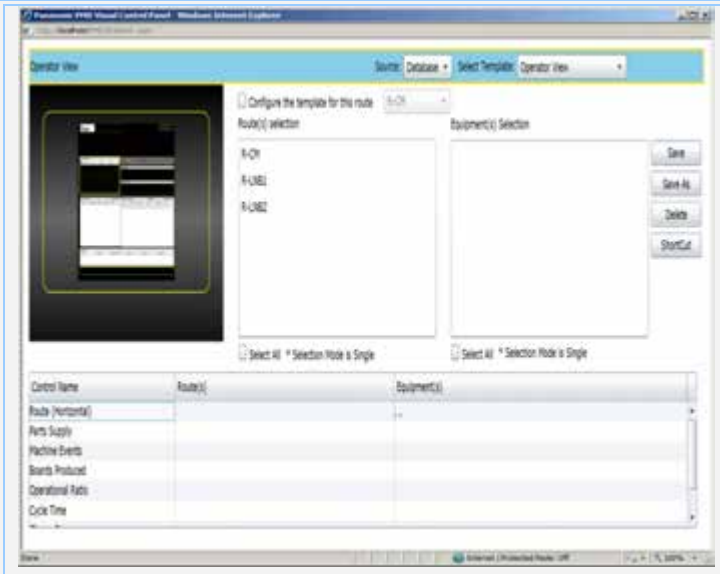
Template Explorer allows users to open existing templates and modify them as necessary.
Additionally, templates can be imported from, or exported to, other users.

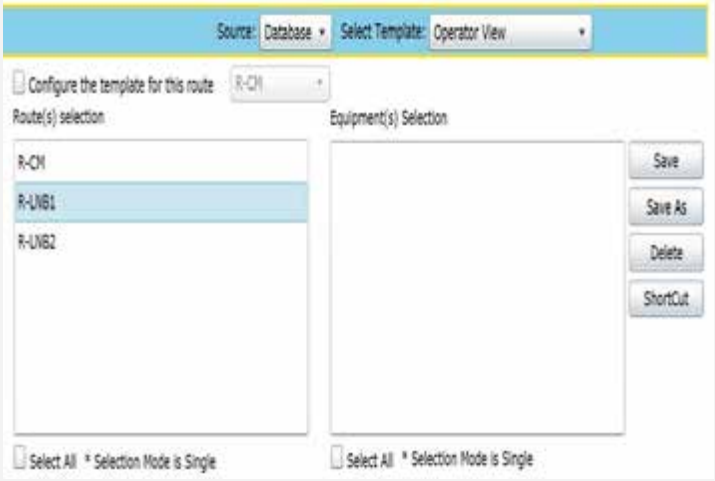
11.2.1 Template Explorer Import/Export

Refer to the action table below to map previously created templates to different route(s) and equipment(s) using Template Explorer. You may overwrite existing templates, or use the Save As feature to add a new template.

Note: Template Explorer does not provide design capabilities. Modifications can be made using the template designer mentioned already.

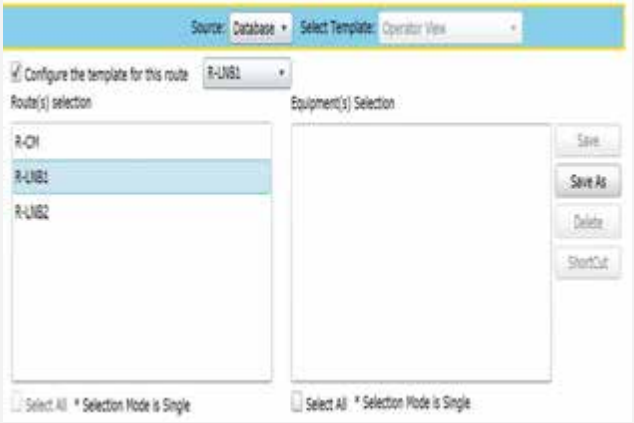
Step	Procedure
1	(To install templates created by other users) Copy the folder containing the template data file to the directory below on the application server. C:¥Program Files (x86)¥Panasonic¥PanaCIM¥FW¥PMD¥TemplatesSL¥Resources
2	Select the **Template Explorer** Application from the Production Monitoring & Dispatch link in the PanaCIM EE Launchpad. 
3	Template Explorer appears. Select the target template from the Source / Select Template drop-down boxes at the top-right conrer of the page. The selected template is displayed as a thumbnail. Source: Database (within this application) / Folder (files imported from other PCs)



4	Select one of the following. • When configuring the template for a single route a. Mark the checkbox for “Configure the template for this route.” b. Select the route to be configured from the **drop-down**. c. The Route(s) Selection box will be disabled. The Equipment(s) Selection box for the selected route will be displayed.  • When configuring the template for multiple routes and equipment types Conflicting setting information is displayed in the Route(s) selection / Equipment Selection Control grid. The Control grid considers the following: • There are some that can be configured for a single route or multiple routes. • There are some that can be configured for a single type or multiple types of equipment. • There are some that do not require any route / equipment selection. ***If you select this option, you will have to select the name of each control, and perform steps 5 to 7 below.***

Control Name	Route(s)	Equipment(s)
Route (Horizontal)	R-LNB1	Z-LNB1-M
Parts Supply	R-LNB1	
Machine Events	R-LNB1	
Boards Produced	R-LNB1	
Operational Ratio	R-LNB1	
Cycle Time	R-LNB1	

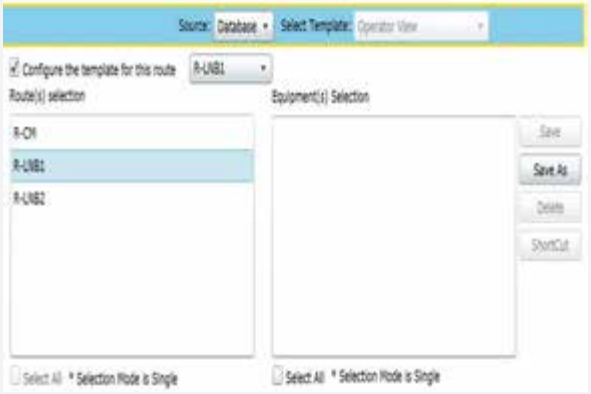
5 The Route(s) Selection box is enabled where you can select the route. When it can be configured for multiple routes, "*Selection Mode Is 'Multiple'." is displayed next to Select All under the box.
In such case, you should select multiple routes while holding down the Ctrl key. When selecting all routes, mark the Select All checkbox under the box.



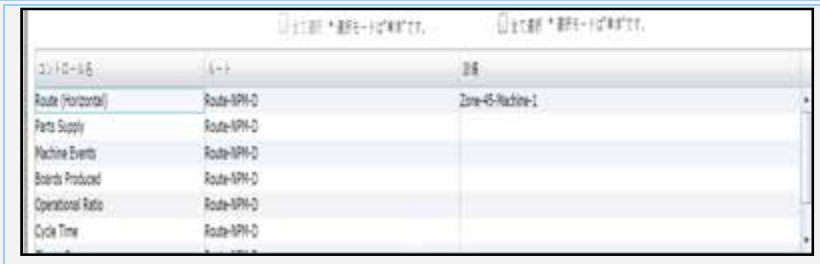
As each new route is selected, it populates into the Control grid.

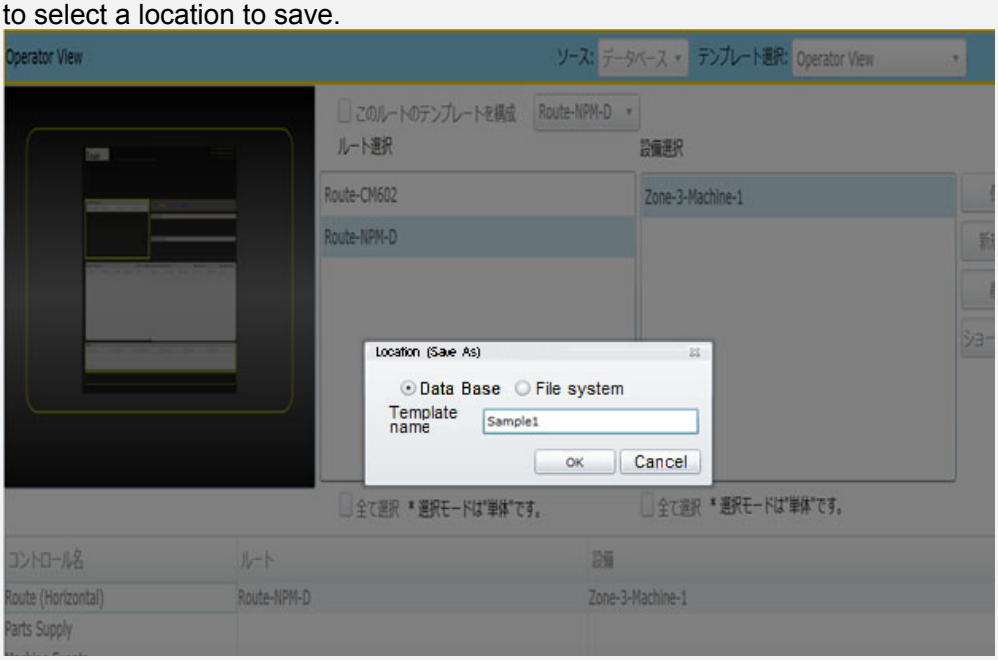
Control Name	Route(s)	Equipment(s)
Route (Horizontal)	R-LNB1	Z-LNB1-M
Parts Supply	R-LNB1	
Machine Events	R-LNB1	
Boards Produced	R-LNB1	
Operational Ratio	R-LNB1	
Cycle Time	R-LNB1	

6 If a control can accommodate more than a single equipment type, the **Equipment(s) Selection** box enables. When it can be configured for multiple types of equipment, "*Selection Mode Is 'Multiple'." is displayed next to Select All under the box.
In such case, select multiple types of equipment while holding down the Ctrl key. When selecting all equipment types, mark the Select All checkbox under the box.



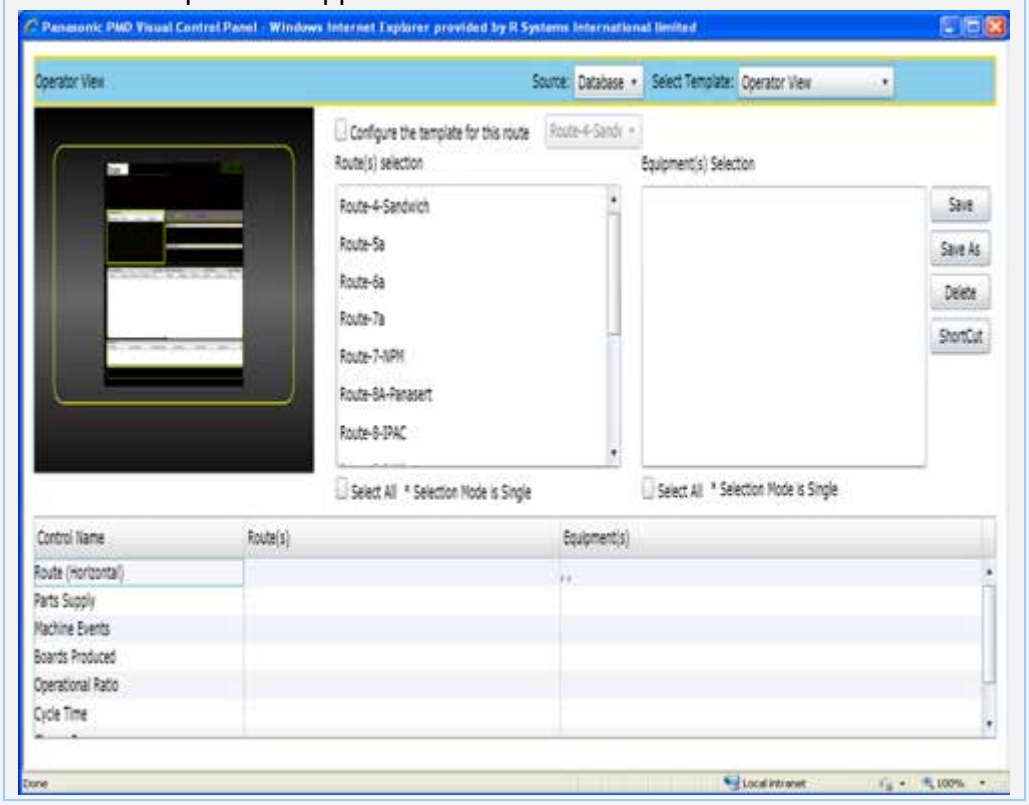
As each new type of equipment is selected, it populates into the Control grid.

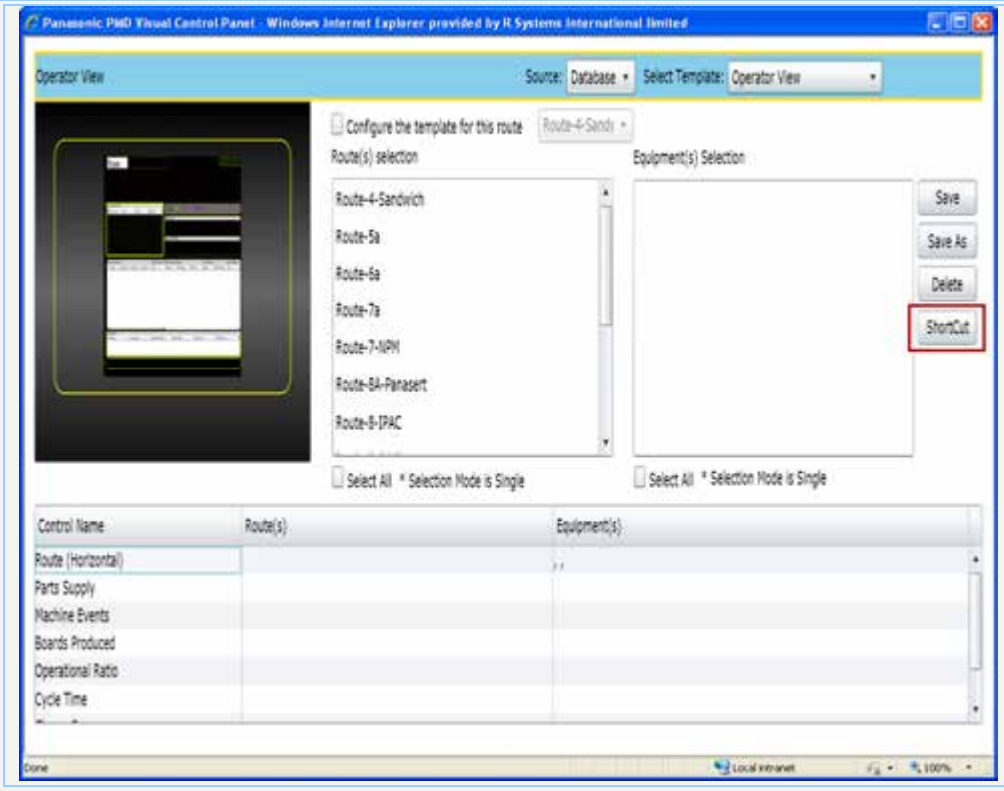


7	**Select the next control name,** and repeat steps 5 and 6. Upon completion of configuring settings for all the controls, proceed to step 8.
8	Upon completion of selecting route(s) / equipment type(s) for each control: Clicking the **Save** button overwrites existing templates (This cannot be selected for the templates imported from folders.); clicking the **OK** button saves your changes; clicking the **Cancel** button discards all changes made; and clicking the **Save As** button saves new templates. The dialog box below appears asking you to select a location to save.  Selecting Database saves it as the template for the Web application in use. Selecting File System generates a folder with the template name in the following directory on the application server and saves it in the folder. C:¥Program Files (x86) ¥Panasonic¥PanaCIM¥FW¥PMD¥TemplatesSL¥Resources Clicking the **OK** button saves your changes. Clicking the **Cancel** button discards all changes made. Clicking the **Delete** button deletes the selected template within the Web application in use. (Any imported file system cannot be deleted.)

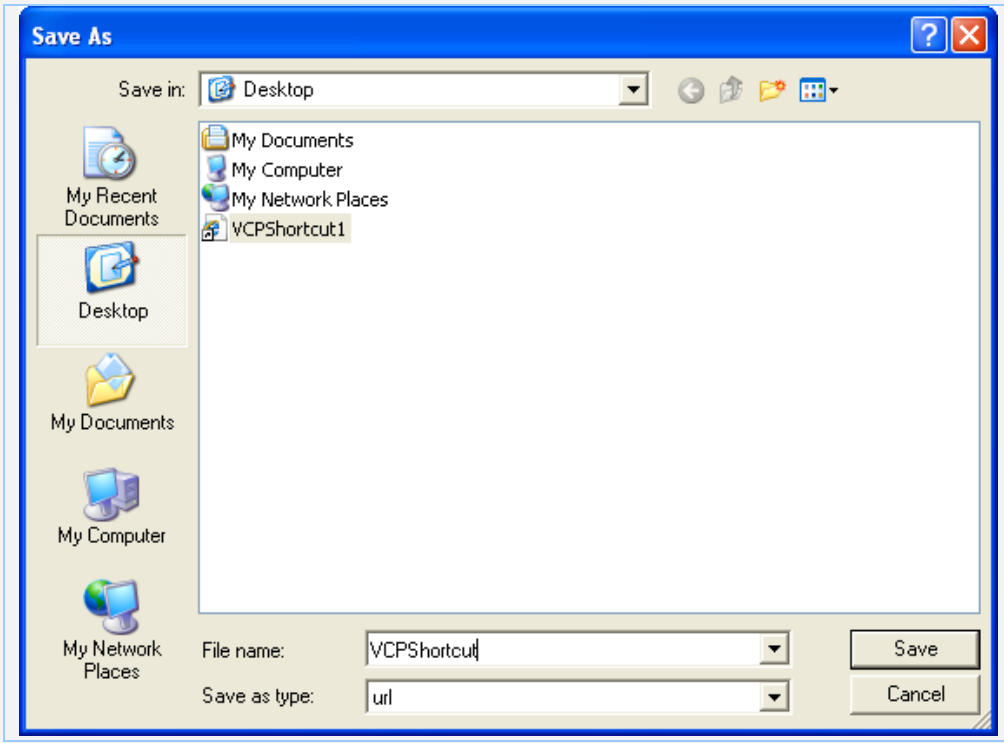
11.2.2 PMD VCP Shortcut

Refer to the action table below to create PMD VCP Shortcut

Step	Procedure
1	Select the **Template Explorer** Application from the Production Monitoring & Dispatch link in the PanaCIM EE Launchpad.
2	The Template Explorer page displays. **Select a template** from the drop-down selection box at the top of the page. A thumbnail view of the selected template will appear. 
3	Click on **Shortcut** button



4	Standard File Save as Dialog pops up



5	Enter the name of the VCP shortcut in the File name textbox and click the **Save** button.
6	As shown in step 4, shortcut is saved as VCPShortcut.
7	You will see the VCPShortcut ICON on your desktop

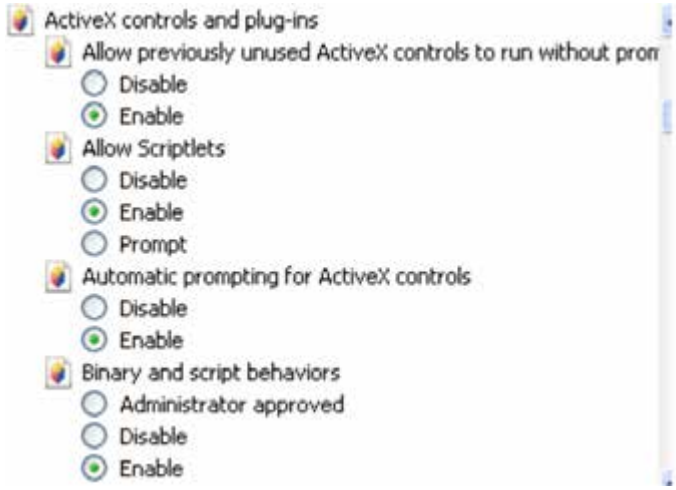


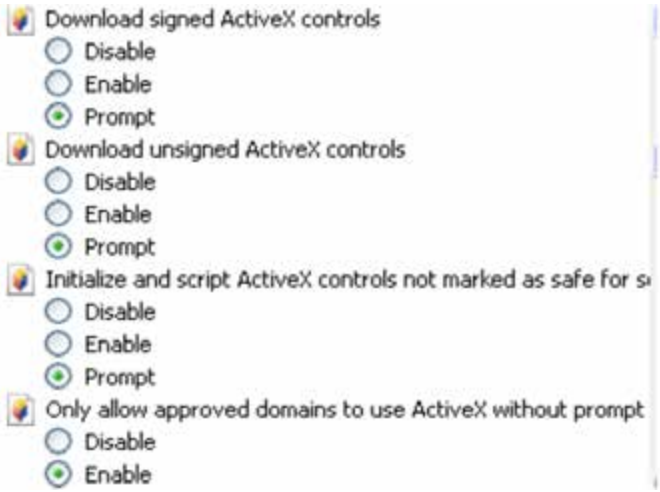
12. APPENDIX

12.1 Setting for ActiveX and JavaScript

■ FOR LOCAL URL: http://localhost/PMD

1. Open Internet option
2. Select Security tab
3. Select Local internet icon
4. Click on custom level
5. Do setting as below







JavaScript



■ FOR CLIENT SERVER MODE : http://ipaddress/pmd

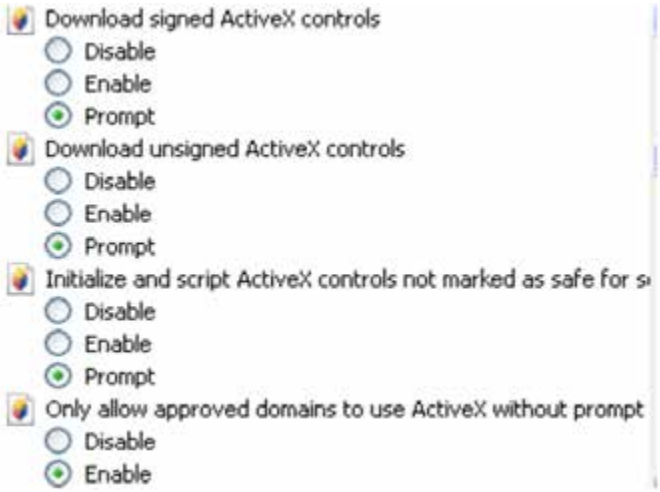
SETTING AT CLIENT PC

1. Open Internet option
2. Select Security tab
3. Select Trusted Site
4. Click on Sites button



5. Write the PMD url
6. Unchecked the check box of Require server verification
7. Click on Add button
8. Click on Close button
9. Click on custom level
10. Do setting as below


ActiveX controls and plug-ins
Allow previously unused ActiveX controls to run without pron
Disable
Enable
Allow Scriptlets
Disable
Enable
Prompt
Automatic prompting for ActiveX controls
Disable
Enable
Binary and script behaviors
Administrator approved
Disable
Enable


Download signed ActiveX controls
Disable
Enable
Prompt
Download unsigned ActiveX controls
Disable
Enable
Prompt
Initialize and script ActiveX controls not marked as safe for s
Disable
Enable
Prompt
Only allow approved domains to use ActiveX without prompt
Disable
Enable


Allow script-initiated windows without size or position constra
Disable
Enable
Allow webpages to use restricted protocols for active conten
Disable
Enable
Prompt
Allow websites to open windows without address or status b
Disable
Enable
Display mixed content
Disable
Enable
Prompt

JavaScript


Scripting
Active scripting
Disable
Enable
Prompt
Allow Programmatic clipboard access
Disable
Enable
Prompt
Allow status bar updates via script
Disable
Enable
Allow websites to prompt for information using scripted windo
Disable
Enable

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